

(12) **United States Patent**
Kalina, Jr. et al.

(10) **Patent No.:** **US 12,414,798 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEMS AND METHODS FOR
DELIVERING MULTIPLE OCULAR
IMPLANTS**

(71) Applicant: **GLAUKOS CORPORATION**, Aliso
Viejo, CA (US)

(72) Inventors: **Charles Raymond Kalina, Jr.**, Irvine,
CA (US); **Todd N. Fjield**, Irvine, CA
(US); **Douglas Daniel Crimaldi**, San
Marcos, CA (US); **Emma Claire
Benjaminson**, Laguna Hills, CA (US);
Tyler Joseph Brown, Dana Point, CA
(US); **Richard William Martin**, San
Marcos, CA (US)

(73) Assignee: **Glaukos Corporation**, Aliso Viejo, CA
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 618 days.

(21) Appl. No.: **17/810,035**

(22) Filed: **Jun. 30, 2022**

(65) **Prior Publication Data**

US 2022/0330979 A1 Oct. 20, 2022

Related U.S. Application Data

(62) Division of application No. 16/132,252, filed on Sep.
14, 2018, now Pat. No. 11,376,040.
(Continued)

(51) **Int. Cl.**
A61B 17/34 (2006.01)
A61B 17/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61B 17/3468** (2013.01); **A61F 9/0017**
(2013.01); **A61F 9/007** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC A61B 17/3468; A61B 2017/00367; A61F
9/007; A61F 9/0017; A61F 9/00781;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,269,963 A 1/1942 Frederick
3,416,530 A 12/1968 Ness
(Continued)

FOREIGN PATENT DOCUMENTS

AU 200072059 12/2000
AU 2004264913 12/2011
(Continued)

OTHER PUBLICATIONS

Bucciarelli, Patrice D., "Working Model is Next Step in Team's
Long Journey to Commercial Product", Healthfirst, Business First
of Louisville, louisville.bizjournals.com, Feb. 27, 2004.
(Continued)

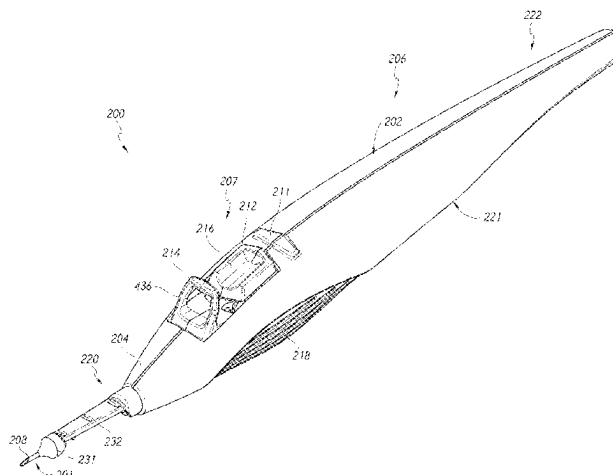
Primary Examiner — Adam Marcetich

Assistant Examiner — Timothy L Flynn

(74) *Attorney, Agent, or Firm* — Knobbe, Martens, Olson
& Bear, LLP

(57) **ABSTRACT**

Systems and methods for delivering multiple ocular
implants to reduce intraocular pressure are disclosed. The
ocular implants can be implanted at multiple sites within a
single human eye without requiring removal of the delivery
apparatus from the eye. A system for delivering multiple
ocular implants can include a plurality of implants, an
external housing, and an introducer assembly. The external
housing can provide access to an implant singulation actua-
tor and an implant delivery actuator. The introducer assem-
bly can comprise an auto-retracting introducer portion. The
(Continued)



delivery apparatus can include an infinite activation mechanism/portion and/or a manual singulation mechanism/portion.

20 Claims, 40 Drawing Sheets

Related U.S. Application Data

- (60) Provisional application No. 62/671,286, filed on May 14, 2018, provisional application No. 62/578,273, filed on Oct. 27, 2017, provisional application No. 62/569,458, filed on Oct. 6, 2017.

(51) Int. Cl.

A61F 9/00 (2006.01)

A61F 9/007 (2006.01)

(52) U.S. Cl.

CPC *A61F 9/00781* (2013.01); *A61B 2017/00367* (2013.01); *A61F 2250/0064* (2013.01)

(58) Field of Classification Search

CPC *A61F 2250/0064*; *A61F 2250/0067*; *A61F 2230/0069*; *A61F 2220/0008*; *A61K 31/215*

See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

3,439,675 A	4/1969	Cohen	5,486,165 A	1/1996	Stegmann
3,948,271 A	4/1976	Akiyama	5,556,400 A	9/1996	Tunis
3,948,871 A	4/1976	Butterfield et al.	5,558,637 A	9/1996	Allonen et al.
3,976,077 A	8/1976	Kerfoot, Jr.	5,626,588 A	5/1997	Sauer et al.
4,113,088 A	9/1978	Binkhorst	5,643,321 A	7/1997	McDevitt
4,299,227 A	11/1981	Lincoff	5,651,782 A	7/1997	Simon et al.
4,366,582 A	1/1983	Faulkner	5,651,783 A	7/1997	Reynard
4,449,529 A	5/1984	Burns et al.	5,653,724 A	8/1997	Imonti
4,501,274 A	2/1985	Skjaerpe	5,669,501 A	9/1997	Hissong et al.
4,560,383 A	12/1985	Leiske	5,676,679 A	10/1997	Simon et al.
4,578,058 A	3/1986	Grandon	5,681,323 A	10/1997	Arick
4,634,418 A	1/1987	Binder	5,695,479 A	12/1997	Jagpal
4,642,090 A	2/1987	Ultrata	5,702,414 A	12/1997	Richter et al.
4,782,819 A	11/1988	Adair	5,702,419 A	12/1997	Berry et al.
4,800,870 A	1/1989	Reid, Jr.	5,725,546 A	3/1998	Samson
4,800,890 A	1/1989	Cramer	5,733,256 A	3/1998	Costin
4,846,172 A	7/1989	Berlin	5,741,282 A	4/1998	Anspach, III et al.
4,846,793 A	7/1989	Leonard et al.	5,741,292 A	4/1998	Mendius
4,867,173 A	9/1989	Leoni	5,762,625 A	6/1998	Igaki
4,870,953 A	10/1989	DonMichael et al.	5,792,099 A	8/1998	DeCamp et al.
4,900,300 A	2/1990	Lee	5,807,244 A	9/1998	Barot
4,905,667 A	3/1990	Foerster et al.	5,817,100 A	10/1998	Gaki
4,991,602 A	2/1991	Amplatz et al.	5,833,694 A	11/1998	Poncet
5,053,040 A	10/1991	Goldsmith, III	5,836,939 A	11/1998	Negus et al.
5,053,044 A	10/1991	Mueller et al.	D402,757 S	12/1998	Davis et al.
5,095,887 A	3/1992	Leon et al.	5,846,199 A	12/1998	Hijlkema et al.
5,129,895 A	7/1992	Vassiliadis et al.	5,865,831 A	2/1999	Cozean et al.
5,169,386 A	12/1992	Becker et al.	5,868,697 A	2/1999	Ritcher et al.
5,180,362 A	1/1993	Worst	5,891,084 A	4/1999	Lee
5,207,685 A	5/1993	Cinberg et al.	5,893,837 A	4/1999	Eagles et al.
5,221,255 A	6/1993	Mahurkar et al.	5,927,585 A	7/1999	Moorman et al.
5,246,451 A	9/1993	Trescony et al.	5,941,250 A	8/1999	Aramant et al.
5,284,476 A	2/1994	Koch	5,984,913 A	11/1999	Kritzinger et al.
5,290,295 A	3/1994	Querals et al.	6,004,302 A	12/1999	Brierley
5,324,306 A	6/1994	Makower et al.	6,030,416 A	2/2000	Huo et al.
5,342,370 A	8/1994	Simon et al.	6,036,678 A	3/2000	Giungo
5,360,399 A	11/1994	Stegmann	6,036,682 A	3/2000	Lange et al.
5,409,457 A	4/1995	del Cerro et al.	6,045,557 A	4/2000	White et al.
5,415,666 A	5/1995	Gourlay et al.	6,050,999 A	4/2000	Paraschac et al.
5,423,330 A	6/1995	Lee	6,071,286 A	6/2000	Mawad
5,445,637 A	8/1995	Bretton	6,074,395 A	6/2000	Trott et al.
5,462,558 A	10/1995	Kolesa et al.	6,135,977 A	10/2000	Drasler et al.
5,472,440 A	12/1995	Beckman	6,142,990 A	11/2000	Burk
			6,146,387 A	11/2000	Trott et al.
			6,187,016 B1	2/2001	Hedges et al.
			6,203,513 B1	3/2001	Yaron et al.
			6,221,078 B1	4/2001	Bylsma
			6,224,570 B1	5/2001	Le et al.
			6,254,612 B1	7/2001	Hieshima
			6,264,668 B1	7/2001	Prywes
			6,287,313 B1	9/2001	Sasso
			6,290,684 B1	9/2001	Herrick
			6,299,603 B1	10/2001	Hecker et al.
			6,342,058 B1	1/2002	Portney
			6,355,033 B1	3/2002	Moorman et al.
			6,361,519 B1	3/2002	Knudson et al.
			6,363,938 B2	4/2002	Saadat
			6,375,642 B1	4/2002	Grieshaber et al.
			6,378,526 B1	4/2002	Bowman et al.
			6,402,734 B1	6/2002	Weiss
			6,405,732 B1	6/2002	Edwards et al.
			6,428,501 B1	8/2002	Reynard
			6,428,566 B1	8/2002	Holt
			6,450,937 B1	9/2002	Mercereau et al.
			6,450,984 B1	9/2002	Lynch et al.
			6,454,787 B1	9/2002	Maddalo et al.
			6,464,724 B1	10/2002	Lynch et al.
			6,524,275 B1	2/2003	Lynch et al.
			6,530,896 B1	3/2003	Elliott
			6,544,249 B1	4/2003	Yu et al.
			6,582,426 B2	6/2003	Moorman et al.
			6,585,680 B2	7/2003	Bugge
			6,589,203 B1	7/2003	Mitrev
			6,607,542 B1	8/2003	Wild
			6,613,343 B2	9/2003	Dillingham et al.
			6,620,154 B1	9/2003	Amirkhanlan et al.
			6,626,858 B2	9/2003	Lynch et al.
			6,629,981 B2	10/2003	Bui et al.
			6,638,239 B1	10/2003	Bergheim et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,666,841	B2	12/2003	Gharib et al.	8,333,742	B2	12/2012	Bergheim et al.
6,676,607	B2	1/2004	De Juan, Jr. et al.	8,337,445	B2	12/2012	Tu et al.
6,699,272	B2	3/2004	Slepian et al.	8,348,877	B2	1/2013	Tu et al.
6,726,676	B2	4/2004	Stegmann et al.	8,388,568	B2	3/2013	Lynch et al.
D490,152	S	5/2004	Myall et al.	8,506,515	B2	8/2013	Burns et al.
6,730,056	B1	5/2004	Ghaem et al.	8,540,659	B2	9/2013	Berlin
6,736,791	B1	5/2004	Tu et al.	8,579,846	B2	11/2013	Tu et al.
6,763,833	B1	7/2004	Khera et al.	8,617,094	B2	12/2013	Smedley et al.
6,764,439	B2	7/2004	Schaaf et al.	8,679,089	B2	3/2014	Berlin
6,767,346	B2	7/2004	Damasco et al.	8,771,217	B2	7/2014	Lynch et al.
6,780,164	B2	8/2004	Bergheim et al.	8,801,648	B2	8/2014	Bergheim et al.
6,780,165	B2	8/2004	Kadziauskas et al.	8,801,649	B2	8/2014	De Juan, Jr. et al.
6,783,544	B2	8/2004	Lynch et al.	8,808,219	B2	8/2014	Bergheim et al.
6,827,699	B2	12/2004	Lynch et al.	8,814,820	B2	8/2014	Bergheim et al.
6,827,700	B2	12/2004	Lynch et al.	8,852,137	B2	10/2014	Horvath et al.
6,827,738	B2	12/2004	Willis et al.	8,852,266	B2	10/2014	Brooks et al.
6,902,577	B2	6/2005	Lipshitz et al.	8,882,781	B2	11/2014	Smedley et al.
6,939,298	B2	9/2005	Brown et al.	8,998,983	B2	4/2015	Auld
6,955,656	B2	10/2005	Bergheim et al.	9,066,782	B2	6/2015	Tu et al.
6,966,888	B2	11/2005	Cullen	9,155,654	B2	10/2015	Tu et al.
6,969,384	B2	11/2005	De Juan, Jr. et al.	9,155,656	B2	10/2015	Schaller et al.
6,981,958	B1	1/2006	Gharib et al.	9,173,775	B2	11/2015	Haffner et al.
7,077,821	B2	7/2006	Durgin	9,220,632	B2	12/2015	Smedley et al.
7,077,848	B1	7/2006	de Juan et al.	9,301,875	B2	4/2016	Tu et al.
7,090,681	B2	8/2006	Weber et al.	9,492,320	B2	11/2016	Lynch et al.
7,094,225	B2	8/2006	Tu et al.	9,554,940	B2	1/2017	Haffner et al.
7,135,009	B2	11/2006	Tu et al.	9,561,131	B2	2/2017	Tu et al.
7,135,016	B1	11/2006	Asia et al.	9,572,963	B2	2/2017	Tu et al.
7,163,543	B2	1/2007	Smedley et al.	9,592,151	B2	3/2017	Rangel-Friedman et al.
7,186,232	B1	3/2007	Smedley et al.	9,597,230	B2	3/2017	Haffner et al.
7,192,412	B1	3/2007	Zhou et al.	9,603,738	B2	3/2017	Haffner et al.
7,217,263	B2	5/2007	Humayun et al.	9,636,255	B2	5/2017	Haffner et al.
7,220,238	B2	5/2007	Lynch et al.	9,668,915	B2	6/2017	Haffner et al.
7,273,475	B2	9/2007	Tu et al.	9,693,899	B2	7/2017	Wardle et al.
7,297,130	B2	11/2007	Bergheim et al.	9,730,638	B2	8/2017	Haffner et al.
7,331,984	B2	2/2008	Tu et al.	9,789,001	B2	10/2017	Tu et al.
7,344,528	B1	3/2008	Tu et al.	9,827,143	B2	11/2017	Lynch et al.
7,431,710	B2	10/2008	Tu et al.	9,849,027	B2	12/2017	Highley et al.
7,458,953	B2	12/2008	Peyman	9,962,290	B2	5/2018	Burns et al.
7,468,065	B2	12/2008	Weber et al.	9,987,472	B2	6/2018	Tu et al.
7,488,303	B1	2/2009	Haffner et al.	9,993,368	B2	6/2018	Bergheim et al.
7,520,876	B2	4/2009	Reesemann et al.	D833,008	S	11/2018	Kalina, Jr. et al.
D592,746	S	5/2009	Highley et al.	10,159,601	B2	12/2018	Berlin
7,563,241	B2	7/2009	Tu et al.	10,188,551	B2	1/2019	Rangel-Friedman et al.
D606,190	S	12/2009	Pruitt et al.	10,206,813	B2	2/2019	Haffner et al.
7,678,065	B2	3/2010	Haffner et al.	D846,738	S	4/2019	Kalina, Jr. et al.
7,708,711	B2	5/2010	Tu et al.	10,245,178	B1	4/2019	Heitzmann et al.
7,713,228	B2	5/2010	Robin	10,271,989	B2	4/2019	Haffner et al.
7,758,624	B2	7/2010	Dorn et al.	10,285,853	B2	5/2019	Rangel-Friedman et al.
7,771,388	B2	8/2010	Olsen et al.	10,285,856	B2	5/2019	Tu et al.
7,850,637	B2	12/2010	Lynch et al.	10,406,029	B2	9/2019	Tu et al.
7,857,782	B2	12/2010	Tu et al.	10,485,701	B2	11/2019	Haffner et al.
7,867,186	B2	1/2011	Haffner et al.	10,485,702	B2	11/2019	Bergheim et al.
7,867,205	B2	1/2011	Bergheim et al.	10,492,950	B2	12/2019	Lynch et al.
7,879,001	B2	2/2011	Haffner et al.	10,499,809	B2	12/2019	Kalina, Jr. et al.
7,879,079	B2	2/2011	Tu et al.	10,517,759	B2	12/2019	Crimaldi et al.
7,905,904	B2	3/2011	Stone et al.	10,568,762	B2	2/2020	Lynch et al.
7,931,660	B2	4/2011	Aranyi et al.	D886,997	S	6/2020	Kalina, Jr. et al.
7,945,336	B2	5/2011	Sauter/Starace et al.	10,674,906	B2	6/2020	Kalina, Jr. et al.
7,951,155	B2	5/2011	Smedley et al.	10,780,218	B2	9/2020	Novakovic et al.
7,959,632	B2	6/2011	Fugo	10,813,789	B2	10/2020	Haffner et al.
7,967,772	B2	6/2011	McKenzie et al.	D901,683	S	11/2020	Kalina, Jr. et al.
8,007,459	B2	8/2011	Haffner et al.	10,828,195	B2	11/2020	Burns et al.
D645,489	S	9/2011	Gille et al.	10,828,473	B2	11/2020	Haffner et al.
D645,490	S	9/2011	Gille et al.	10,959,941	B2	3/2021	Haffner
8,062,244	B2	11/2011	Tu et al.	11,019,996	B2	6/2021	Kalina, Jr. et al.
8,070,290	B2	12/2011	Gille et al.	11,019,997	B2	6/2021	Kalina, Jr. et al.
8,075,511	B2	12/2011	Tu et al.	11,116,625	B2	9/2021	Kalina, Jr.
8,118,768	B2	2/2012	Tu et al.	D938,585	S	12/2021	Kalina, Jr. et al.
8,142,364	B2	3/2012	Haffner et al.	11,197,780	B2	12/2021	Haffner et al.
8,152,752	B2	4/2012	Lynch et al.	11,253,394	B2	2/2022	Haffner et al.
8,197,418	B2	6/2012	Lal et al.	11,318,043	B2	5/2022	Heitzmann et al.
8,267,882	B2	9/2012	Euteneuer et al.	11,376,040	B2	7/2022	Kalina, Jr. et al.
8,273,050	B2	9/2012	Bergheim et al.	11,426,306	B2	8/2022	Haffner et al.
8,307,701	B2	11/2012	Laricchiuta	11,523,938	B2	12/2022	Rangel-Friedman et al.
				11,559,430	B2	1/2023	Crimaldi et al.
				11,564,833	B2	1/2023	Burns et al.
				11,744,458	B2	9/2023	Kalina, Jr. et al.
				2001/0000527	A1	4/2001	Yaron et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2001/0025150 A1	9/2001	de Juan et al.	2006/0241749 A1	10/2006	Tu et al.
2002/0052640 A1	5/2002	Bigus et al.	2007/0021653 A1	1/2007	Hattenbach et al.
2002/0072673 A1	6/2002	Yamamoto et al.	2007/0031473 A1	2/2007	Peyman
2002/0111608 A1	8/2002	Baerveldt et al.	2007/0073275 A1	3/2007	Conston et al.
2002/0120284 A1	8/2002	Schachar et al.	2007/0078471 A1	4/2007	Schachar et al.
2002/0120285 A1	8/2002	Schachar et al.	2007/0088432 A1	4/2007	Solovay et al.
2002/0133168 A1	9/2002	Smedley et al.	2007/0118065 A1	5/2007	Pinchuk et al.
2002/0143284 A1	10/2002	Tu et al.	2007/0118066 A1	5/2007	Pinchuk et al.
2002/0165522 A1	11/2002	Holmen	2007/0123812 A1	5/2007	Pinchuk et al.
2002/0177856 A1	11/2002	Richter et al.	2007/0123919 A1	5/2007	Schachar et al.
2003/0014021 A1	1/2003	Holmen	2007/0149927 A1	6/2007	Itou et al.
2003/0014092 A1	1/2003	Neuhann	2007/0161981 A1	7/2007	Sanders et al.
2003/0055372 A1	3/2003	Lynch et al.	2007/0179471 A1	8/2007	Christian et al.
2003/0060752 A1	3/2003	Bergheim et al.	2007/0191863 A1	8/2007	De Juan, Jr. et al.
2003/0074008 A1	4/2003	Ou	2007/0276315 A1	11/2007	Haffner
2003/0079329 A1	5/2003	Yaron et al.	2007/0282244 A1	12/2007	Tu et al.
2003/0093084 A1	5/2003	Nissan et al.	2007/0282245 A1	12/2007	Tu et al.
2003/0097117 A1	5/2003	Buono	2007/0287958 A1	12/2007	McKenzie et al.
2003/0097151 A1	5/2003	Smedley et al.	2007/0293807 A1	12/2007	Lynch et al.
2003/0105456 A1	6/2003	Lin	2007/0293872 A1	12/2007	Peyman
2003/0109907 A1	6/2003	Shadduck	2007/0293873 A1	12/2007	Chang
2003/0135149 A1	7/2003	Cullen et al.	2008/0027304 A1	1/2008	Pardo et al.
2003/0139729 A1	7/2003	Stegmann et al.	2008/0033351 A1	2/2008	Trogden et al.
2003/0181848 A1	9/2003	Bergheim et al.	2008/0045878 A1	2/2008	Bergheim et al.
2003/0187384 A1	10/2003	Bergheim et al.	2008/0051681 A1	2/2008	Schwartz
2003/0195438 A1	10/2003	Petillo	2008/0058704 A1	3/2008	Hee et al.
2003/0208217 A1	11/2003	Dan	2008/0082078 A1	4/2008	Berlin
2003/0212383 A1	11/2003	Cote et al.	2008/0091224 A1	4/2008	Griffis, III et al.
2003/0229303 A1	12/2003	Haffner et al.	2008/0097214 A1	4/2008	Meyers et al.
2003/0236484 A1	12/2003	Lynch et al.	2008/0097335 A1	4/2008	Trogden et al.
2004/0024345 A1	2/2004	Gharib et al.	2008/0108933 A1	5/2008	Yu et al.
2004/0088048 A1	5/2004	Richter et al.	2008/0109037 A1	5/2008	Steiner et al.
2004/0098122 A1	5/2004	Lee et al.	2008/0114440 A1	5/2008	Hlavka et al.
2004/0102729 A1	5/2004	Haffner et al.	2008/0125691 A1	5/2008	Yaron et al.
2004/0102832 A1	5/2004	Doty	2008/0140059 A1	6/2008	Schachar et al.
2004/0111050 A1	6/2004	Smedley et al.	2008/0147083 A1	6/2008	Vold et al.
2004/0147870 A1	7/2004	Burns et al.	2008/0183289 A1	7/2008	Werblin
2004/0154946 A1	8/2004	Solovay et al.	2008/0188860 A1	8/2008	Vold
2004/0210185 A1	10/2004	Tu et al.	2008/0200860 A1	8/2008	Tu et al.
2004/0216749 A1	11/2004	Tu	2008/0200923 A1	8/2008	Beckman et al.
2004/0236343 A1	11/2004	Taylor et al.	2008/0208176 A1	8/2008	Loh
2004/0243227 A1	12/2004	Starksen et al.	2008/0215062 A1	9/2008	Bowen et al.
2004/0249404 A1	12/2004	Haefliger	2008/0221501 A1	9/2008	Cote et al.
2004/0254517 A1	12/2004	Quiroz/Mercado et al.	2008/0228127 A1	9/2008	Burns et al.
2004/0254519 A1	12/2004	Tu et al.	2008/0243156 A1	10/2008	John
2004/0254520 A1	12/2004	Porteous et al.	2008/0255545 A1	10/2008	Mansfield et al.
2004/0260228 A1	12/2004	Lynch et al.	2008/0269730 A1	10/2008	Dotson
2005/0049578 A1	3/2005	Tu et al.	2008/0281250 A1	11/2008	Bergsneider et al.
2005/0096639 A1	5/2005	Slatkine et al.	2008/0306429 A1	12/2008	Shields et al.
2005/0125003 A1	6/2005	Pinchuk et al.	2009/0043242 A1	2/2009	Bene et al.
2005/0165385 A1	7/2005	Simon	2009/0043321 A1	2/2009	Conston et al.
2005/0171562 A1	8/2005	Crisuolo et al.	2009/0043365 A1	2/2009	Friedland et al.
2005/0203542 A1	9/2005	Weber et al.	2009/0076436 A2	3/2009	Gharib et al.
2005/0209549 A1	9/2005	Bergheim et al.	2009/0112245 A1	4/2009	Haefliger
2005/0209672 A1	9/2005	George et al.	2009/0124973 A1	5/2009	D'Agostino et al.
2005/0240222 A1	10/2005	Shipp	2009/0132040 A1	5/2009	Frion et al.
2005/0250788 A1	11/2005	Tu et al.	2009/0137989 A1	5/2009	Kataoka
2005/0267478 A1	12/2005	Corradi et al.	2009/0182421 A1	7/2009	Silvestrini et al.
2005/0277864 A1	12/2005	Haffner et al.	2009/0198213 A1	8/2009	Tanaka
2005/0288619 A1	12/2005	Gharib et al.	2009/0204053 A1	8/2009	Nissan et al.
2006/0032507 A1	2/2006	Tu	2009/0227933 A1	9/2009	Karageozian
2006/0067978 A1	3/2006	Heller et al.	2009/0227934 A1	9/2009	Eutenever et al.
2006/0074375 A1	4/2006	Bergheim et al.	2009/0264813 A1	10/2009	Chang
2006/0084907 A1	4/2006	Bergheim et al.	2009/0280158 A1	11/2009	Butuner
2006/0106370 A1	5/2006	Baerveldt et al.	2009/0287233 A1	11/2009	Huculak
2006/0116626 A1	6/2006	Smedley et al.	2010/0004581 A1	1/2010	Brigatti et al.
2006/0155300 A1	7/2006	Stamper et al.	2010/0010416 A1	1/2010	Juan, Jr. et al.
2006/0173397 A1	8/2006	Tu et al.	2010/0010452 A1	1/2010	Paques et al.
2006/0195055 A1	8/2006	Bergheim et al.	2010/0030150 A1	2/2010	Paques et al.
2006/0195056 A1	8/2006	Bergheim et al.	2010/0056979 A1	3/2010	Smedley et al.
2006/0200113 A1*	9/2006	Haffner A61F 9/00781	2010/0057093 A1	3/2010	Ide et al.
		606/6	2010/0076419 A1	3/2010	Chew et al.
2006/0210605 A1	9/2006	Chang et al.	2010/0121248 A1	5/2010	Yu et al.
2006/0217741 A1	9/2006	Ghannoum	2010/0121249 A1	5/2010	Yu et al.
2006/0241580 A1	10/2006	Mittelstein et al.	2010/0121342 A1	5/2010	Schieber et al.
			2010/0137981 A1	6/2010	Silvestrini et al.
			2010/0152626 A1	6/2010	Schwartz
			2010/0158980 A1	6/2010	Kopczynski et al.
			2010/0173866 A1	7/2010	Hee et al.

Page 5

(56)

References Cited

U.S. PATENT DOCUMENTS

2010/0185138 A1

7/2010

Yaron et al.

2010/0185205 A1

7/2010

Novakovic et al.

2010/0191103 A1

7/2010

Stamper et al.

2010/0225061 A1

9/2010

Bath

2010/0240987 A1

9/2010

Christian et al.

2010/0255061 A1

10/2010

de Juan, Jr. et al.

2010/0262174 A1

10/2010

Sretavan

2010/0274258 A1

10/2010

Silvestrini et al.

2010/0278898 A1

11/2010

Hughes et al.

2010/0280317 A1

11/2010

Silvestrini et al.

2011/0009874 A1

1/2011

Wardle et al.

2011/0009958 A1

1/2011

Wardle et al.

2011/0022065 A1

1/2011

Shipp

2011/0028883 A1

2/2011

Juan, Jr. et al.

2011/0028983 A1

2/2011

Silvestrini et al.

2011/0046536 A1

2/2011

Stegmann et al.

2011/0071524 A1

3/2011

Keller

2011/0077626 A1

3/2011

Baerveldt et al.

2011/0082385 A1

4/2011

Diaz et al.

2011/0092965 A1

4/2011

Slatkine et al.

2011/0098629 A1

4/2011

Juan, Jr. et al.

2011/0098809 A1

4/2011

Wardle et al.

2011/0112546 A1

5/2011

Juan, Jr. et al.

2011/0118649 A1

5/2011

Stegmann et al.

2011/0118835 A1

5/2011

Silvestrini et al.

2011/0144641 A1

6/2011

Dimalanta, Jr. et al.

2011/0202049 A1

8/2011

Jia et al.

2011/0224597 A1

9/2011

Stegmann et al.

2011/0230877 A1

9/2011

Huculak et al.

2011/0257658 A1

10/2011

Chen et al.

2011/0306915 A1

12/2011

De Juan, Jr. et al.

2011/0319793 A1

12/2011

Hyhynen

2011/0319806 A1

12/2011

Wardle

2012/0016286 A1

1/2012

Silvestrini et al.

2012/0022409 A1

1/2012

Gertner et al.

2012/0022424 A1

1/2012

Yamamoto et al.

2012/0022429 A1

1/2012

Silvestrini et al.

2012/0035524 A1

2/2012

Silvestrini

2012/0035525 A1

2/2012

Silvestrini

2012/0065570 A1

3/2012

Yeung et al.

2012/0071908 A1

3/2012

Sorensen et al.

2012/0078158 A1

3/2012

Haffner et al.

2012/0078281 A1

3/2012

Cox et al.

2012/0078362 A1

3/2012

Haffner et al.

2012/0123439 A1

5/2012

Romoda et al.

2012/0123440 A1

5/2012

Horvath et al.

2012/0165721 A1

6/2012

Grabner et al.

2012/0165722 A1

6/2012

Horvath et al.

2012/0165723 A1

6/2012

Horvath et al.

2012/0165933 A1

6/2012

Haffner et al.

2012/0197175 A1

8/2012

Horvath et al.

2012/0197217 A1

8/2012

Coldren

2012/0203262 A1

8/2012

Connors et al.

2012/0220917 A1

8/2012

Silvestrini et al.

2012/0232570 A1

9/2012

Jenson et al.

2012/0238994 A1

9/2012

Nazzaro et al.

2012/0257167 A1

10/2012

Gille et al.

2012/0259195 A1

10/2012

Haffner et al.

2012/0271272 A1

10/2012

Hammack et al.

2012/0283557 A1

11/2012

Berlin

2012/0310137 A1

12/2012

Silvestrini

2012/0323159 A1

12/2012

Wardle et al.

2013/0006165 A1

1/2013

Eutenener et al.

2013/0018295 A1

1/2013

Haffner et al.

2013/0018296 A1

1/2013

Berghheim et al.

2013/0018412 A1

1/2013

Journey et al.

2013/0079701 A1

3/2013

Schieber et al.

2013/0079759 A1

3/2013

Dotson et al.

2013/0085507 A1

4/2013

Nagasaka

2013/0090534 A1

4/2013

Burns et al.

2013/0110125 A1

5/2013

Silvestrini et al.

2013/0184631 A1

7/2013

Pinchuk

2013/0245532 A1

9/2013

Tu et al.

2013/0253404 A1

9/2013

Tu

2013/0253405 A1

9/2013

Tu

2013/0253528 A1 *

2013/0281908 A1

2013/0281910 A1

2013/0289467 A1

2013/0310930 A1

2014/0052046 A1

2014/0081194 A1

2014/0135916 A1

2014/0155803 A1

2014/0207137 A1

2014/0276901 A1

2015/0223981 A1

2015/0238687 A1

2015/0342875 A1

2016/0045363 A1

2016/0256323 A1

2016/0287438 A1

2016/0354309 A1

2017/0105750 A1

2017/0135857 A1

2018/0021170 A1

2018/0028361 A1

2018/0085065 A1

2018/0104102 A1

2018/0161205 A1

2018/0256393 A1

2018/0280194 A1

2018/0303665 A1

2018/0333296 A1

2019/0000673 A1

2019/0021991 A9

2019/0053704 A1

2019/0083307 A1

2019/0083313 A1

2019/0104936 A1

2019/0224046 A1

2019/0314199 A1

2019/0321220 A1

2019/0321225 A1

2019/0374380 A1

2020/0000635 A1

2020/0155349 A1

2020/0179171 A1

2020/0367745 A1

2021/0015662 A1

2021/0137737 A1

2021/0154449 A1

2021/0298948 A1

2021/0315806 A1

2021/0369447 A1

2022/0000663 A1

2022/0015628 A1

2022/0119350 A1

2022/0233349 A1

2022/0233354 A1

2022/0313486 A1

2023/0053931 A1

2023/0090539 A1

2023/0157868 A1

2023/0210694 A1

2023/0293344 A1

2023/0301830 A1

2023/0372303 A1

9/2013

10/2013

10/2013

10/2013

11/2013

2/2014

3/2014

5/2014

6/2014

7/2014

9/2014

8/2015

8/2015

12/2015

2/2016

9/2016

10/2016

12/2016

4/2017

5/2017

1/2018

2/2018

3/2018

4/2018

6/2018

9/2018

10/2018

10/2018

11/2018

1/2019

1/2019

2/2019

3/2019

3/2019

4/2019

7/2019

10/2019

10/2019

10/2019

12/2019

1/2020

5/2020

6/2020

11/2020

1/2021

5/2021

5/2021

9/2021

10/2021

12/2021

1/2022

1/2022

4/2022

7/2022

7/2022

10/2022

2/2023

3/2023

5/2023

7/2023

9/2023

9/2023

11/2023

Haffner

Schaller et al.

Tu

Haffner et al.

Tu et al.

Peartree et al.

Burns et al.

Clauson et al.

Silvestrini

Keller

Auld

Smedley et al.

Novakovic et al.

Haffner

Haffner et al.

Horvath et al.

Badawi et al.

Heitzmann et al.

Cote et al.

Haffner et al.

Haffner et al.

Haffner et al.

Haffner et al.

Lynch et al.

Tu et al.

Yamamoto et al.

Heitzmann et al.

Heitzmann et al.

Heitzmann et al.

Fjield et al.

Heitzmann et al.

Burns et al.

Burns et al.

Berlin

Gunn et al.

Heitzmann et al.

Haffner et al.

Rangel-Friedman et al.

Smedley et al.

Navratil et al.

Lerner

Haffner et al.

Crimaldi et al.

Kalina, Jr. et al.

Haffner et al.

Burns et al.

Haffner et al.

Haffner et al.

Haffner et al.

Haffner

Kalina, Jr.

Haffner et al.

Kalina, Jr. et al.

Murphy et al.

Haffner et al.

Haffner et al.

Heitzmann et al.

Haffner et al.

Haffner et al.

Crimaldi et al.

Burns et al.

Heitzmann et al.

Liu et al.

A61F 9/00781

606/107

FOREIGN PATENT DOCUMENTS

CA

2273331

6/1998

CA

2244646

2/1999

CA

2311244

6/1999

CA

2643357

11/1999

CA

2442652

1/2011

CA

2766131

1/2011

CA

2762536

11/2020

CH

92111244

7/1993

CN

1767871

5/2006

CN

1976732

6/2007

CN

102781338

11/2012

CN

104540472

4/2015

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	106029019	10/2016
CN	106456372	2/2017
CN	107126315	9/2017
CN	107205815	9/2017
CN	107498340	12/2017
CN	107530190	1/2018
CN	109156098	1/2019
EP	0436232	7/1991
EP	0550791	7/1993
EP	1592407	11/2005
EP	2088976	8/2009
FR	2349469	11/1977
JP	2005-533619	11/2005
JP	2007-500063	1/2007
JP	2009-542370	12/2009
JP	2010-533565	10/2010
JP	2011-522575	8/2011
JP	2014-236980	12/2014
JP	2016-511108	4/2016
RU	2143250	12/1999
WO	WO 1991/08784	6/1991
WO	WO 1992/08406	5/1992
WO	WO 1998/23237	6/1998
WO	WO 1998/37831	9/1998
WO	WO 1999/26567	6/1999
WO	WO 2000/64389	11/2000
WO	WO 2000/64390	11/2000
WO	WO 2000/64391	11/2000
WO	WO 2000/64393	11/2000
WO	WO 2001/68016	9/2001
WO	WO 2001/085065	11/2001
WO	WO 2001/97727	12/2001
WO	WO 2002/36052	5/2002
WO	WO 2002/080811	10/2002
WO	WO 2002/087418	11/2002
WO	WO 2002/089699	11/2002
WO	WO 2003/041622	5/2003
WO	WO 2003/073968	9/2003
WO	WO 2004/014218	2/2004
WO	WO 2004/043231	5/2004
WO	WO 2005/016418	2/2005
WO	WO 2005/105197	11/2005
WO	WO 2005/107845	11/2005
WO	WO 2006/036715	4/2006
WO	WO 2007/130393	11/2007
WO	WO 2008/083118	7/2008
WO	WO 2009/012406	1/2009
WO	WO 2009/126569	10/2009
WO	WO 2009/151543	12/2009
WO	WO 2010/006053	1/2010
WO	WO 2010/077987	7/2010
WO	WO 2010/078063	7/2010

WO	WO 2010/093945	8/2010
WO	WO 2011/020633	2/2011
WO	WO 2011/084550	7/2011
WO	WO 2013/040079	3/2013
WO	WO 2014/150292	9/2014
WO	WO 2014/151070	9/2014
WO	WO 2014/164569	10/2014
WO	WO 2015/073571	5/2015
WO	WO 2015/184173	12/2015
WO	WO 2016/019160	2/2016
WO	WO 2016/023116	2/2016
WO	WO 2016/154066	9/2016
WO	WO 2016/187355	11/2016
WO	WO 2017/015633	1/2017
WO	WO 2017/040853	3/2017
WO	WO 2017/040855	3/2017
WO	WO 2017/053885	3/2017
WO	WO 2017/087713	5/2017
WO	WO 2017/184881	10/2017
WO	WO 2019/036025	2/2019
WO	WO 2019/070385	4/2019
WO	WO 2019/200336	10/2019
WO	WO 2020/172615	8/2020

OTHER PUBLICATIONS

Chen et al., "Implantable Unpowered Parylene MEMS Intraocular Pressure Sensor", Microtechnologies in Medicine and Biology, 2006 International Conference on Publication Date: May 9-12, 2006, 5pp., downloaded from http://ieeexplore.ieee.org/xp/freeabs_all.jsp?arnumber=4281361.

De Juan et al., "Refinements in microinstrumentation for vitreous surgery," Am. J. Ophthalmol. 109:218-20 (1990).
<https://entokey.com/gonioscopy-2/> Uploaded Oct. 2016.

Jordan et al., "A Novel Approach to Suprachoroidal Drainage for the Surgical Treatment of Intractable Glaucoma", J Glaucoma, vol. 15, No. 3, Jun. 2006, pp. 200-205.

Katuri et al., "Intraocular Pressure Monitoring Sensors", IEEE Sensors Journal, vol. 8, No. 1, Jan. 2008, 8 pp.

Online encyclopedia article "Hyaluronan," section on "Medical Applications" accessed Monday, Sep. 27, 2010. http://en.wikipedia.org/wiki/Hyaluronic_acid.

Rizq, et al., "Intraocular Pressure measurement at the Choroid Surface: A Feasibility Study with Implications for Implantable Microsystems", Br J Ophthalmol 2001; 85:868-871, Jul. 2001.

Ianchulev et al., "Minimally Invasive Ab-Interno Suprachoroidal Device (CyPass) for IOP Control in Open-Angle Glaucoma," presented at the Annual Meeting of the American Academy of Ophthalmology Oct. 16-19, 2010, Chicago, IL.

Walter et al., "Development of a Completely Encapsulated Intraocular Pressure Sensor", Ophthalmic Res 2000; 32:278-284. Nov. 5, 1999.

* cited by examiner

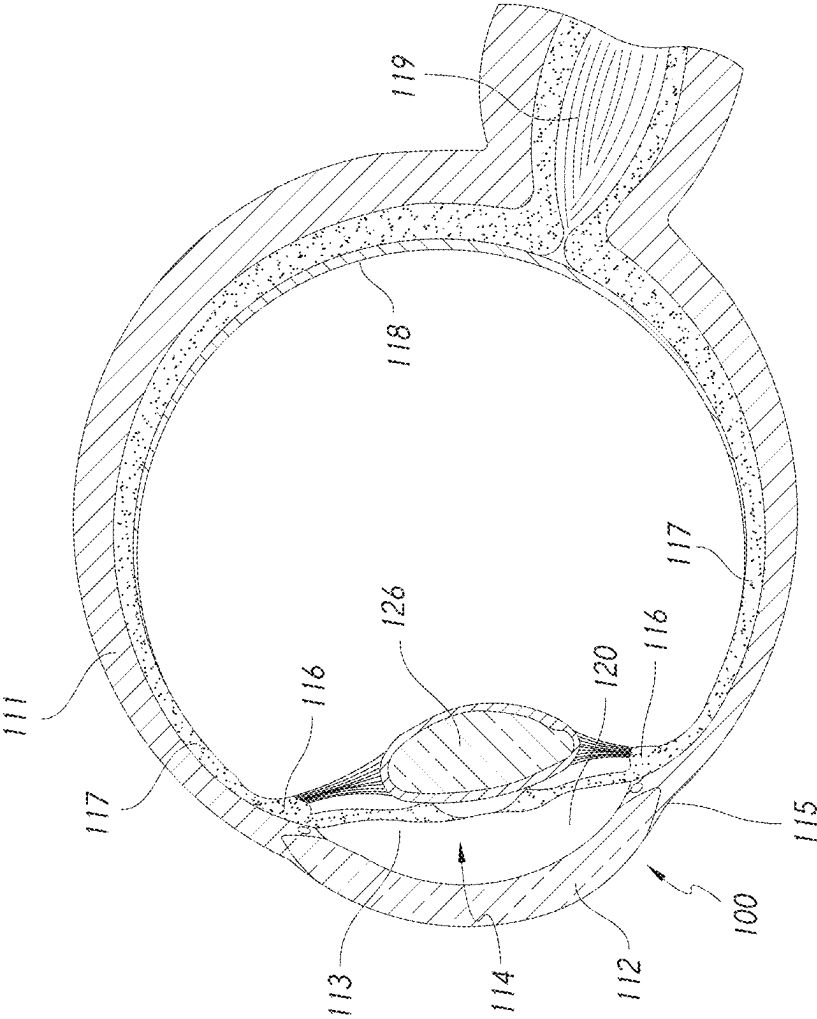


FIG. 1A

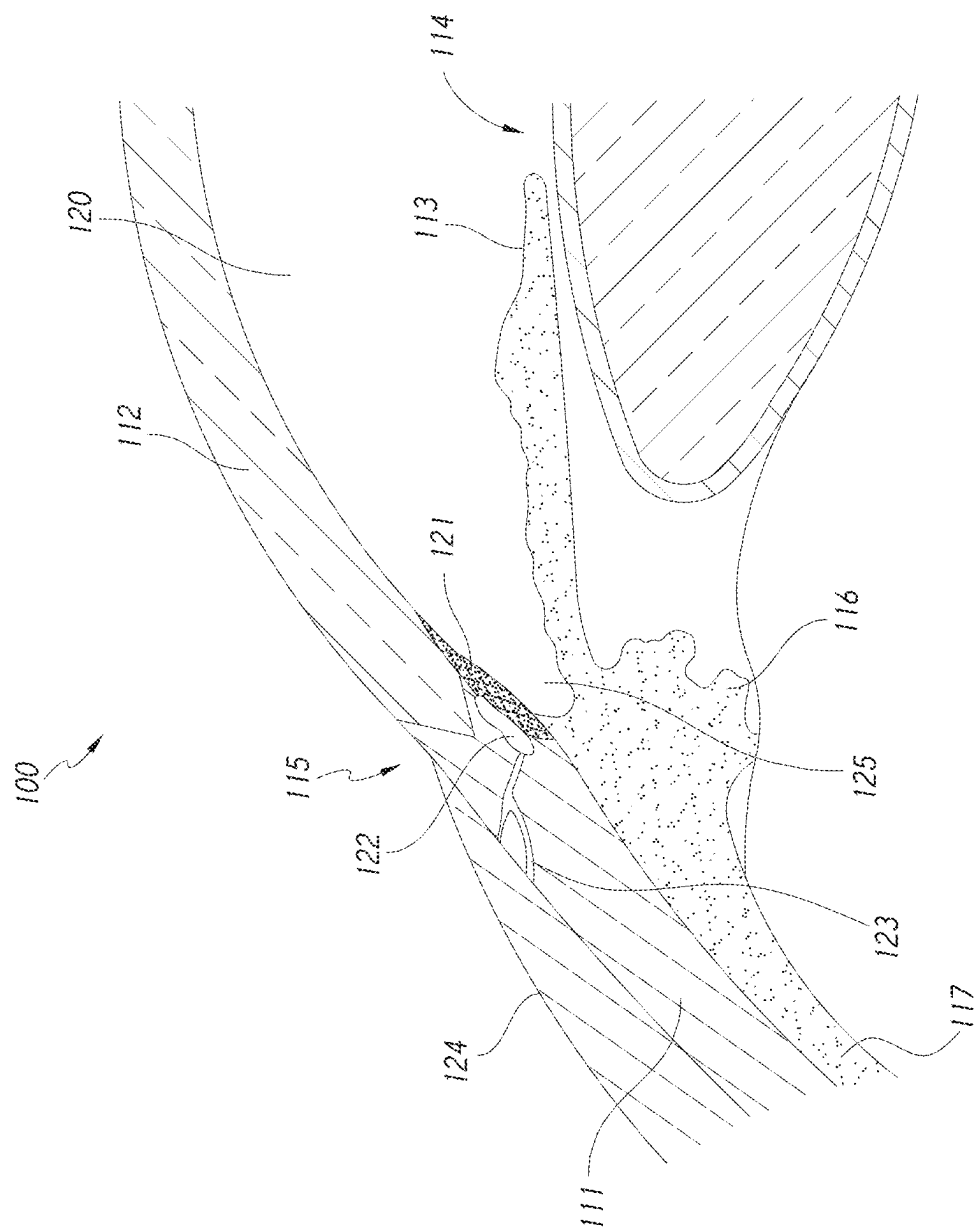
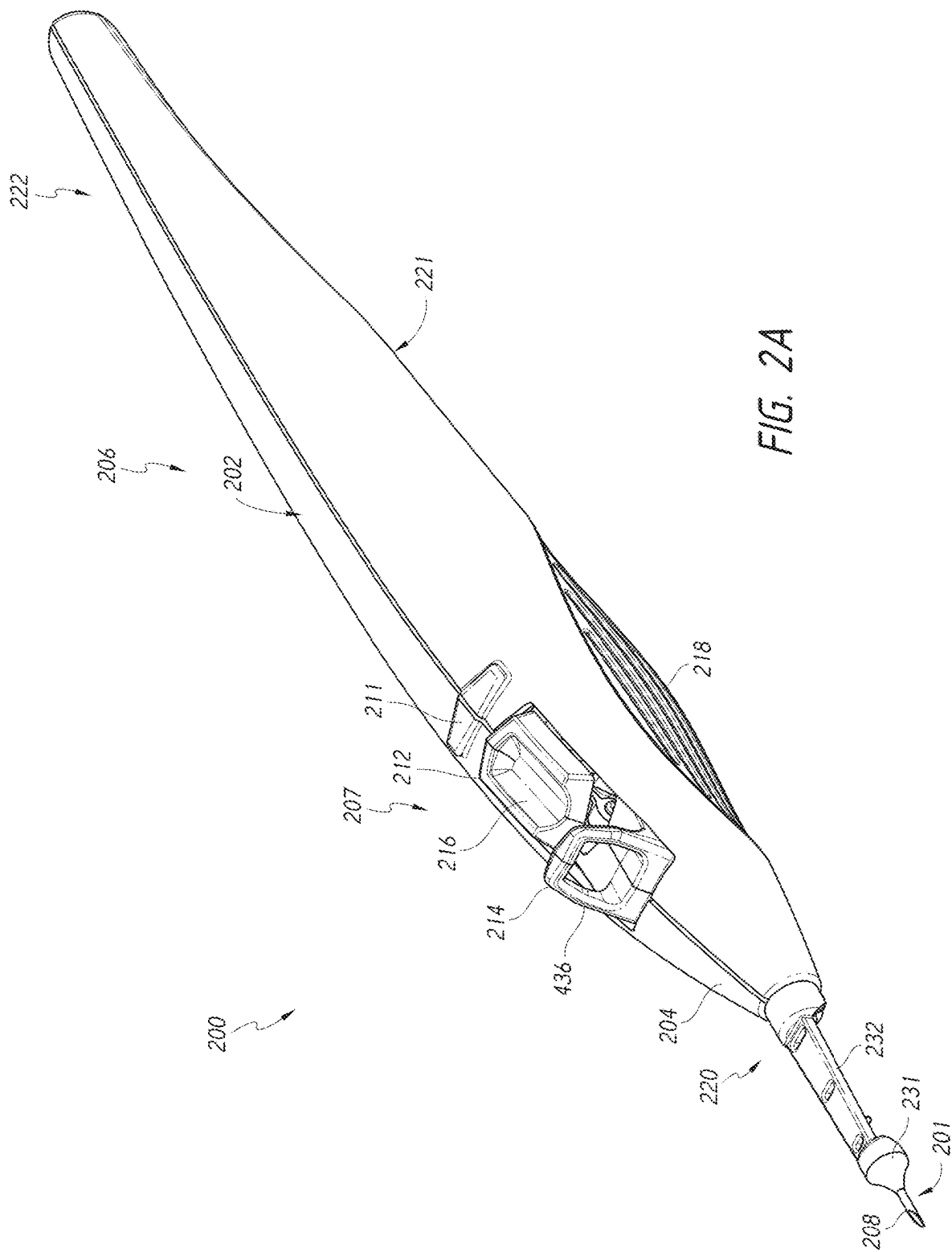
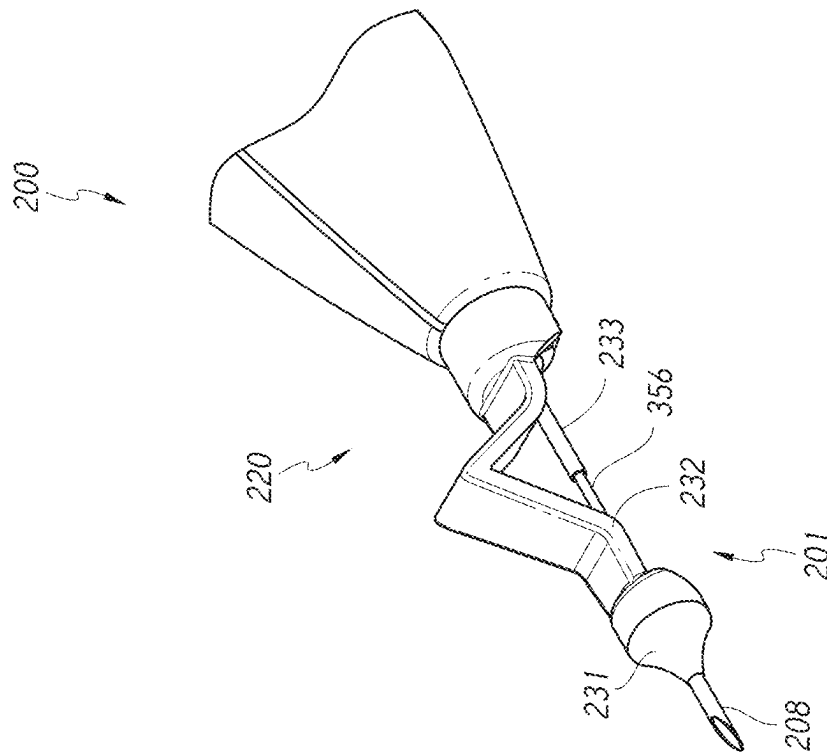


FIG. 1B





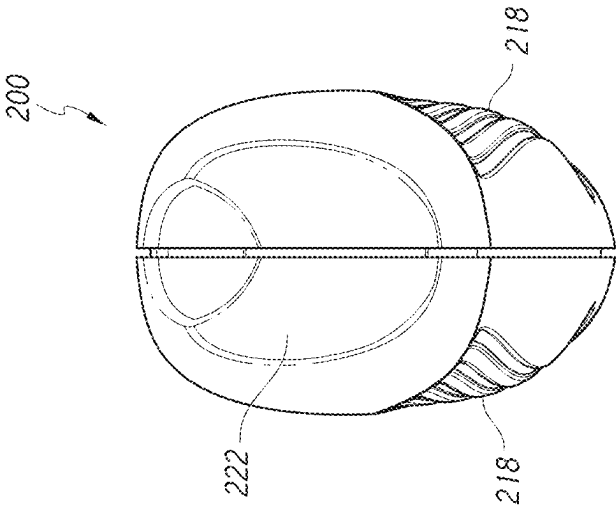


FIG. 2D

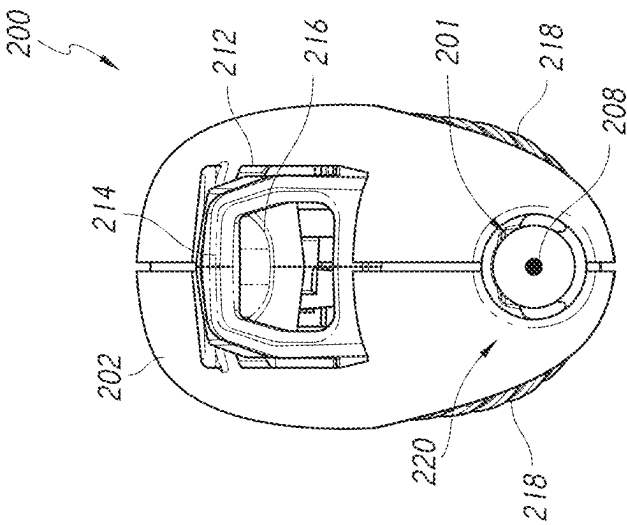
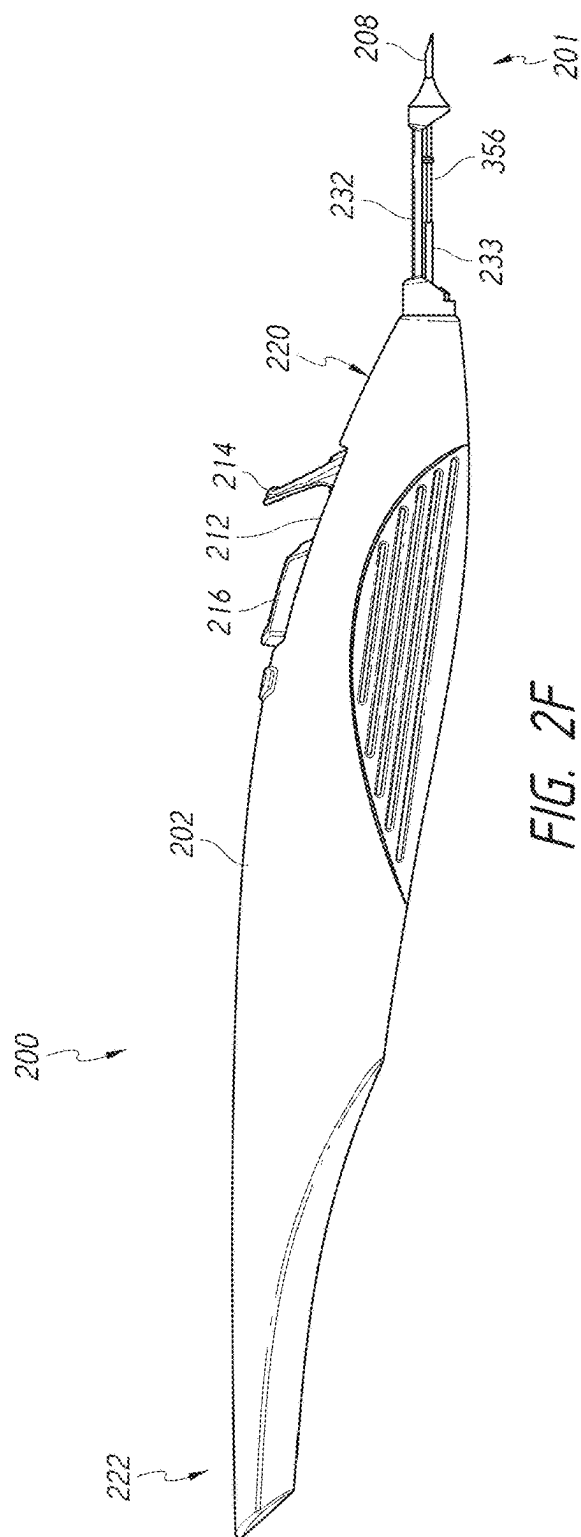
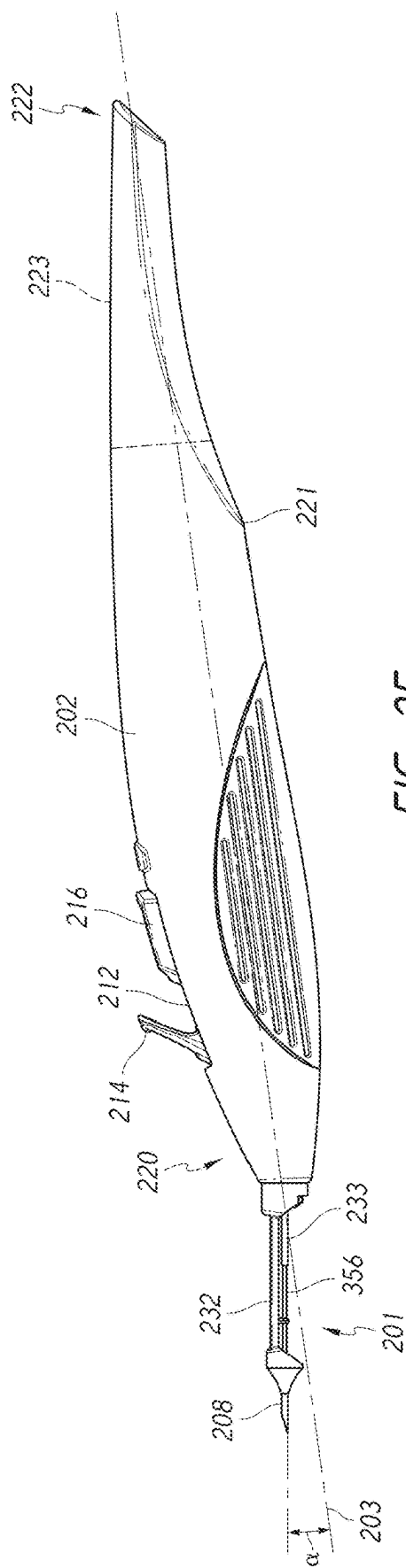
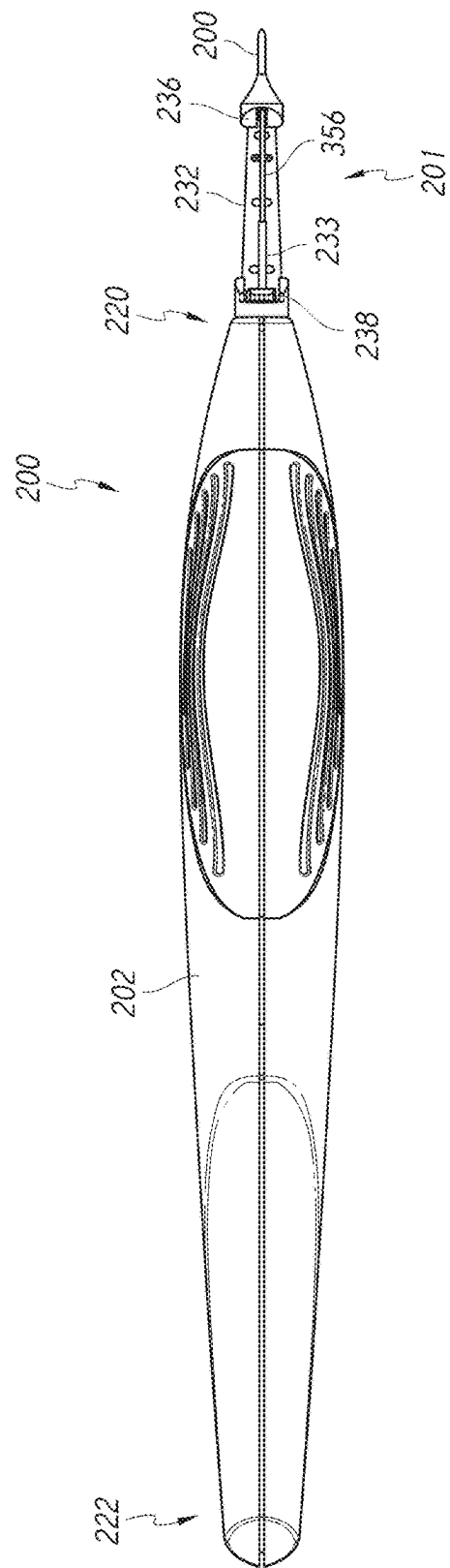
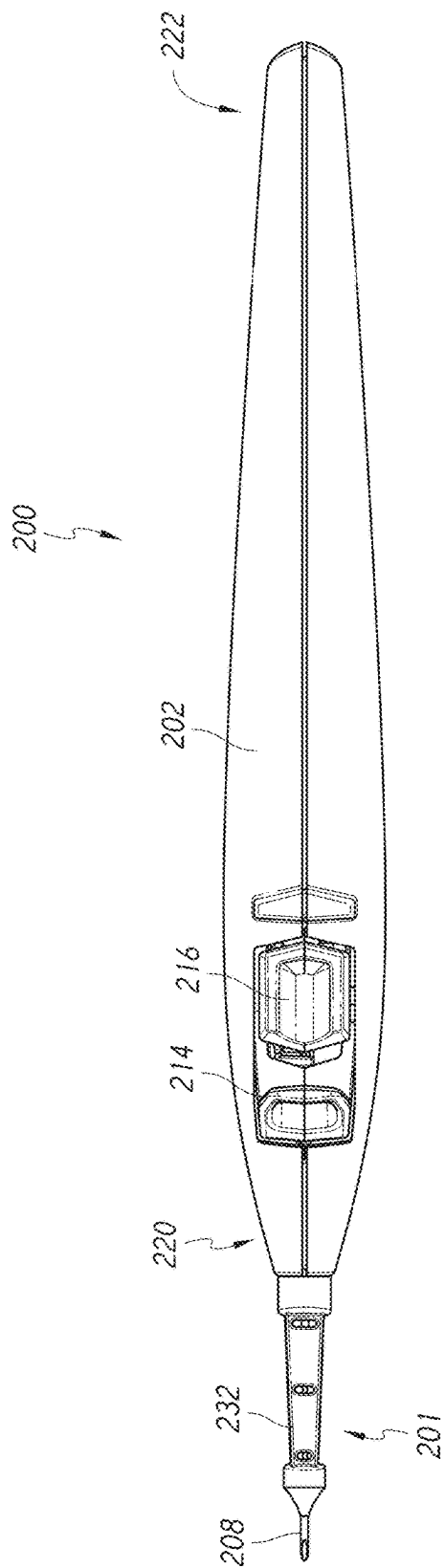
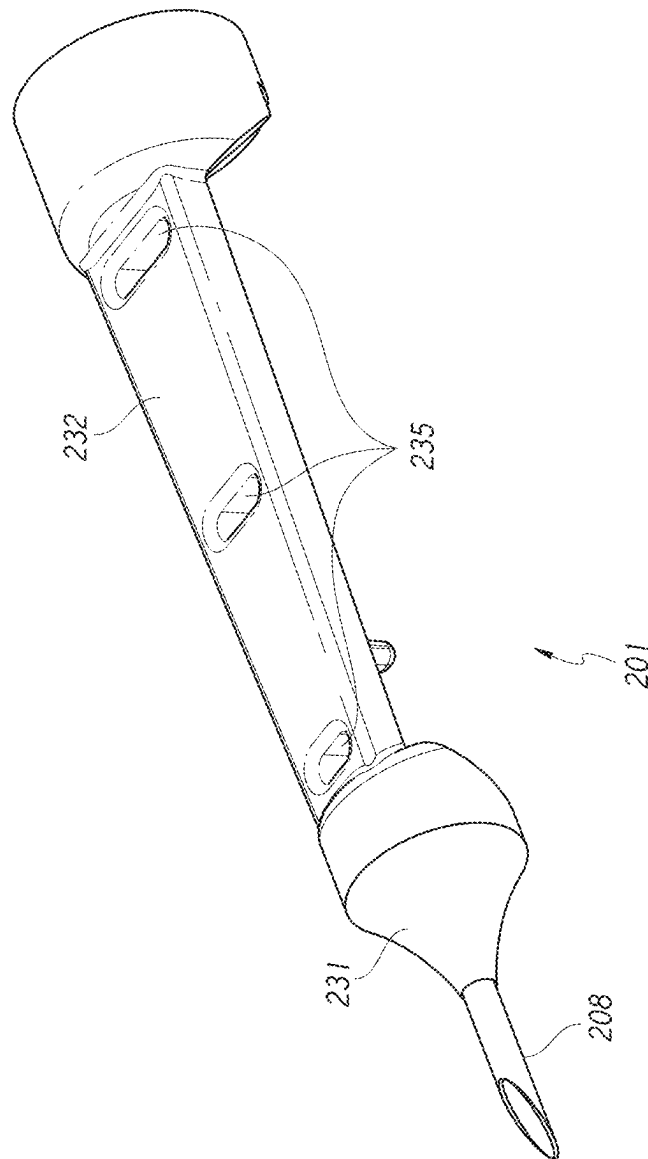


FIG. 2C







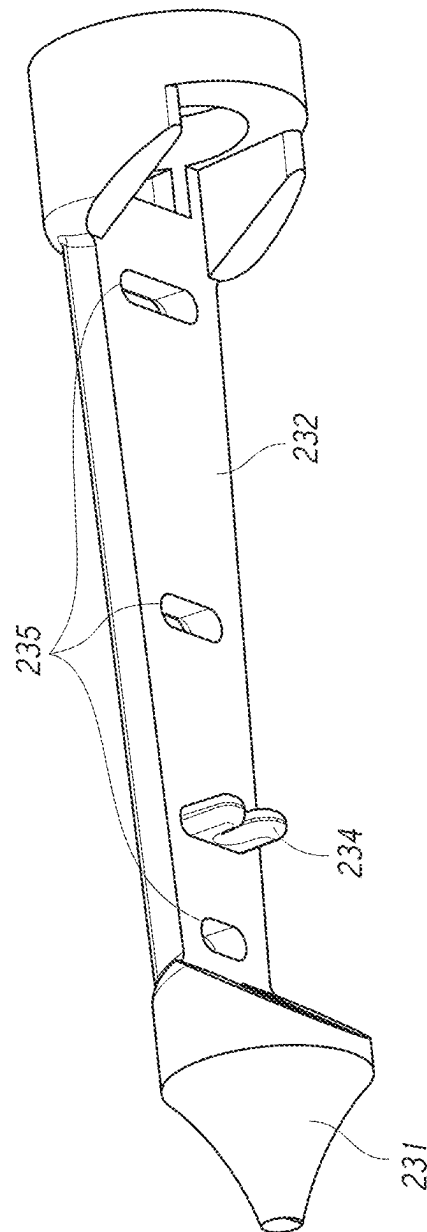
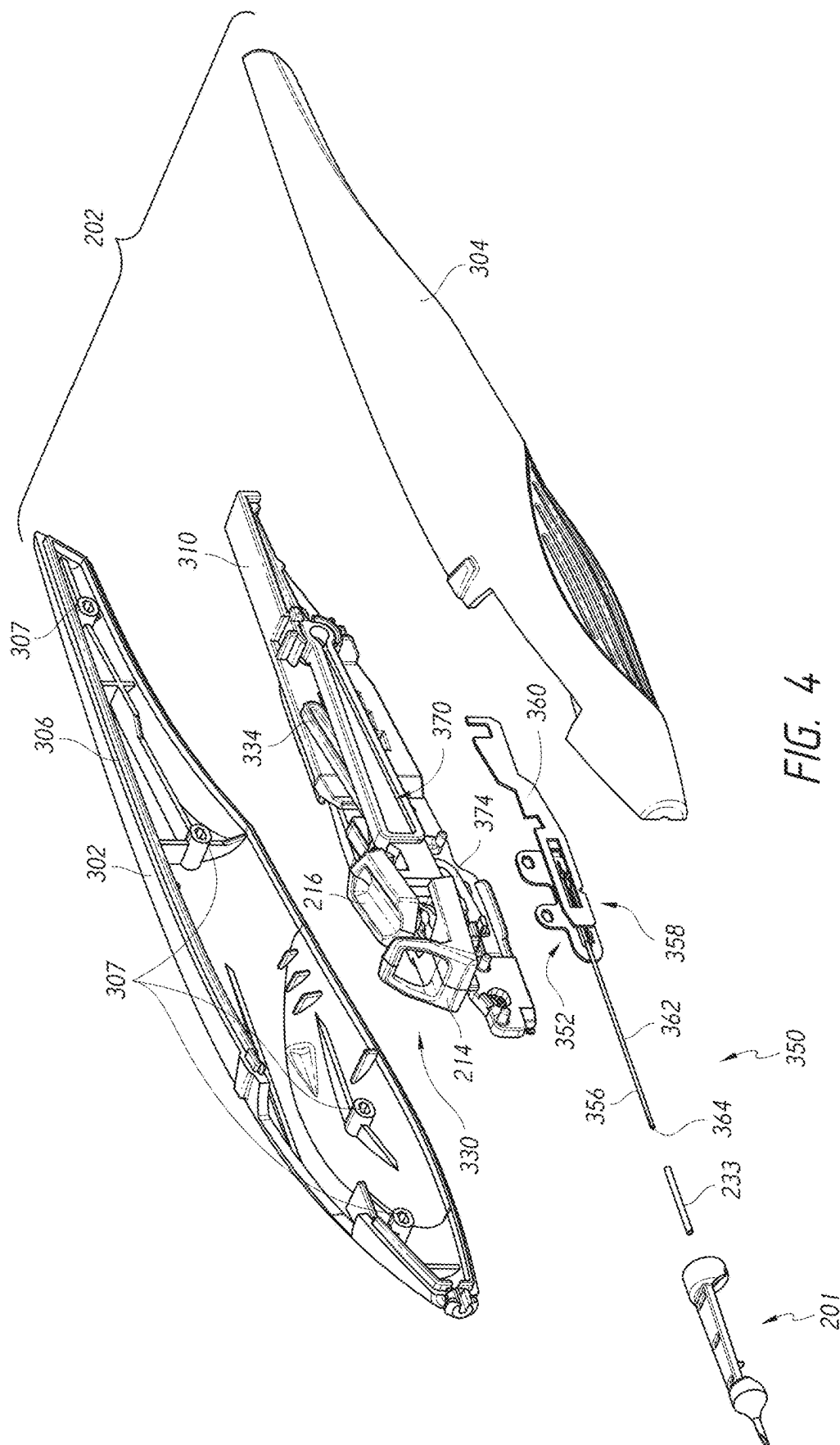


FIG. 3B



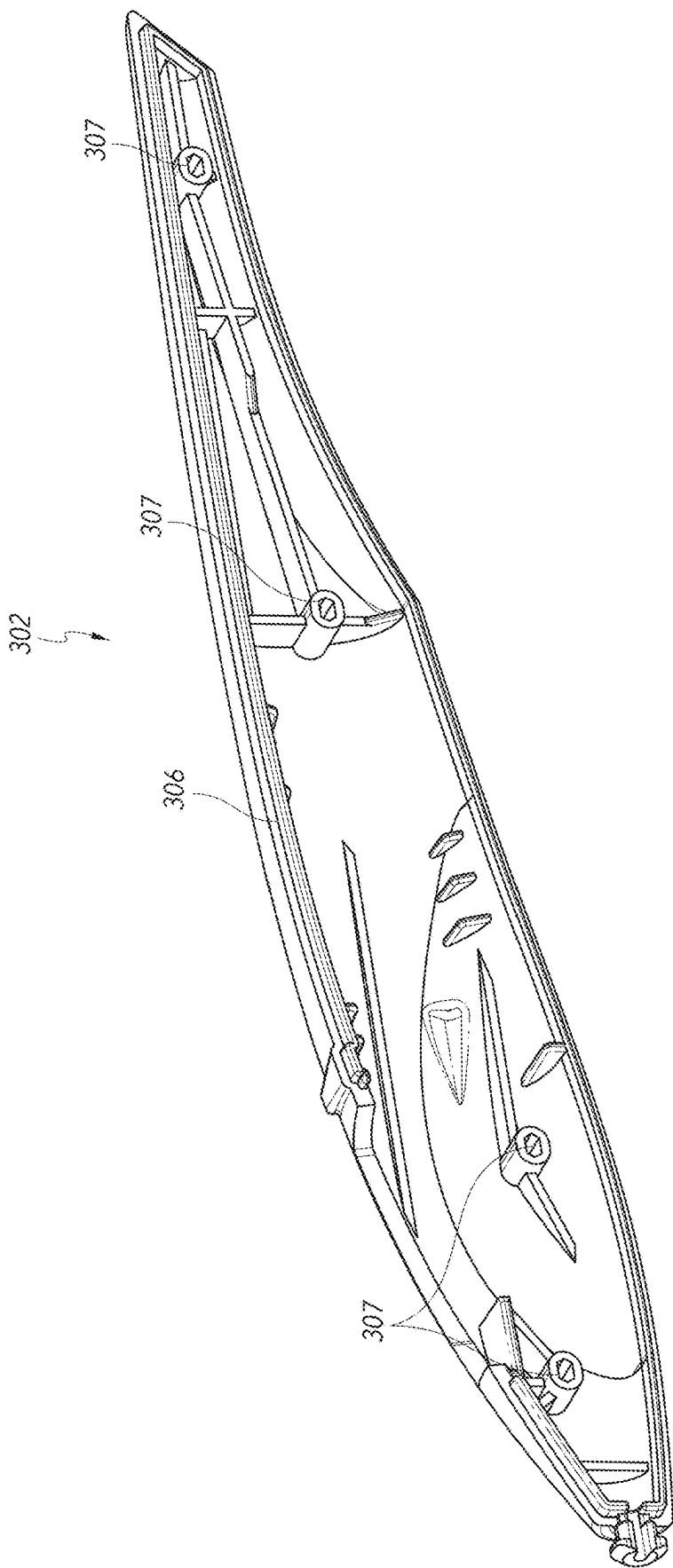


FIG. 5A

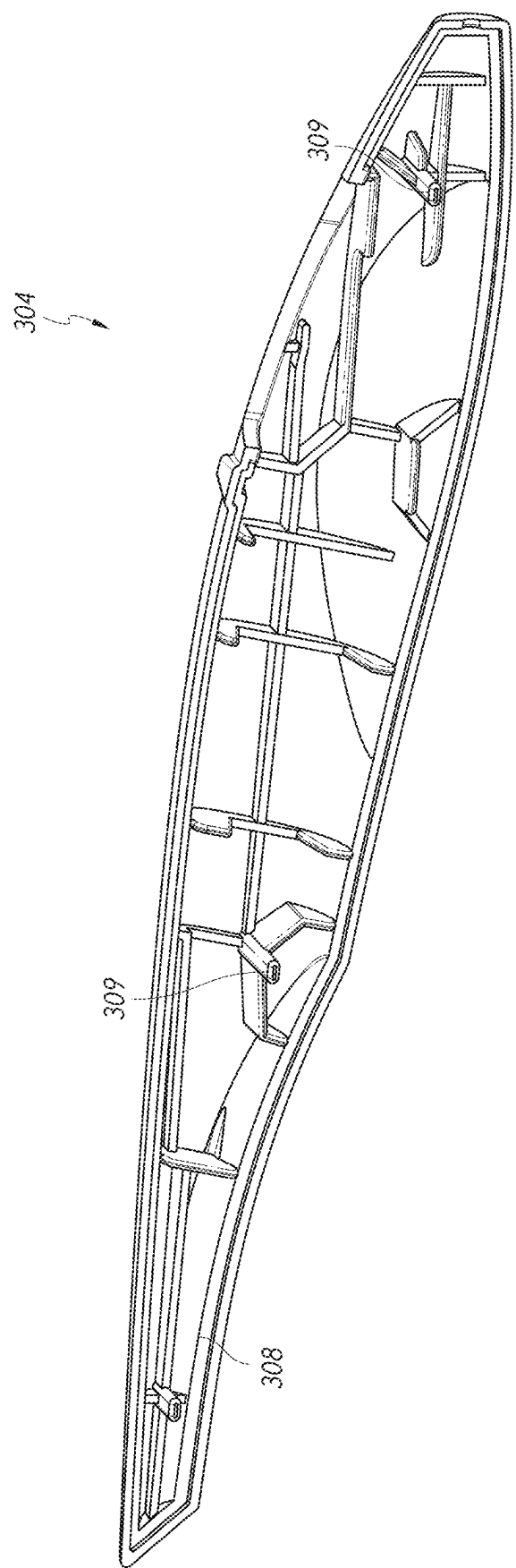
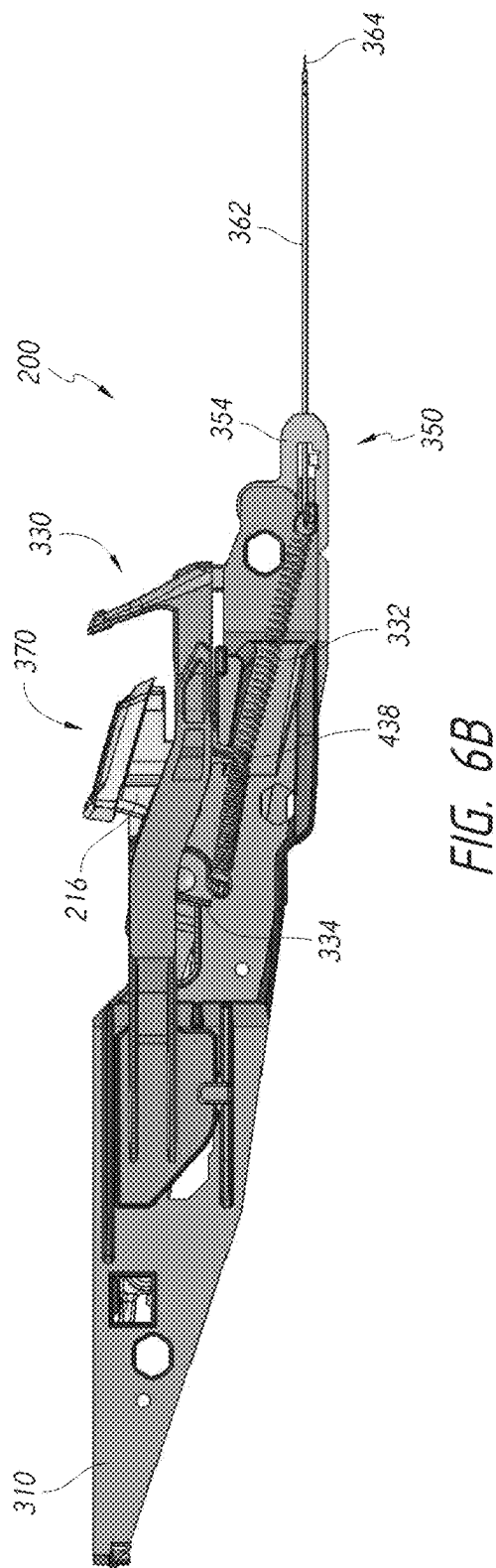
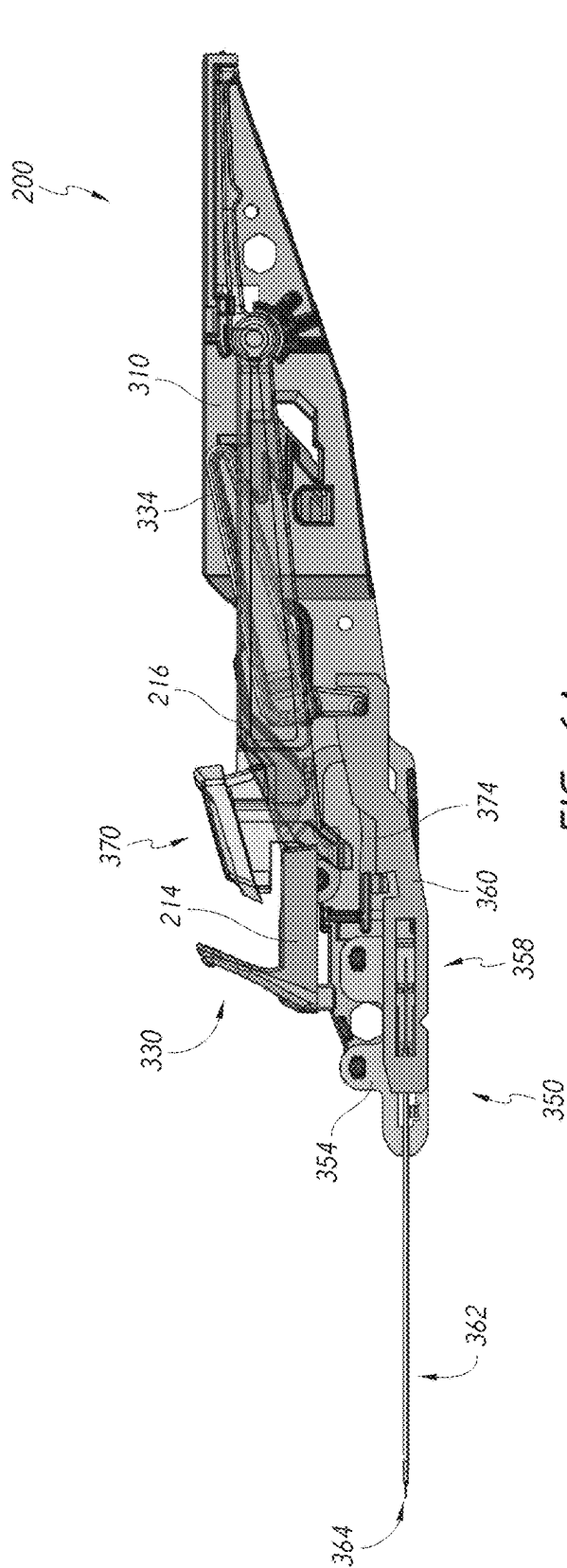
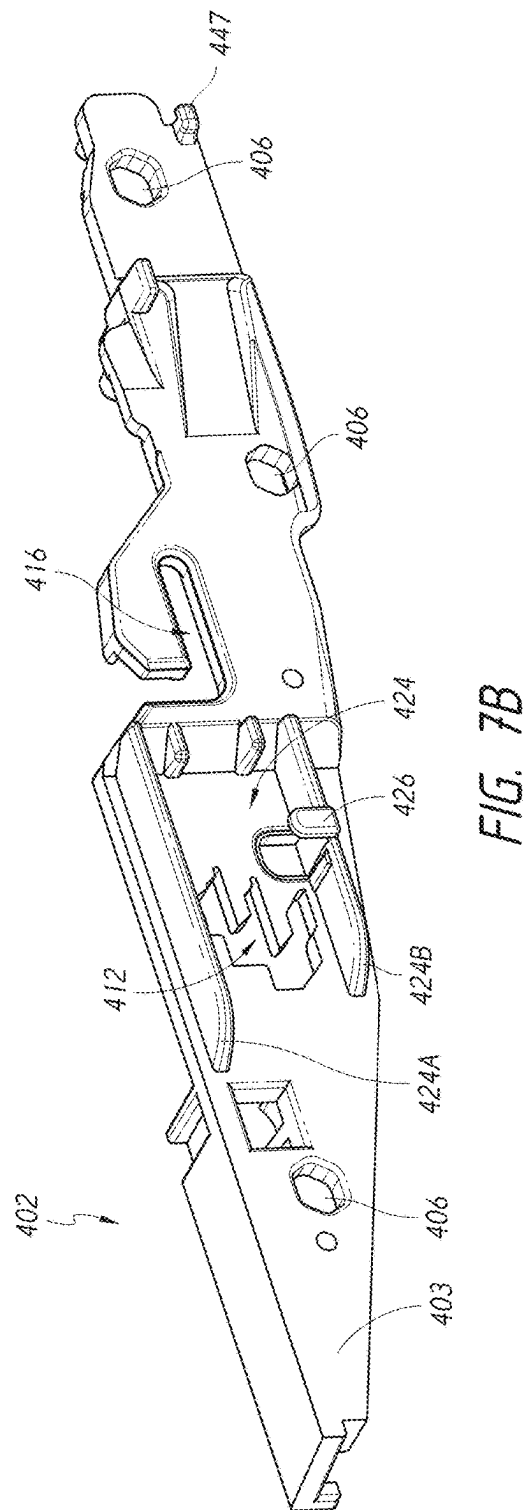
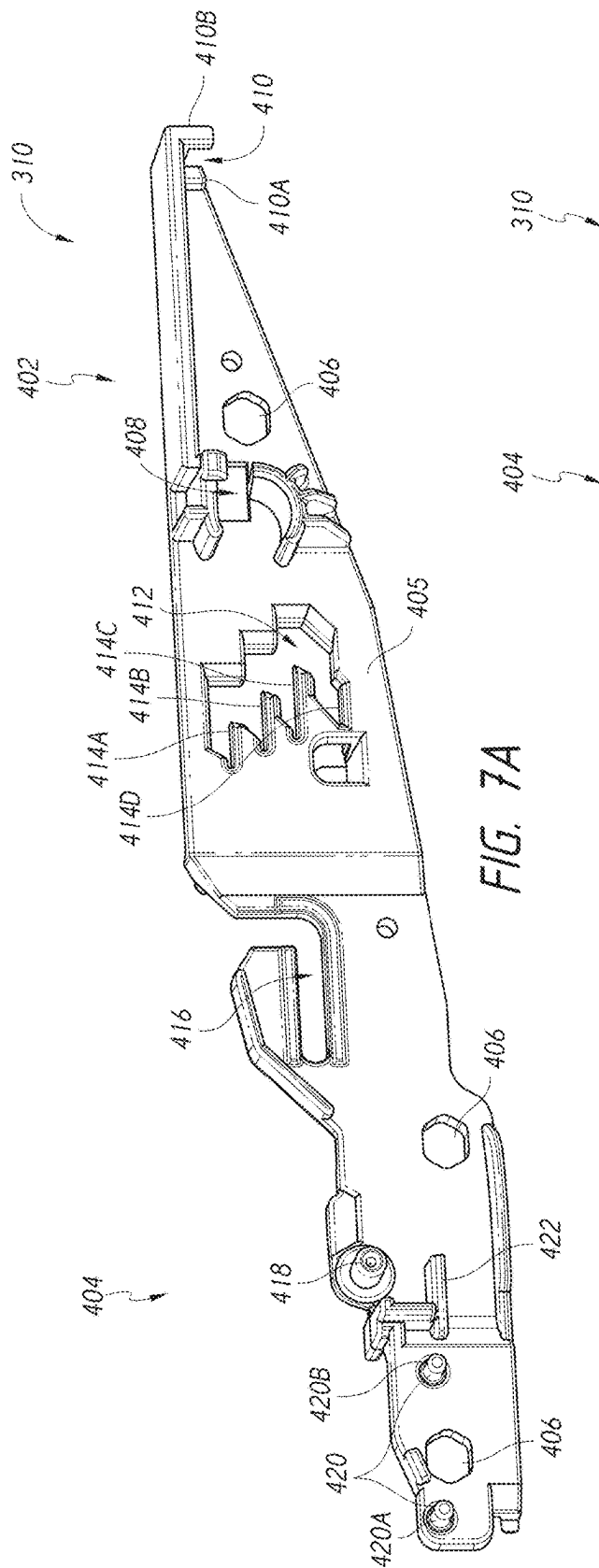
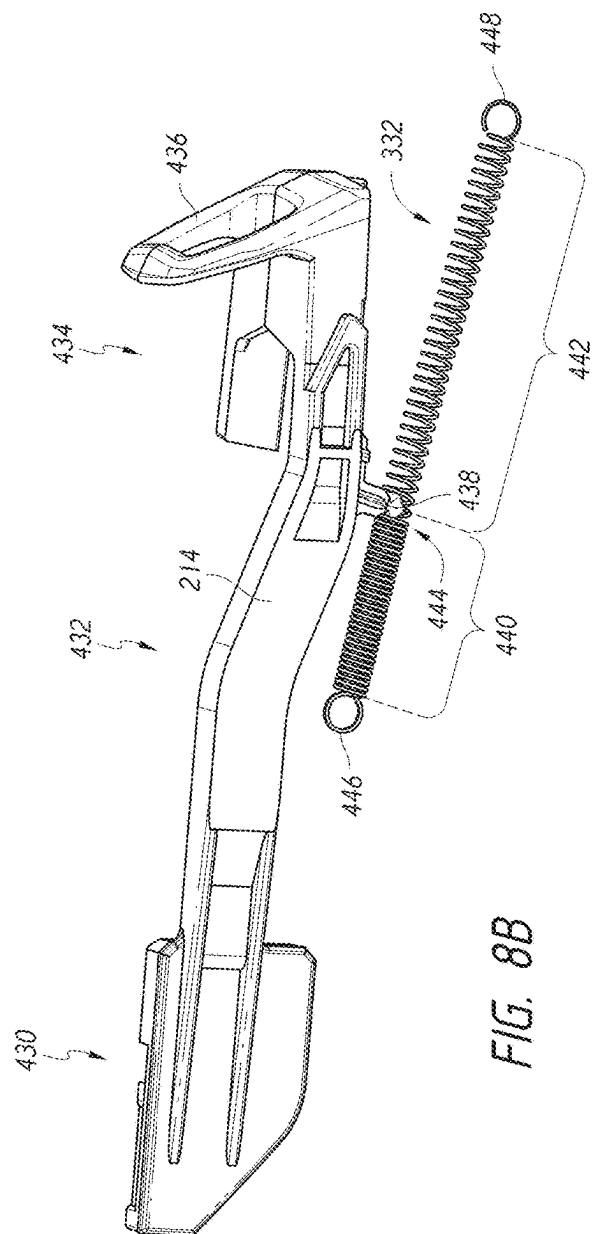
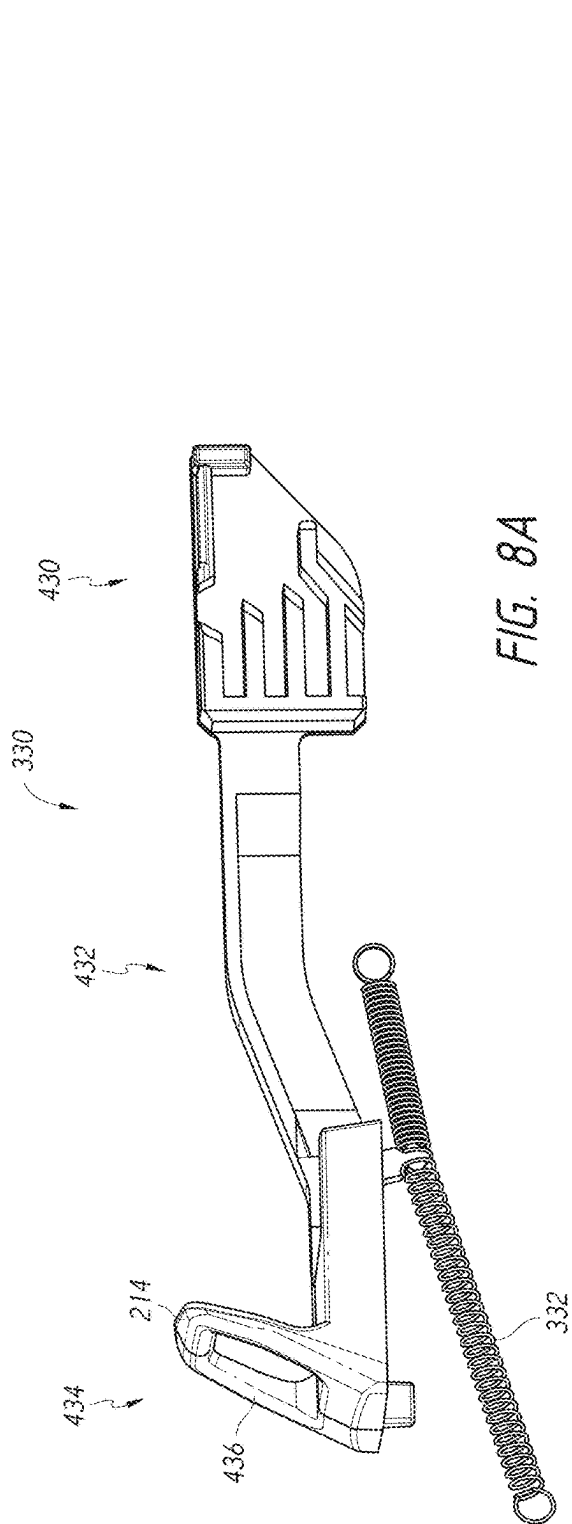


FIG. 5B







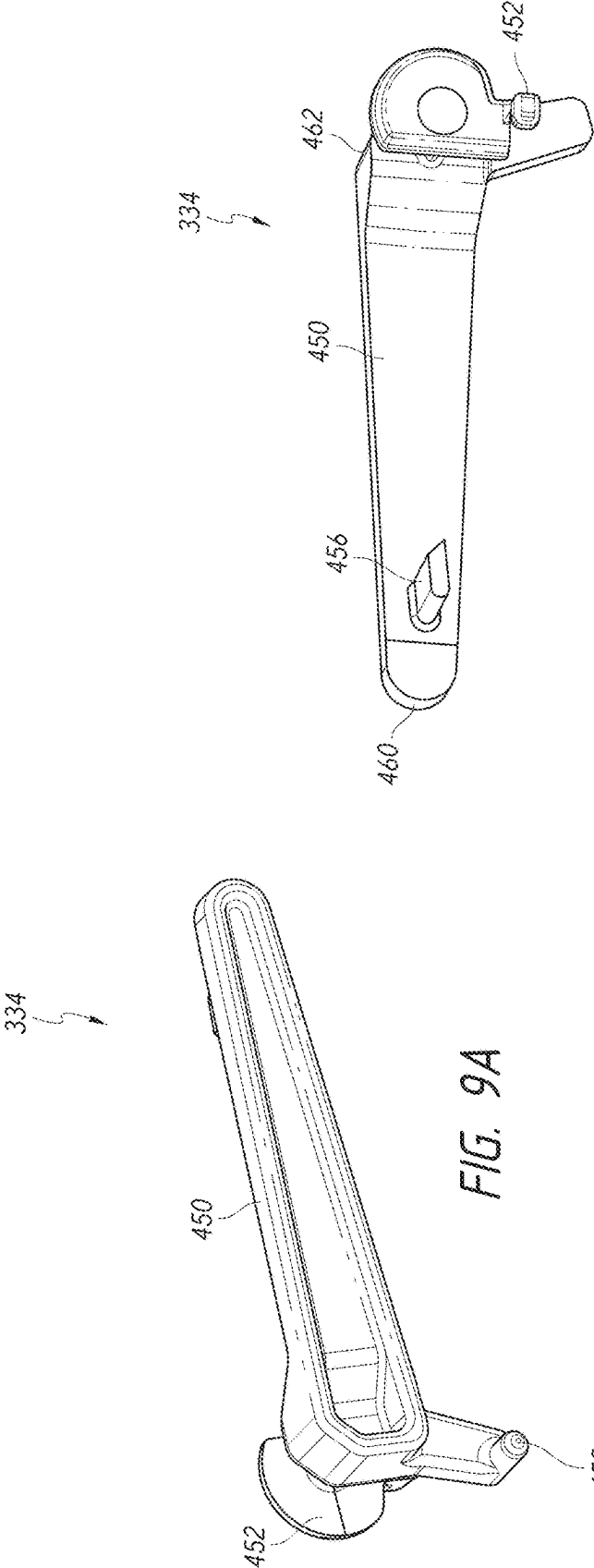


FIG. 9B

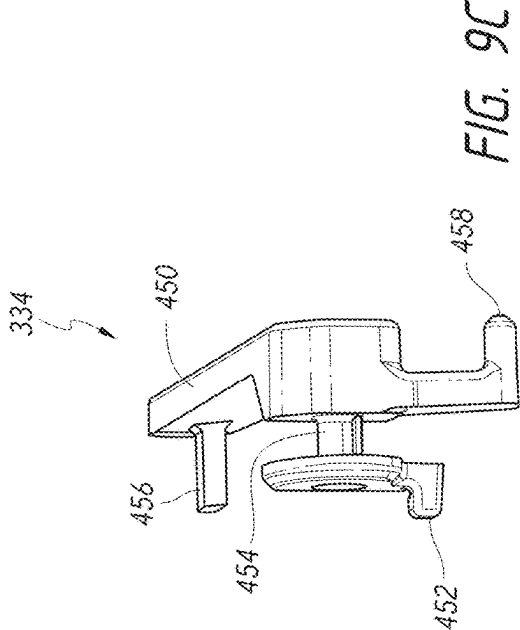
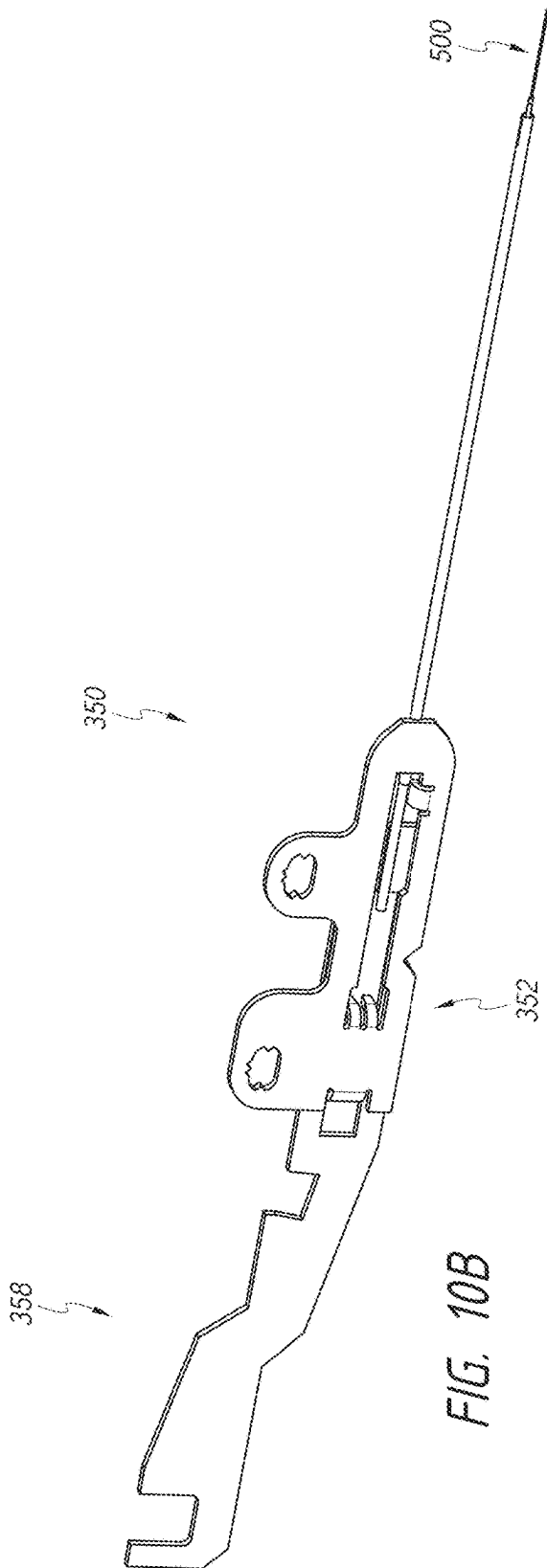
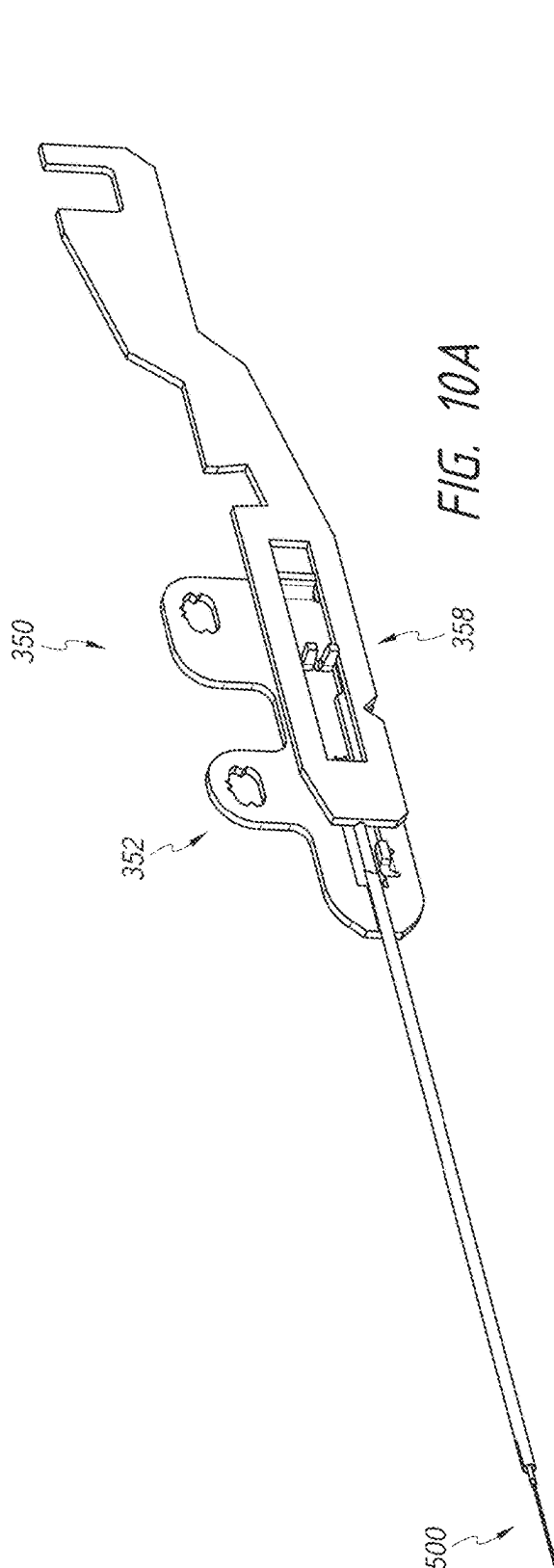
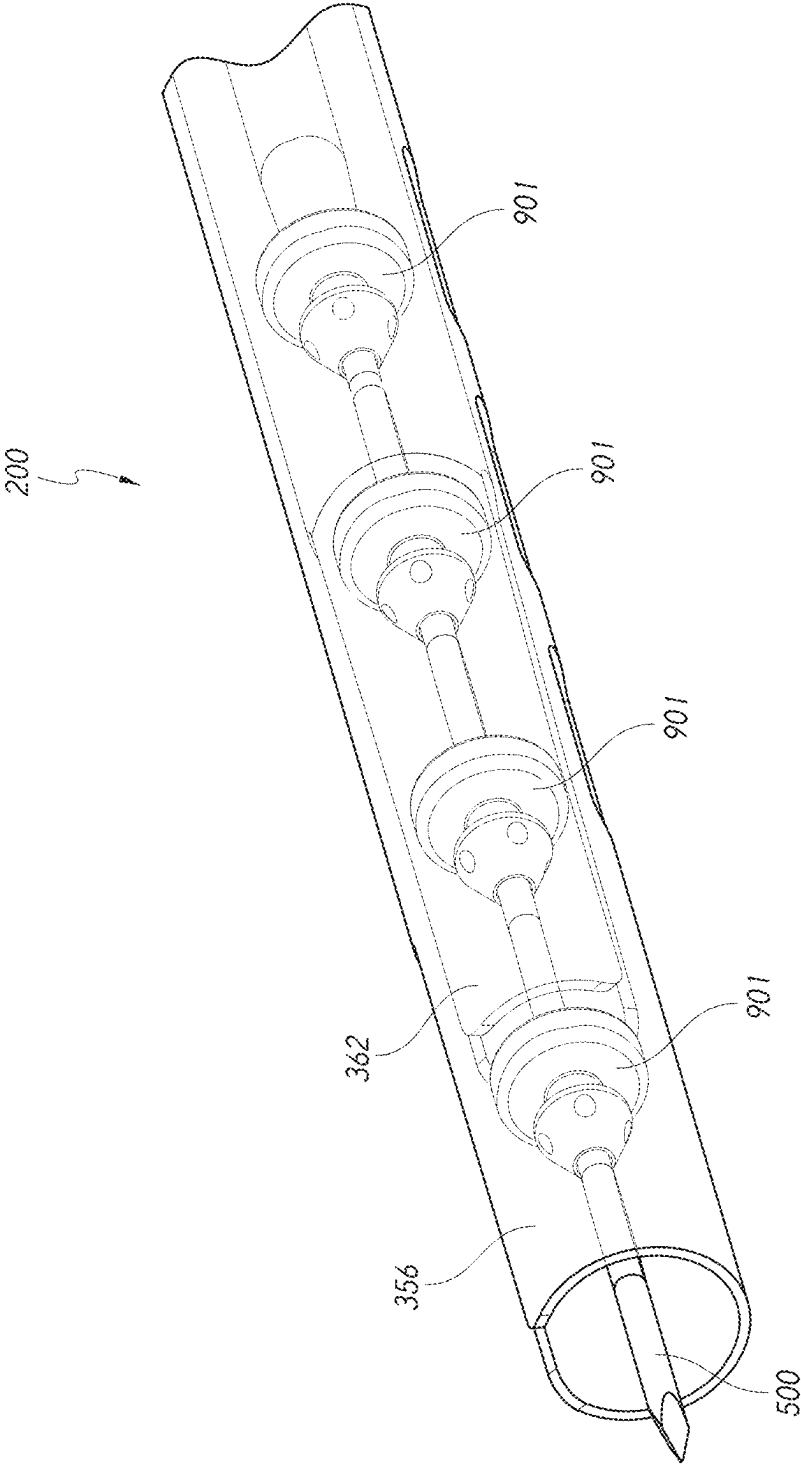
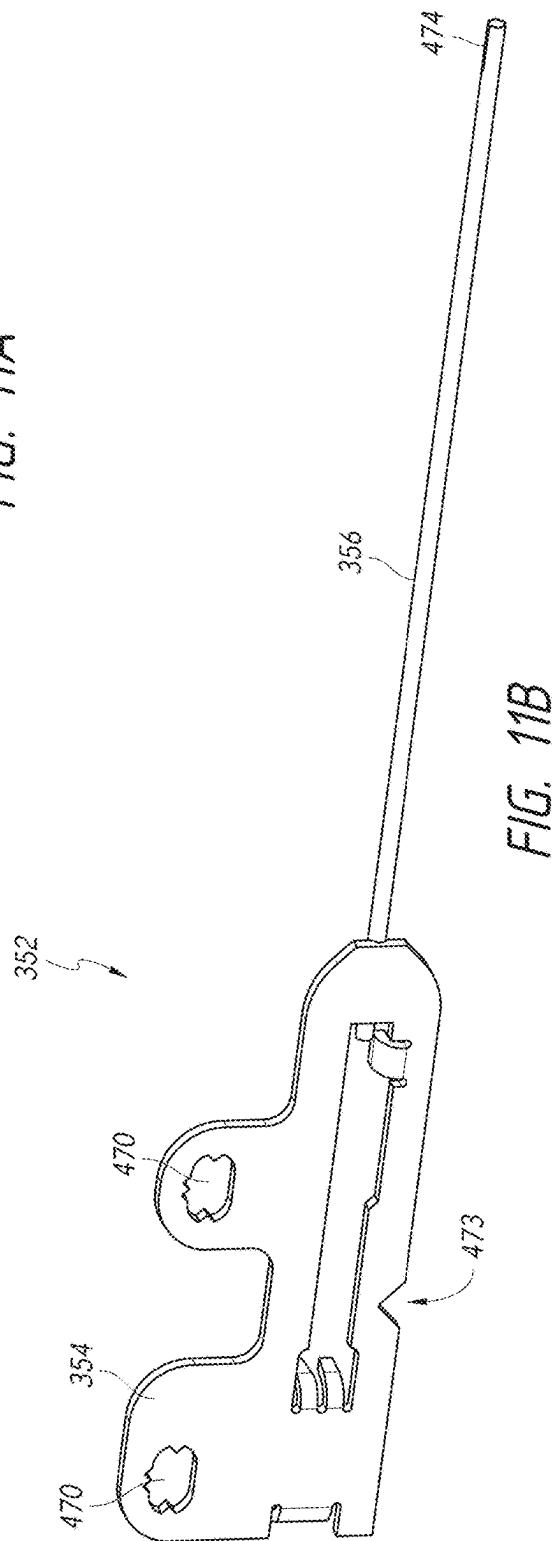
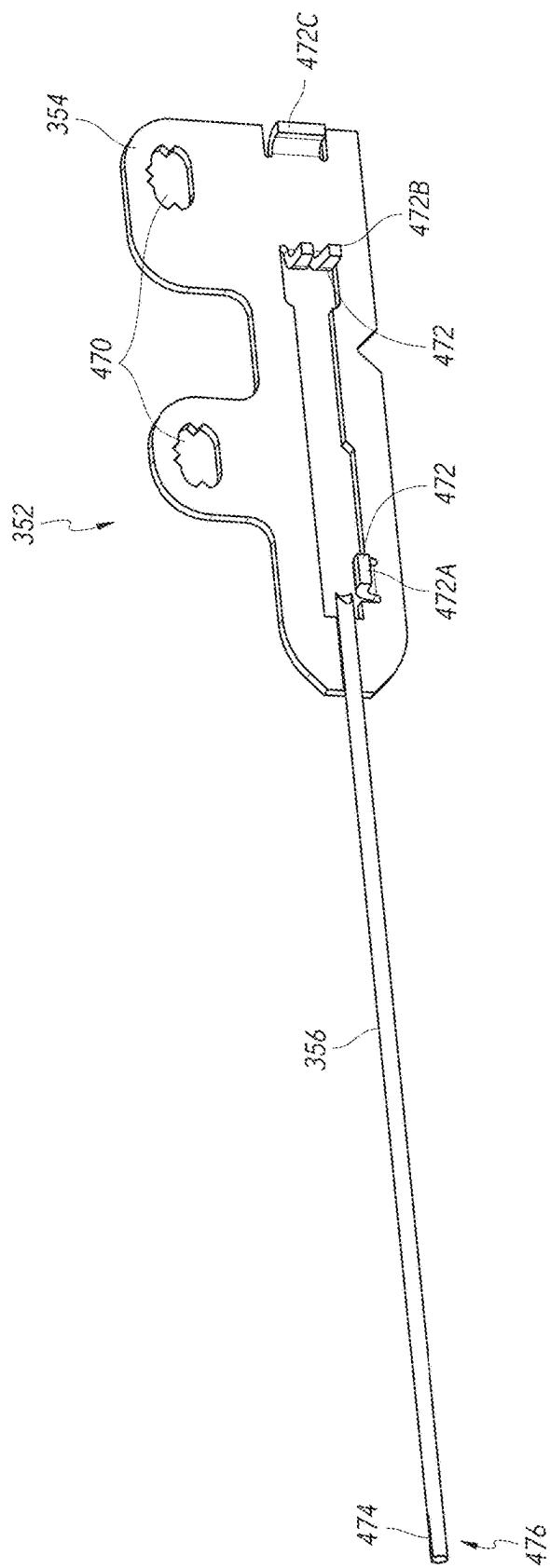
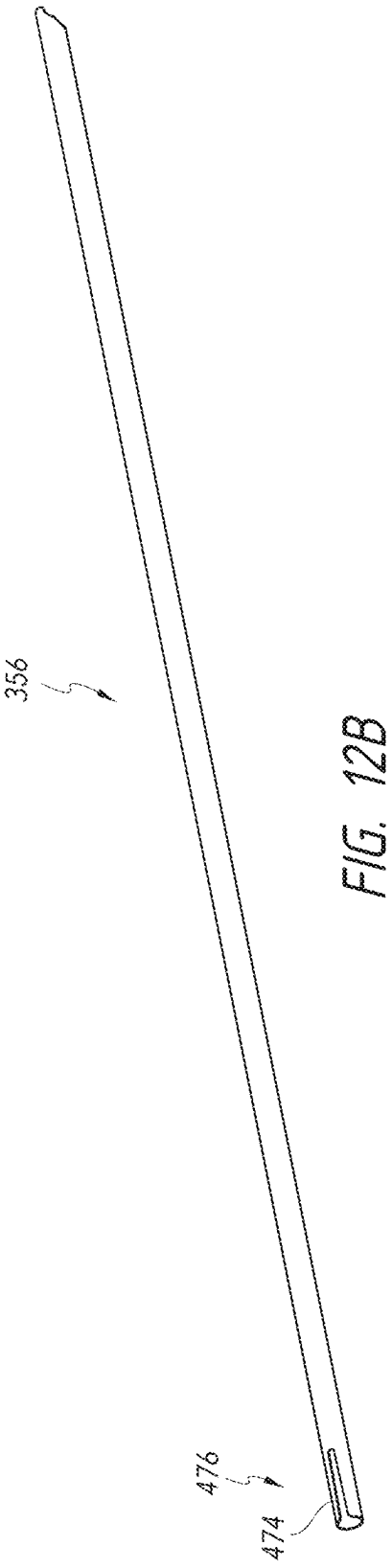
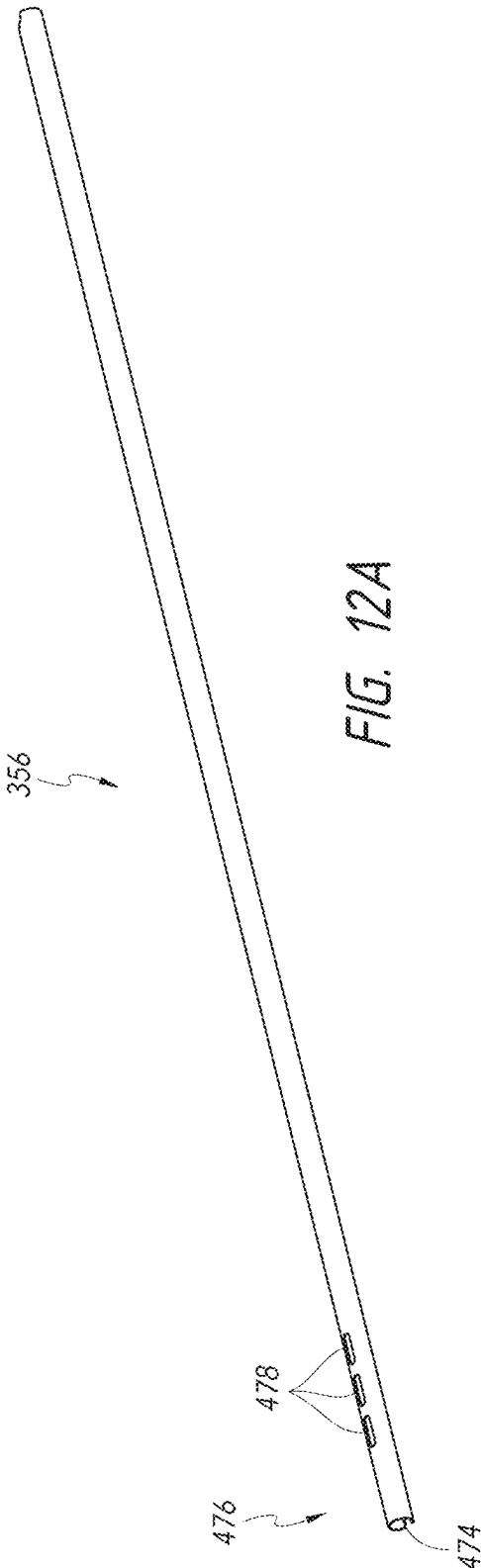


FIG. 9C









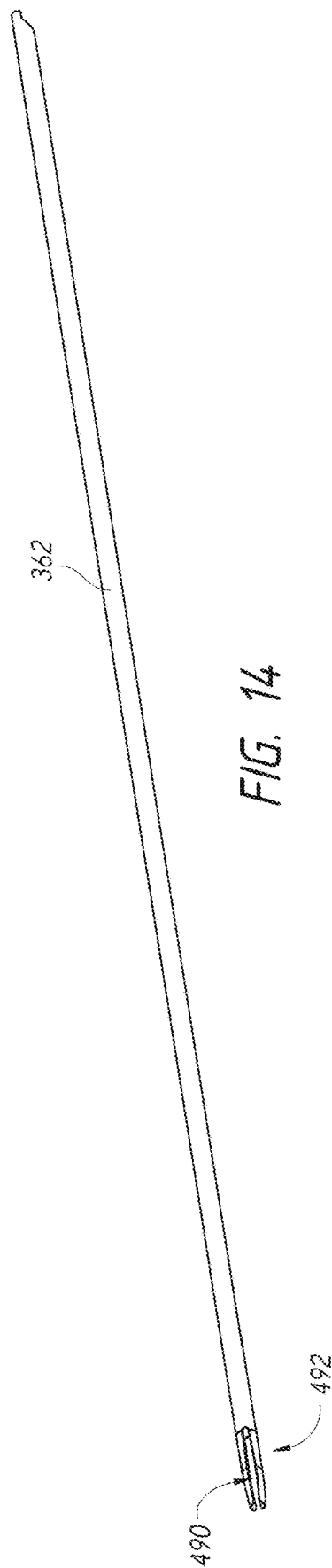
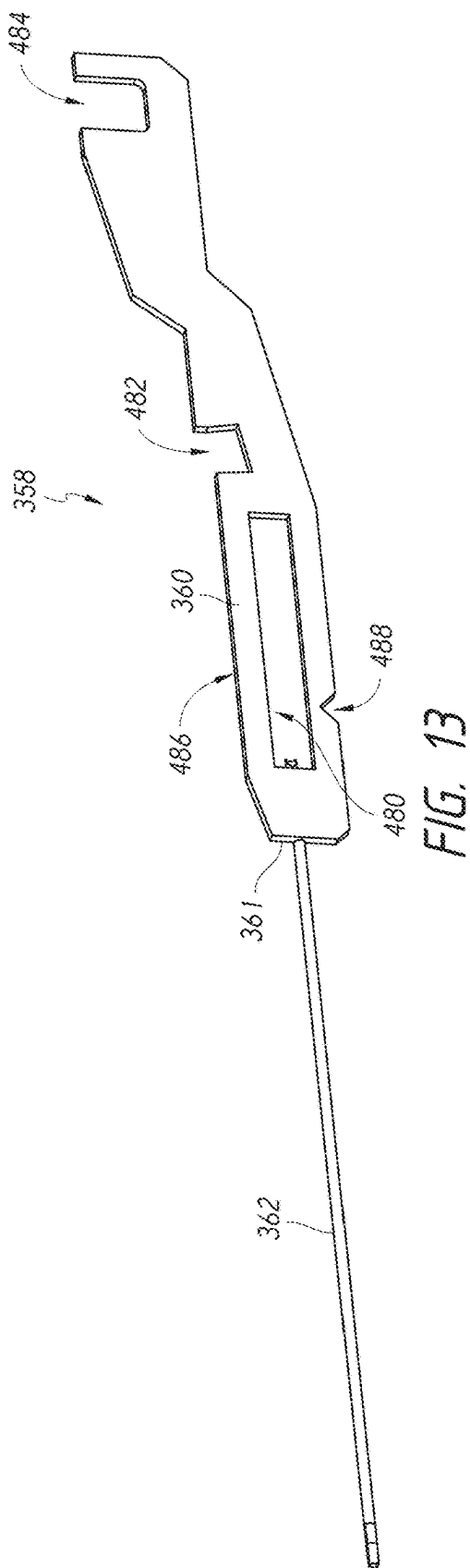




FIG. 15A

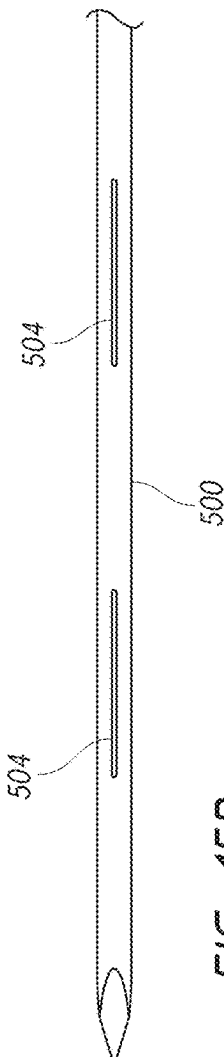


FIG. 15B

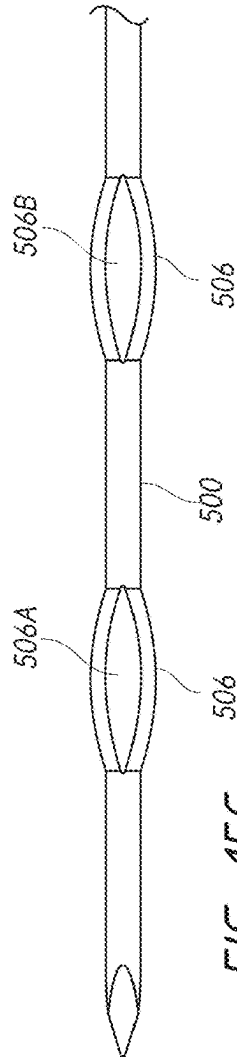


FIG. 15C

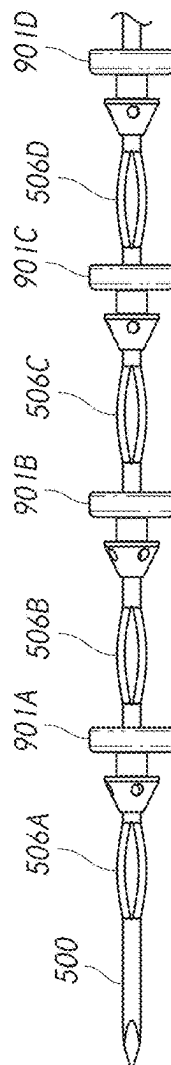


FIG. 15D

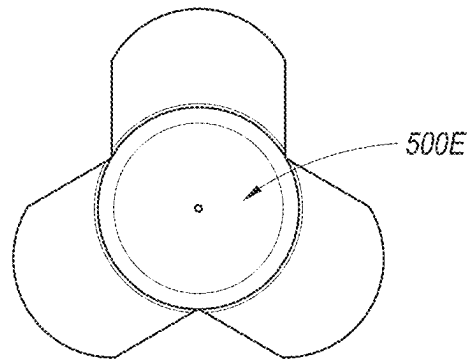


FIG. 15E-1

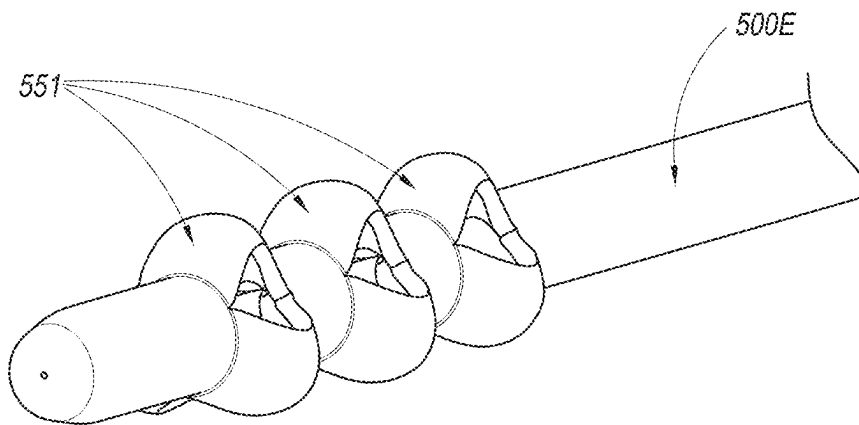


FIG. 15E-2

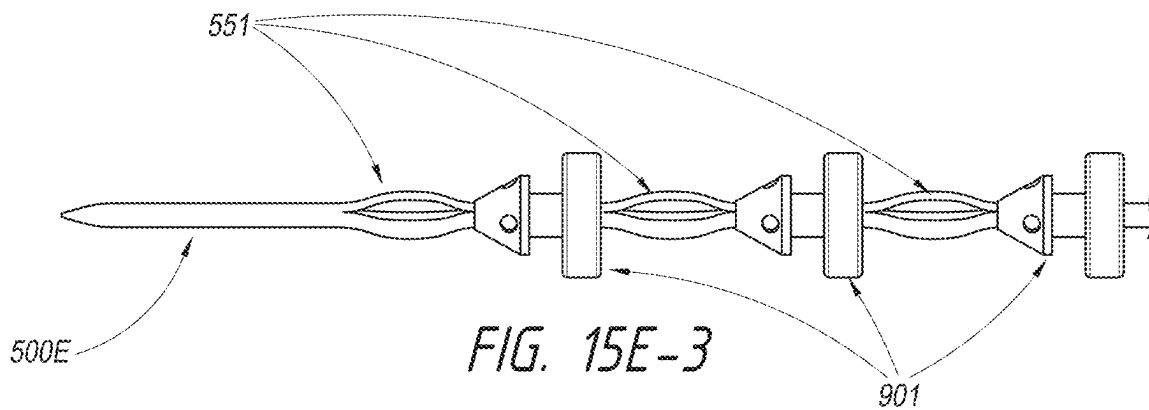
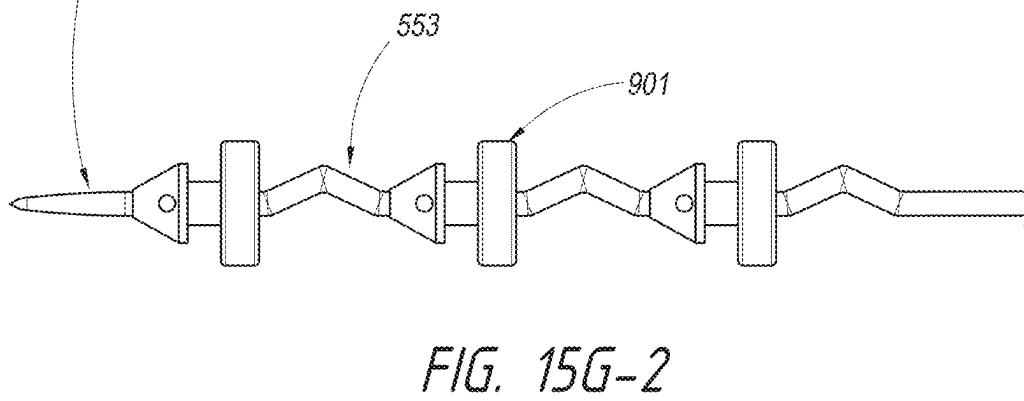
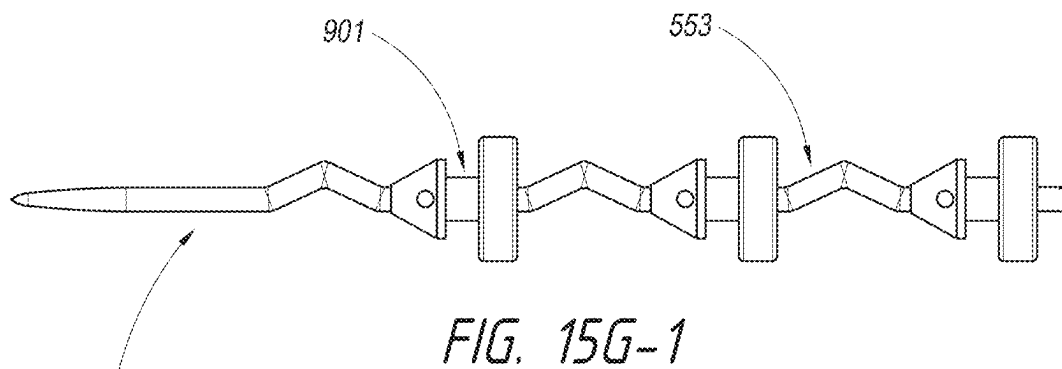
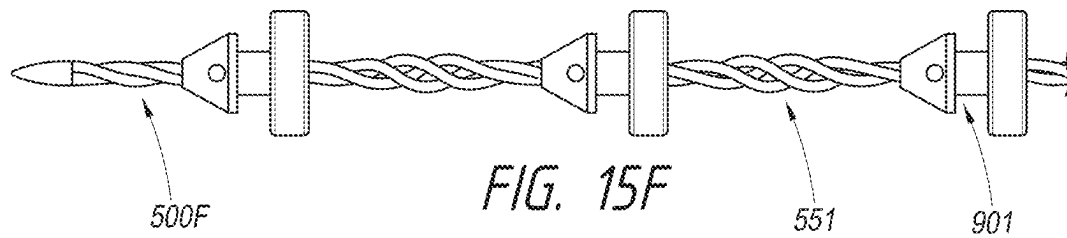
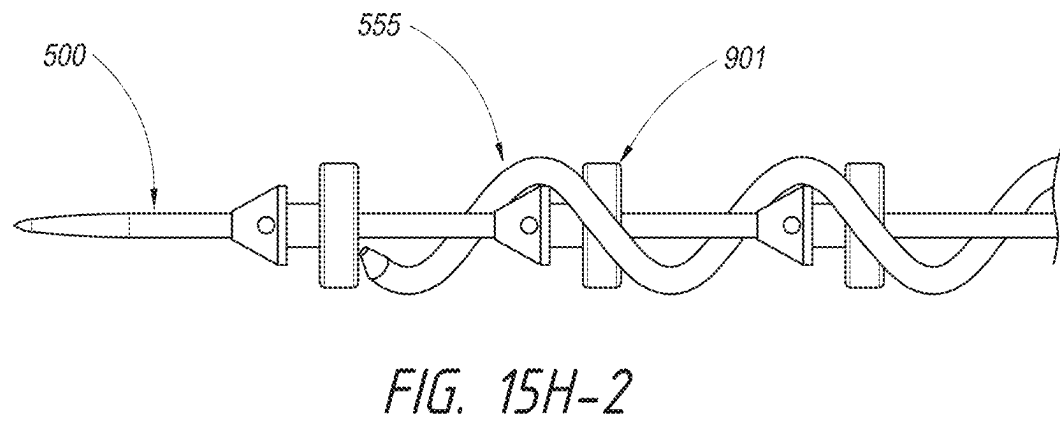
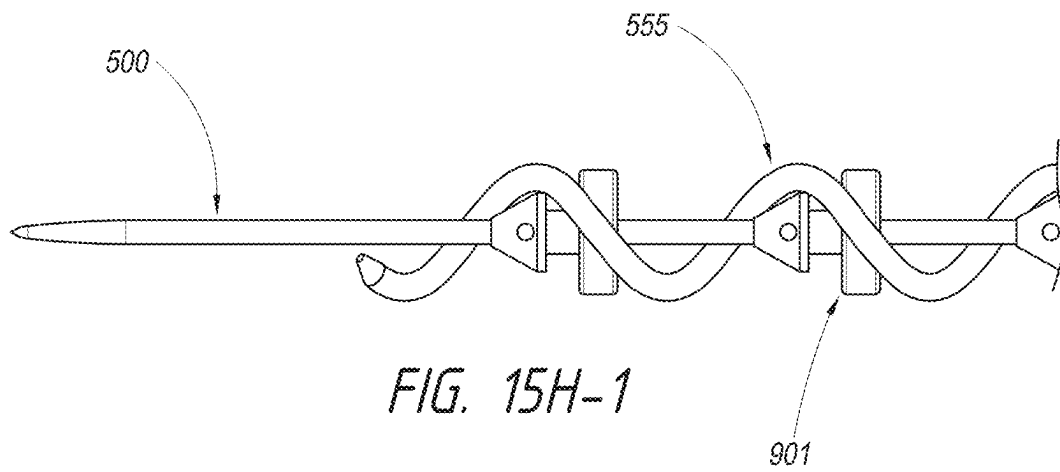
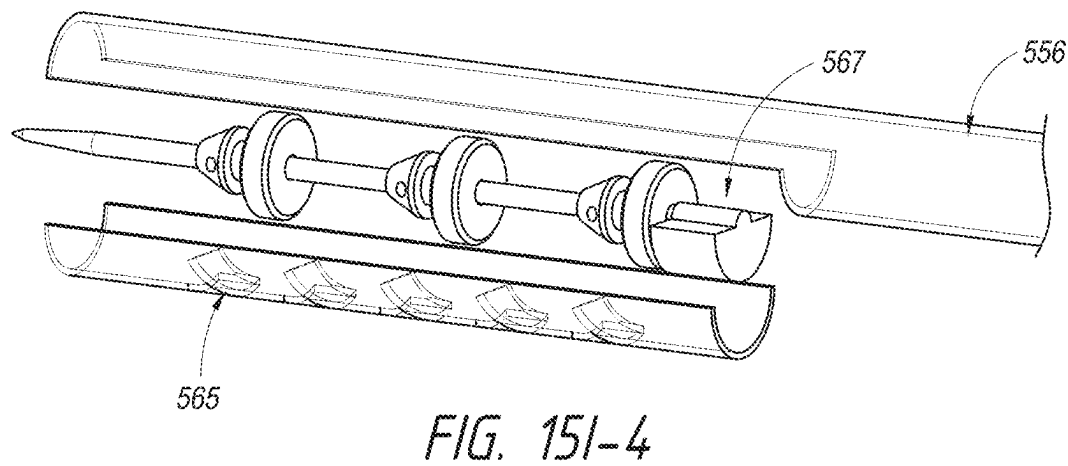
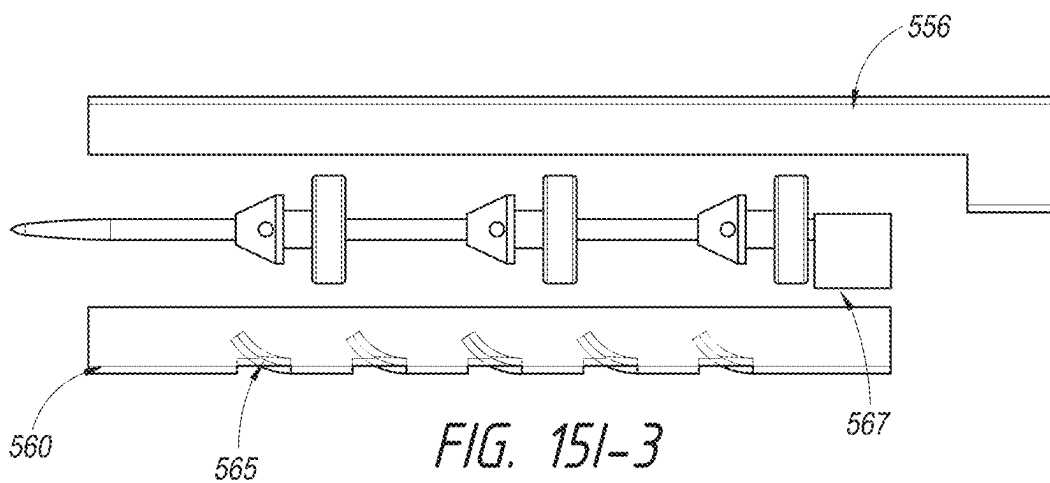
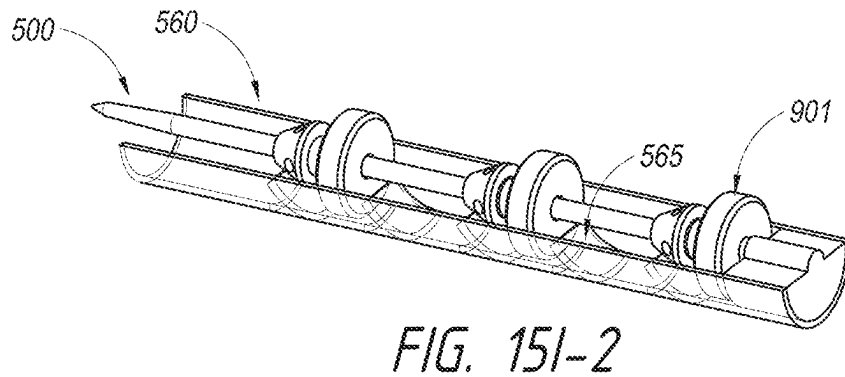
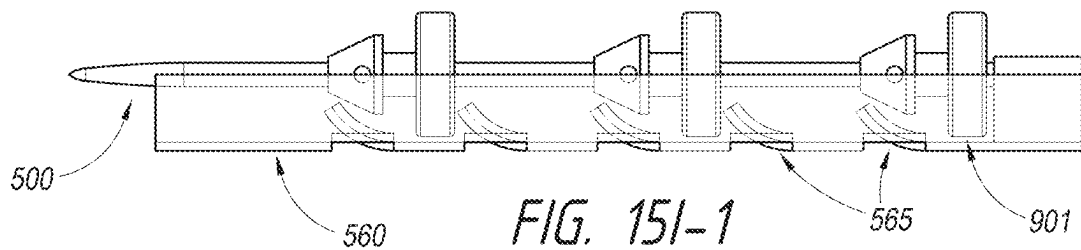


FIG. 15E-3







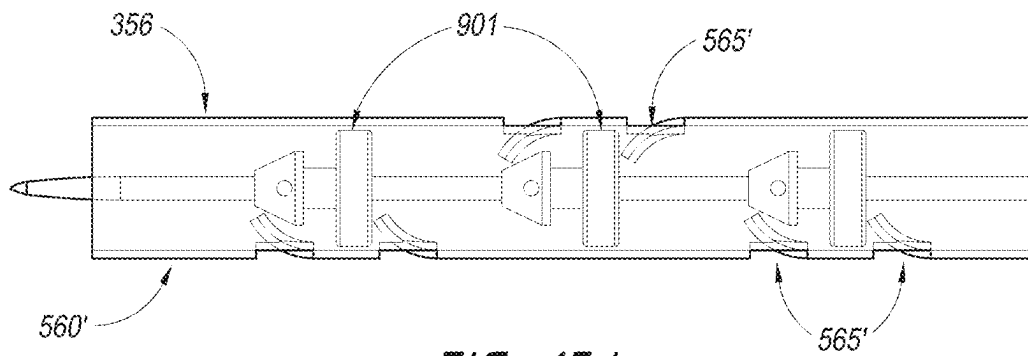


FIG. 15J

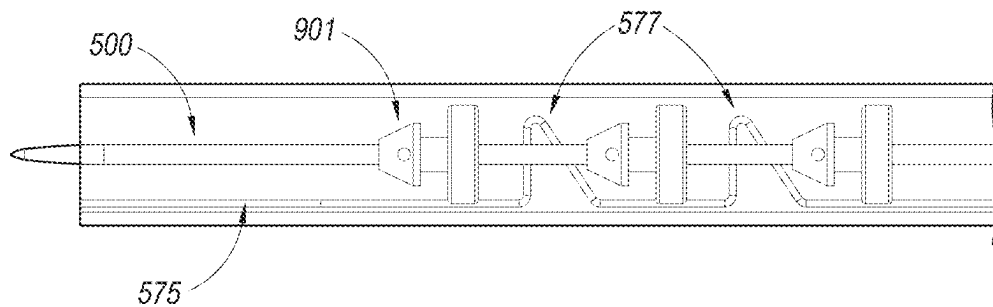


FIG. 15K-1

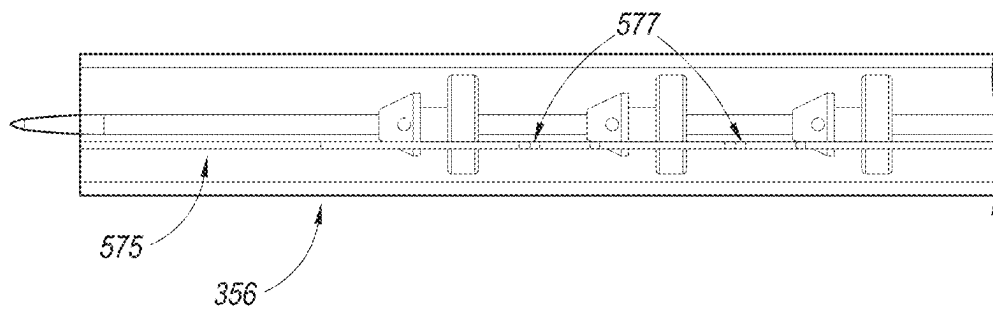


FIG. 15K-2

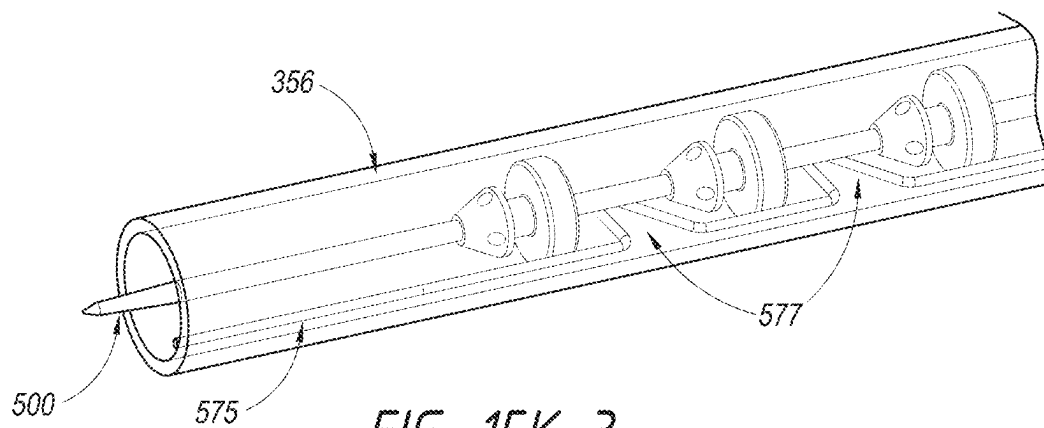
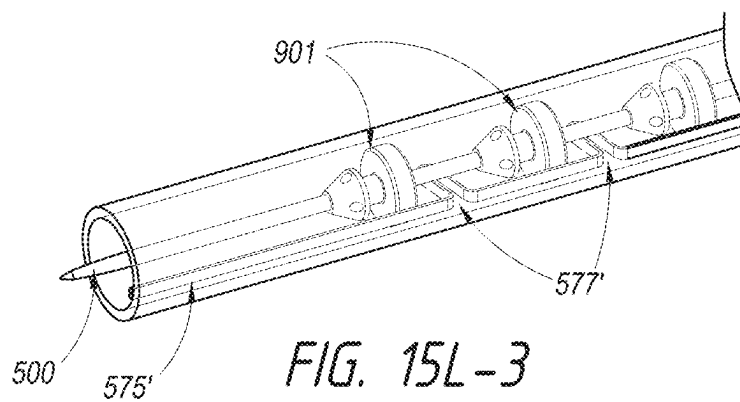
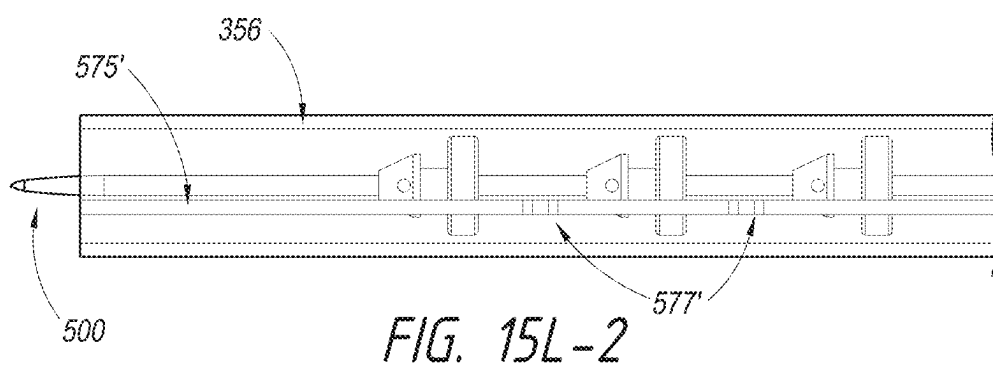
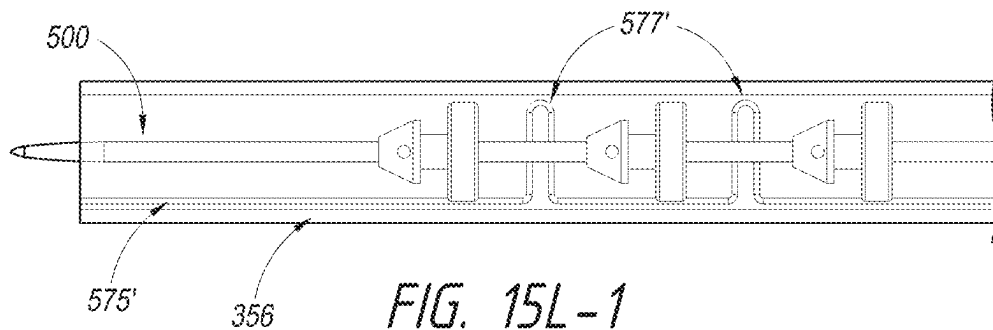


FIG. 15K-3



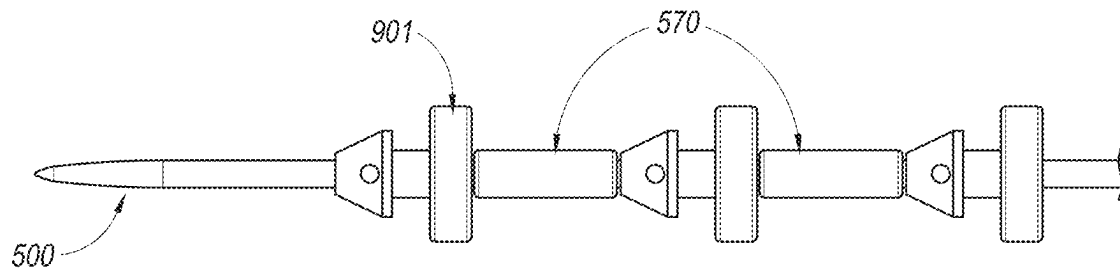


FIG. 15M

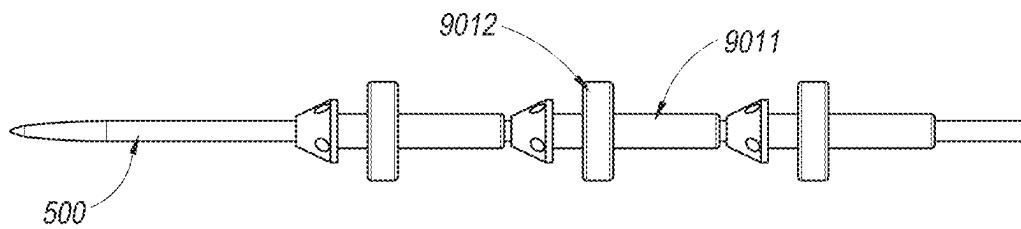


FIG. 15N-1

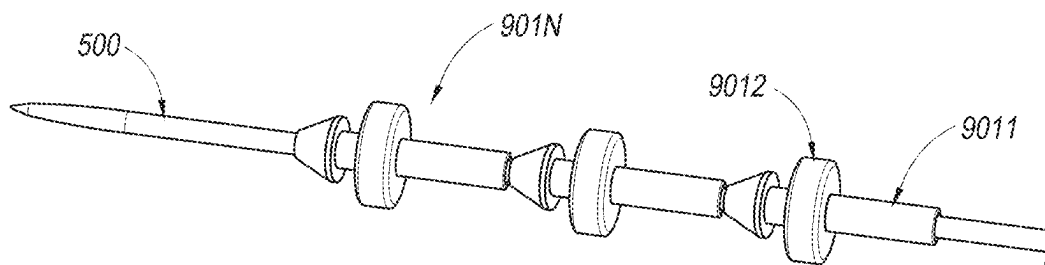


FIG. 15N-2

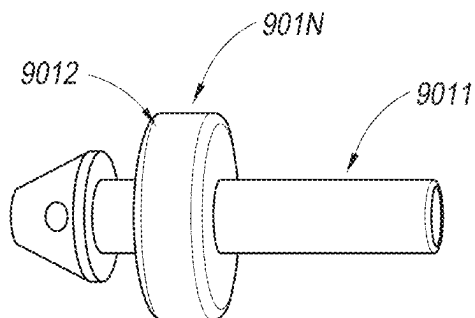
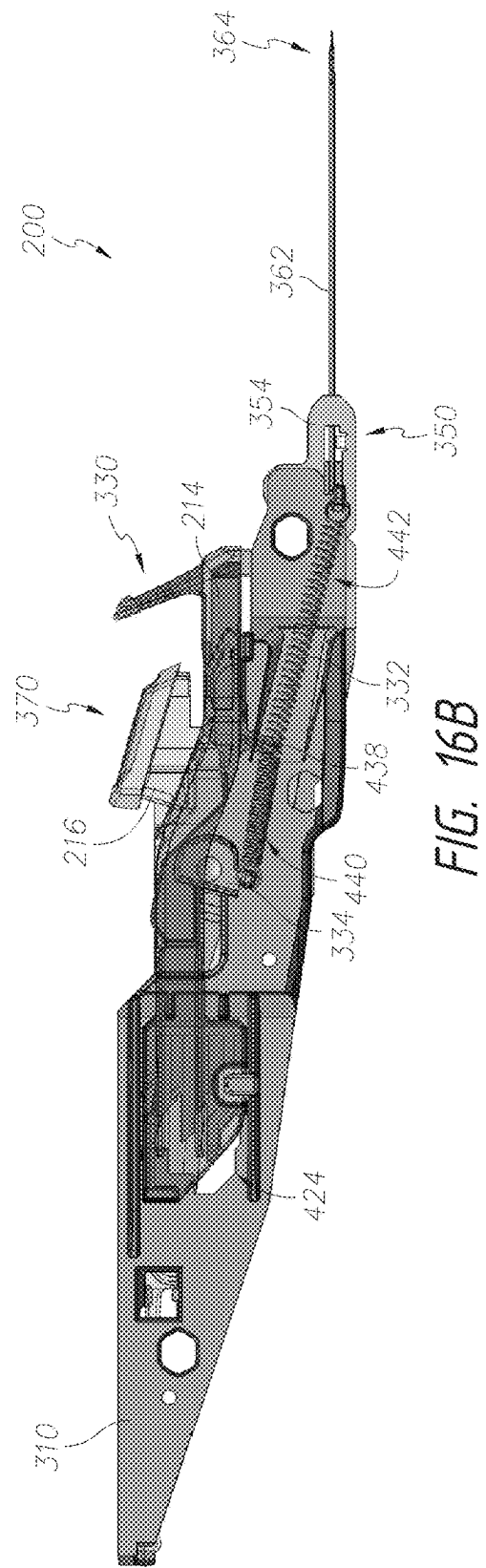
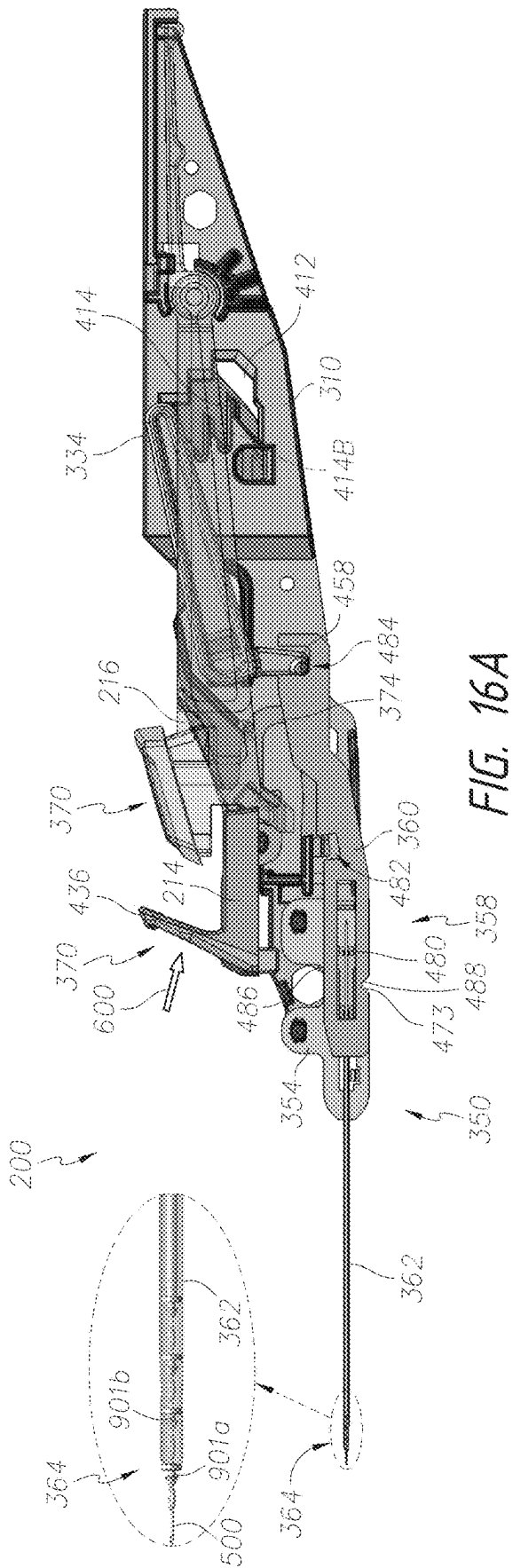
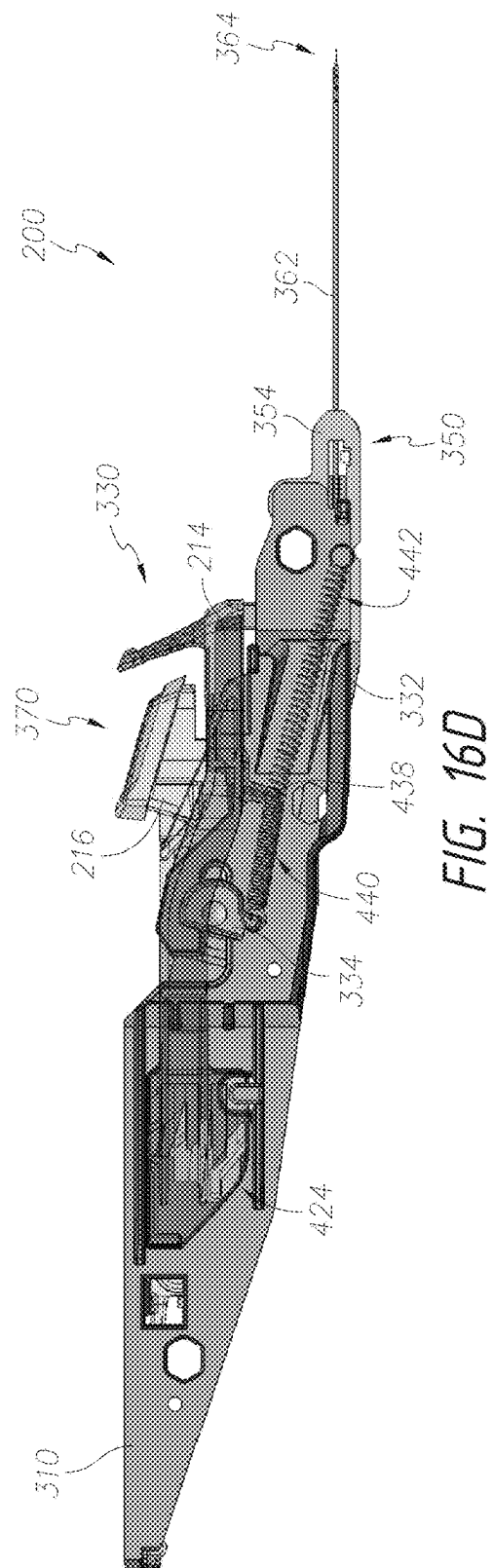
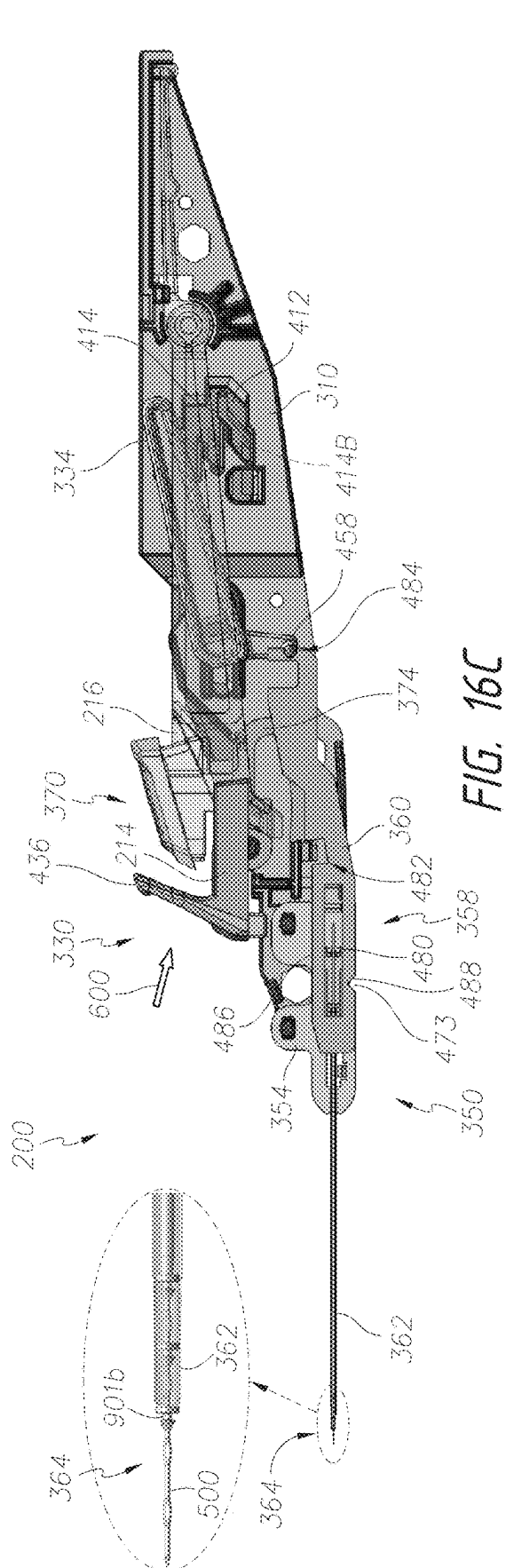
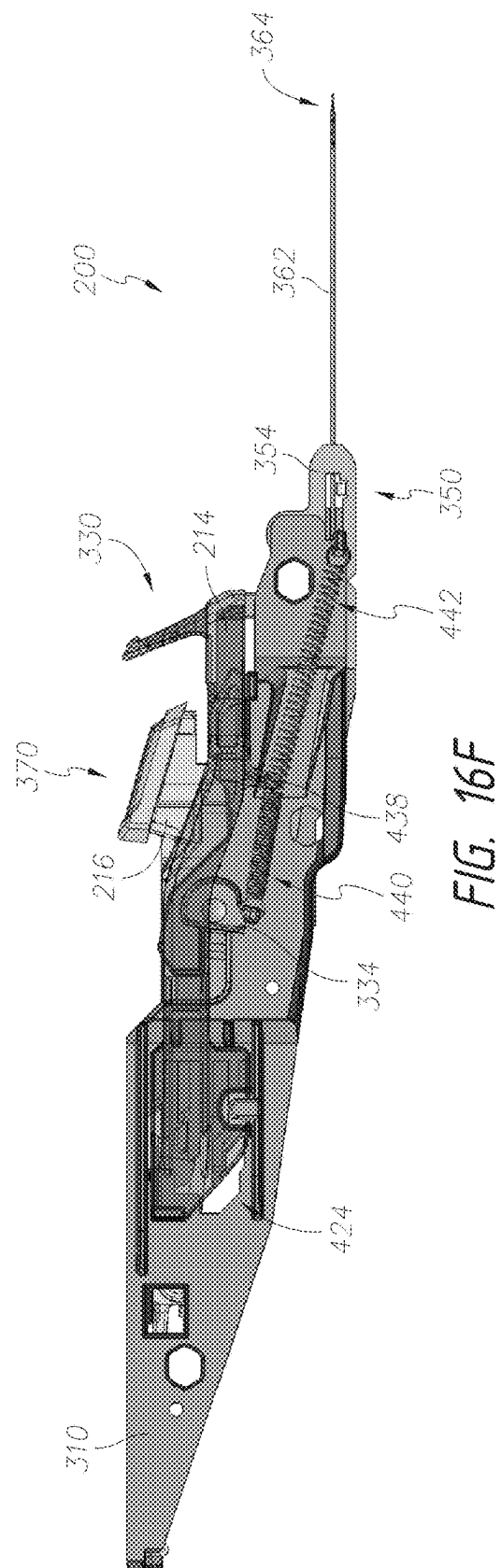
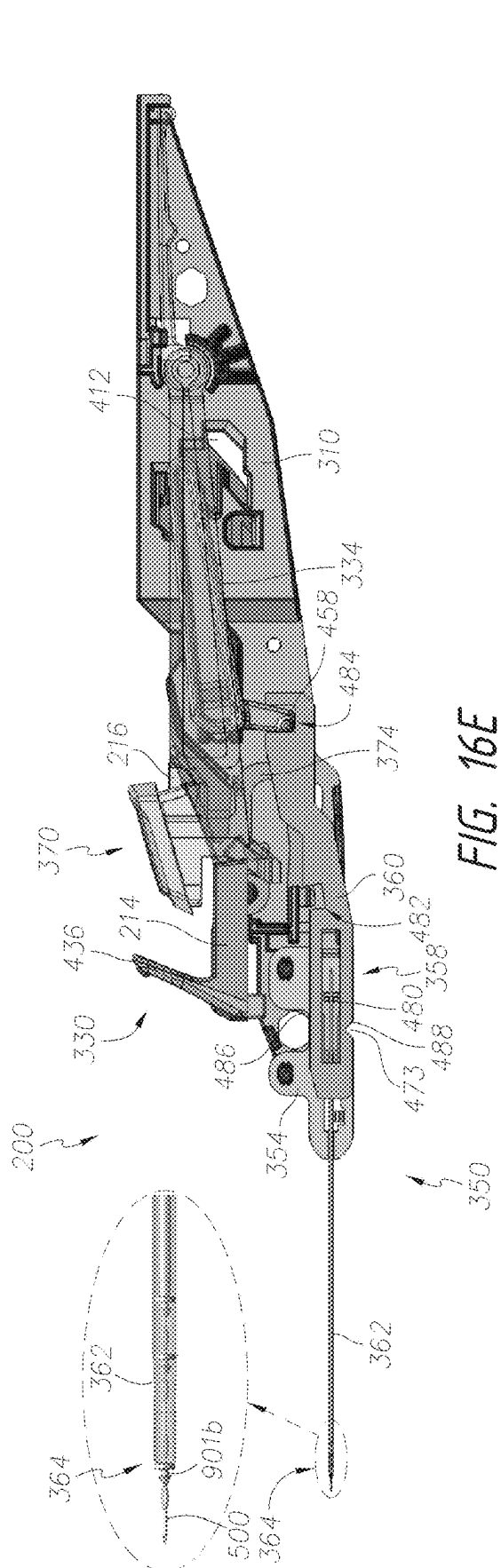


FIG. 15N-3







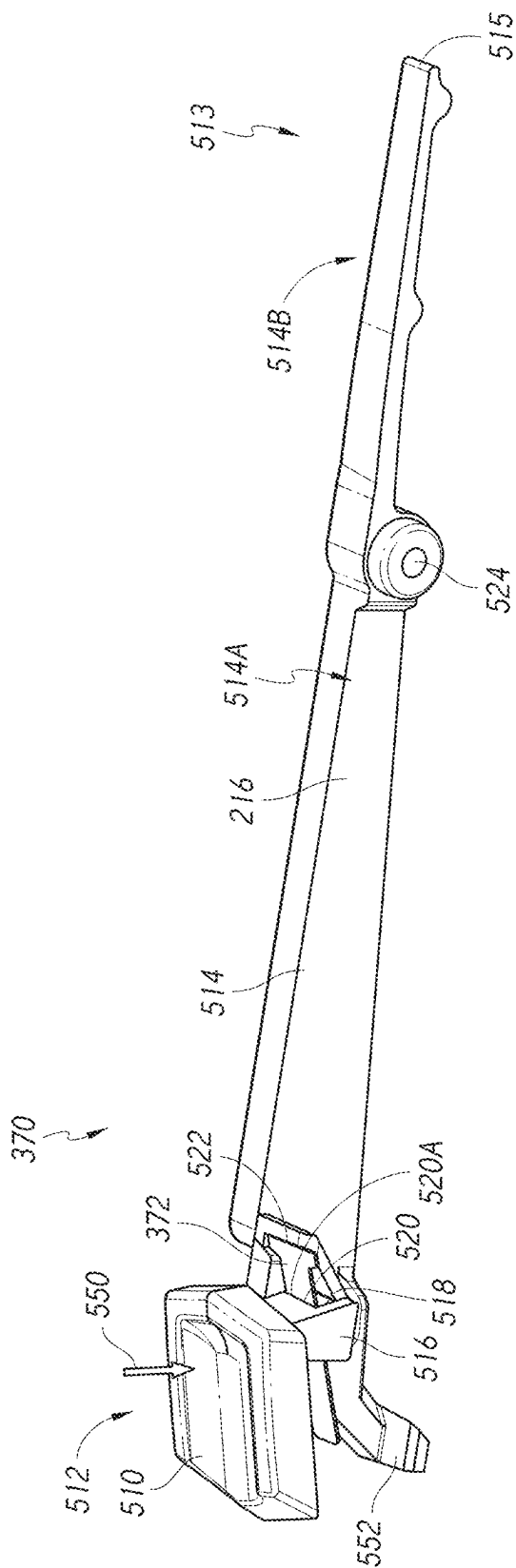


FIG. 17

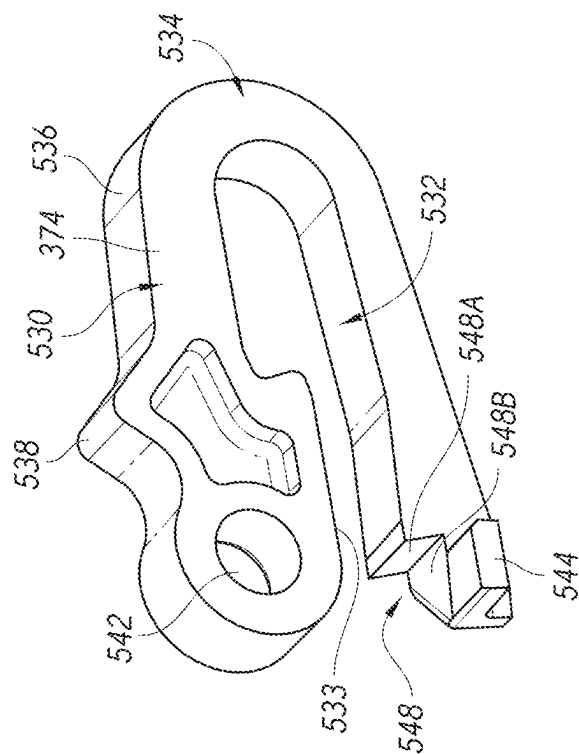


FIG. 19

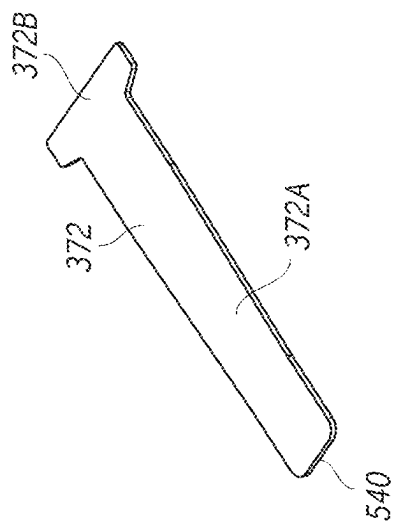


FIG. 18

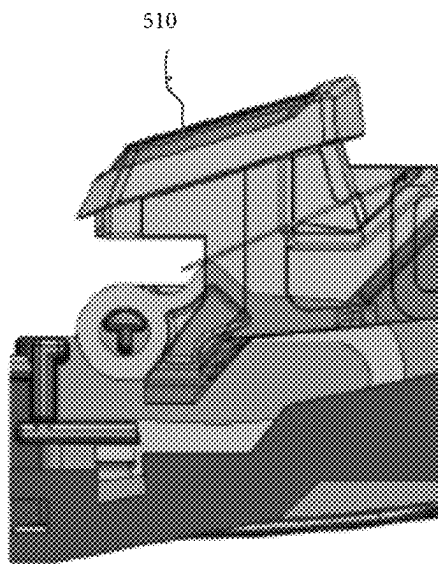


FIG. 20A

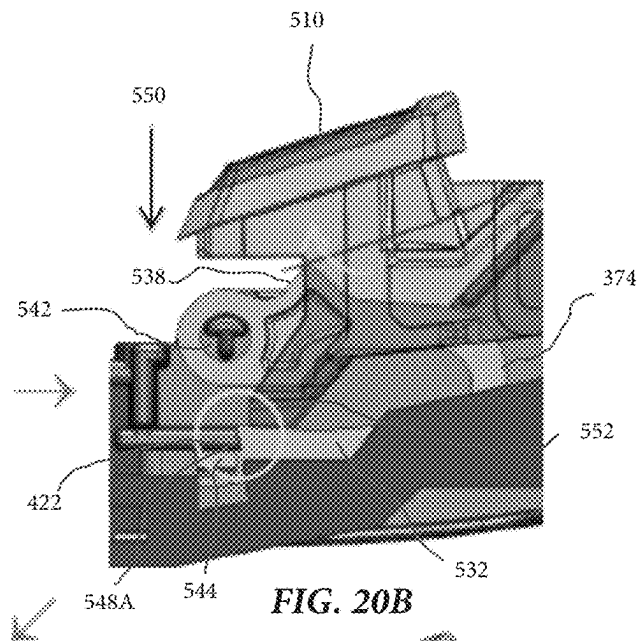


FIG. 20B

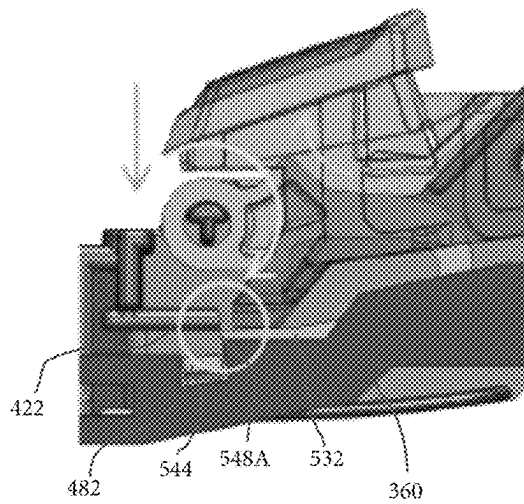


FIG. 20C

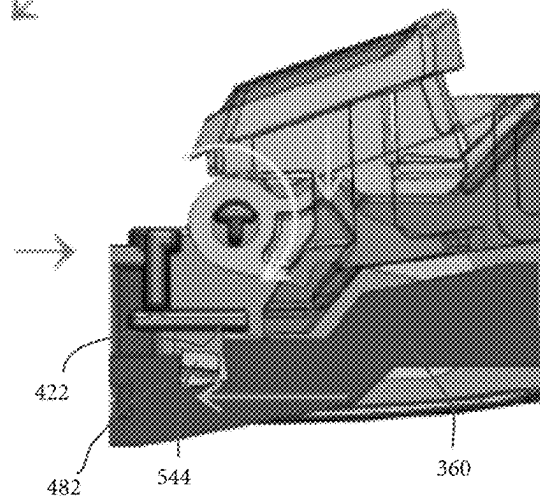


FIG. 20D

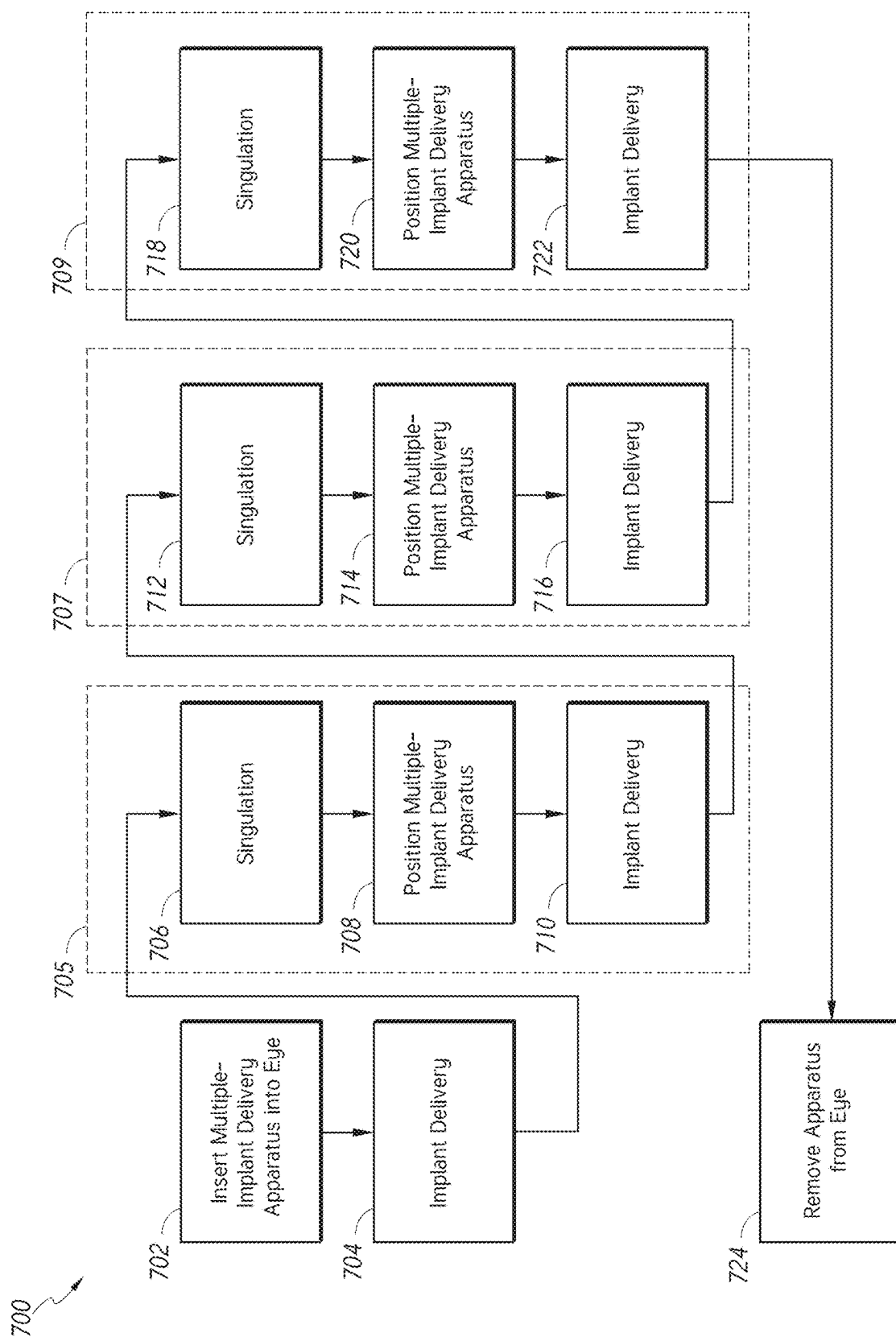


FIG. 21A

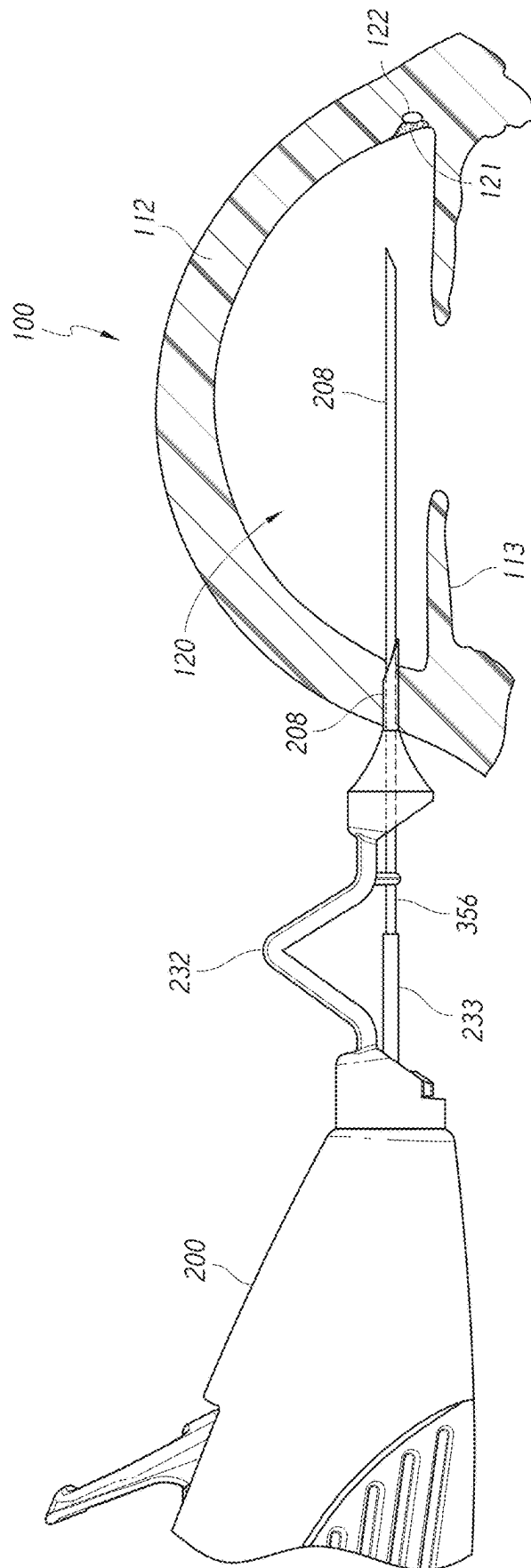


FIG. 21B

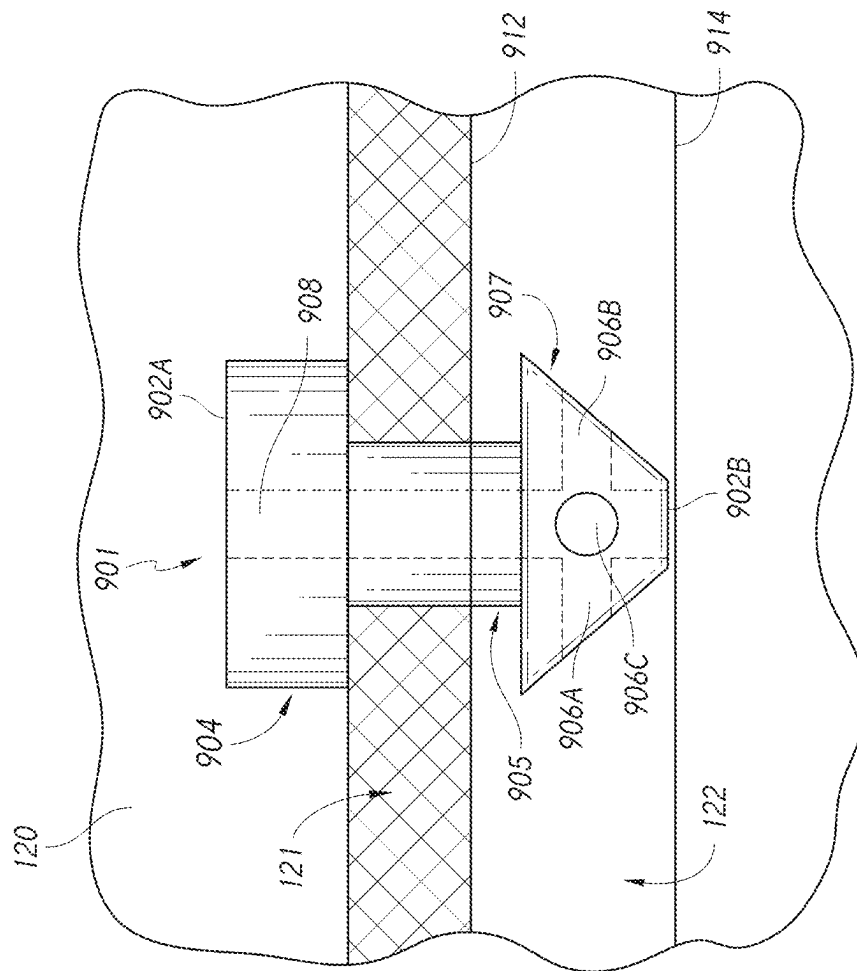


FIG. 22

SYSTEMS AND METHODS FOR DELIVERING MULTIPLE OCULAR IMPLANTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 16/132,252, filed Sep. 14, 2018, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Nos. 62/569,458 filed Oct. 6, 2017; 62/578,273 filed Oct. 27, 2017; and 62/671,286 filed May 14, 2018; each of which is hereby incorporated by reference in its entirety and made a part of this specification for all that it discloses.

FIELD

Embodiments of the inventions generally relate to devices and methods for delivering multiple implants using a single delivery apparatus without having to remove the apparatus from a body of the subject between implantations.

BACKGROUND

A human eye is a specialized sensory organ capable of light reception and is able to receive visual images. Aqueous humor (hereinafter referred to as “aqueous”) is a transparent liquid that fills at least the region between the cornea, at the front of the eye, and the lens. Aqueous is continuously secreted by ciliary processes of a ciliary body to the posterior chamber of the eye and the aqueous flows to the anterior chamber by crossing the pupil, so there is a constant flow of aqueous humor from the ciliary body to the anterior chamber of the eye. The aqueous fluid supplies nutrients to the avascular structures of the eye (for example, the cornea and the lens) and maintains intraocular pressure. Pressure within the eye is determined by a balance between the production of aqueous and its exit through canalicular outflow, uveoscleral outflow, or other outflow routes or pathways.

Many open-angle glaucomas are caused by an increase in the resistance to aqueous drainage through the trabecular meshwork and/or Schlemm’s canal (e.g., the canalicular outflow pathways). The tissue of the trabecular meshwork normally allows the aqueous to enter Schlemm’s canal, which then empties into aqueous collector channels in the posterior wall of Schlemm’s canal and then into aqueous veins, which form the episcleral venous system. The uveoscleral outflow pathways can refer to the aqueous leaving the anterior chamber by diffusion through intercellular spaces among ciliary muscle fibers or through a supraciliary and/or suprachoroidal space.

Intraocular implants (for example, shunts or stents) can be implanted within the eye to facilitate the outflow of aqueous, thereby reducing intraocular pressure. Typical methods of implantation require relatively invasive surgical procedures, pose a risk of excessive trauma to the eye, and require excessive handling of the implant. For example, in a typical method of implantation, an incision is made through the sclera or cornea and the implant is inserted into the desired implantation location using forceps or another like manual grasping device. These forceps are configured for holding, and introducing into the eye only one implant at a time. This requires reloading and repositioning of the forceps prior to inserting each implant into the eye. Once the implants are deposited, the grasping device is removed and the incision is sutured closed.

SUMMARY

According to some embodiments, an implant delivery apparatus for treating an ocular disorder can include an external housing and an introducer assembly. The external housing can include an opening, a singulation portion, and an activation portion. The singulation portion can be actuated by a user. The activation portion can be actuated by the user. The auto-retracting introducer assembly can include a distal introducer tip and a flexible proximal retraction member. The distal introducer tip can extend from a distal end portion of the flexible retraction member. The introducer assembly can surround and be guided by at least a portion of an insertion tube. The introducer tip (and/or an introducer tube surrounded by the introducer tip) can extend from the external housing at an angle relative to a longitudinal axis of the implant delivery apparatus. The singulation portion is configured to be manually actuated by a user. The activation portion may be configured to be an infinite activation portion and the singulation portion may be configured to be manually actuated.

In accordance with several embodiments, a method of treating an ocular disorder includes advancing at least a portion of an implant delivery apparatus through an incision in an eye. The implant delivery apparatus may include an introducer assembly including an insertion tube defining a lumen and a trocar assembly pre-loaded with a plurality of implants, the trocar assembly configured to be positioned within the lumen, and an actuation assembly configured to facilitate delivery of a first implant of the plurality of implants, the actuation assembly comprising an actuator trigger portion configured to be accessible by a user. The method further includes piercing ocular tissue with the introducer assembly, positioning the implant delivery apparatus adjacent a desired implantation location and depressing the actuator trigger portion to effect delivery of the first implant by causing the insertion tube to contact the first implant, the actuator trigger portion configured to be depressed an infinite number of times to properly deliver the first implant. The implants may be ocular implants configured to facilitate drainage of aqueous humor from an anterior chamber of an eye to a physiologic outflow pathway (e.g., Schlemm’s canal, collector channel, suprachoroidal space, supraciliary space) of the eye. The plurality of implants may consist of any one of two implants, three implants, and four implants without requiring reconfiguration. In other words, the apparatus operates in the same manner regardless of how many implants are loaded therein. In some embodiments, energy required to deliver each respective implant of the plurality of implants is generated by an actuation biasing member upon the depressing of the actuator trigger portion such that no energy is pre-stored by the actuation biasing member prior to the depressing of the actuator trigger portion.

In accordance with several embodiments, a method of treating an ocular disorder includes positioning an implant delivery apparatus within an eye. The implant delivery apparatus includes an introducer assembly including a singulation tube and a trocar assembly pre-loaded with a plurality of implants and a singulation assembly configured to facilitate selection of an implant of the plurality of implants, the singulation assembly including a singulation handle configured to be accessible by a user. The method also includes manipulating the singulation handle to effect selection of the implant, the manipulation configured to cause the singulation tube to slide over the at least one

3

implant such that the singulation tube is positioned proximally relative to a proximally facing side of the implant.

In accordance with several embodiments, a method of treating an ocular disorder (e.g., glaucoma) includes positioning an implant delivery apparatus within an eye. The implant delivery apparatus includes a trocar pre-loaded with a plurality of implants and a singulation tube coaxially surrounding the trocar and being configured to move proximally and distally (e.g., rearward and forward) with respect to the trocar. The implant delivery apparatus also includes a singulation assembly configured to facilitate selection of a first implant of the plurality of implants. The singulation assembly includes a singulation handle configured to be accessible by an operator. The method further includes retracting the singulation handle toward the operator to effect selection of the first implant, the retraction configured to cause the singulation tube to slide over the first implant such that the singulation tube is positioned proximally relative to a proximally-facing side of the first implant.

The method may also include causing the singulation tube to propel the first implant toward a distal end of the trocar by actuating an implant delivery actuator of the implant delivery apparatus. The method may further include retracting the singulation handle toward the operator a second time to effect selection of a second implant of the plurality of implants, the retraction configured to cause the singulation tube to slide over the second implant such that the singulation tube is positioned proximally relative to a proximally-facing side of the second implant. The trocar may include a plurality of separation regions formed by slits in the trocar at spaced-apart locations along the length of the trocar. The separation regions may be configured to mechanically separate the plurality of implants from each other until the singulation tube engages the proximal end of a respective one of the plurality of implants and advances the implant to a ready-to-fire position along the trocar. The method may also include repositioning an implant delivery apparatus within an eye at a different location.

In accordance with several embodiments, an implant delivery apparatus configured to deliver a plurality of implants for treating an ocular disorder includes an external housing including an opening on an upper side of the external housing. The apparatus may also include an auto-retracting introducer assembly configured to facilitate introduction of a distal portion of the implant delivery apparatus into an eye of a subject. The introducer assembly includes a distal introducer tip and a flexible proximal retraction member extending from a distal end of the external housing. The apparatus further includes a singulation assembly configured to facilitate on-demand singulation of each of the plurality of implants upon manual actuation by an operator of a lever extending out of the opening of the external housing. The apparatus also includes an implantation actuator assembly configured to effect delivery of each of the plurality of implants following singulation. The implantation actuator assembly includes an implant delivery actuator including a trigger button extending out of the opening of the external housing that is configured to be actuated an infinite (e.g., unlimited) number of times by the operator.

The apparatus may further include an insertion tube extending from a distal end of the external housing at an angle relative to a longitudinal axis of the implant delivery apparatus, the insertion tube configured to retain the plurality of implants therein. The angle may be between 1 and 15 degrees (e.g., between 7 and 9 degrees, between 6 and 10 degrees, between 5 and 12 degrees, between 7 and 11

4

degrees, between 1 and 10 degrees, between 7 and 15 degrees, overlapping ranges thereof, or any value within the recited ranges).

The apparatus may also include a trocar positioned within the external housing, a distal end portion of the trocar being configured to extend within and along a length of a lumen of the insertion tube, wherein the plurality of implants are positioned along the distal end portion of the trocar. The apparatus may further include a collet holder assembly including a collet holder and a singulation tube extending from the collet holder. A distal end of the singulation tube may comprise multiple tines configured to facilitate retraction of the distal end of the singulation tube over a maximum cross-sectional dimension of a respective implant during singulation. The distal end of the singulation tube may be configured to engage a proximal end of the respective implant following singulation and to advance the respective implant to a ready-to-fire position along the trocar. The ready-to-fire position may advantageously be the same position for each successive implant of the plurality of implants.

The singulation assembly may further include a singulation arm coupled to the lever and to the collet holder and a singulation biasing member (e.g., a spring) coupled to the singulation arm and to a fixed frame within the external housing. Proximal retraction of the lever of the singulation assembly may cause the collet holder to retract proximally (e.g., rearwardly). Then, release of the lever may cause the singulation tube to engage the proximal end of the respective implant following singulation and to advance the respective implant to the ready-to-fire position along the trocar. The implantation actuator assembly may further include an actuator arm and an actuator biasing member (e.g., flat spring), wherein the energy sufficient to effect delivery of each respective implant is provided by the actuator biasing member (e.g., bending of the flat spring), and wherein the energy provided by the actuator biasing member is generated from pressing of the trigger button of the implant delivery actuator by the operator.

The trocar may include multiple singulation regions spaced apart along the length of the trocar. The singulation regions may be configured to facilitate mechanical separation of the plurality of implants from each other. The singulation regions may include splayed regions formed by slits in the trocar. The apparatus may be configured to deliver two, three, or four implants without requiring different configurations. In some embodiments, the fixed frame includes a singulation frame slot having a plurality of platforms or slots sized and shaped to facilitate singulation of the plurality of implants through interaction with one or more components of the singulation assembly.

In accordance with several embodiments, an implant delivery apparatus configured to deliver a plurality of implants for treating an ocular disorder includes an external housing including an opening, a trigger button configured to be actuated by a user extending out of the opening, an activation portion configured to be actuated by the user; and an introducer assembly. The introducer assembly includes an introducer tube extending from the external housing at an angle relative to a longitudinal axis of the implant delivery apparatus, the introducer tube configured to retain a plurality of implants therein. The angle may be between 1 and 15 degrees (e.g., between 7 and 9 degrees, between 6 and 10 degrees, between 5 and 12 degrees, between 7 and 11 degrees, between 1 and 10 degrees, between 7 and 15 degrees, overlapping ranges thereof, or any value within the recited ranges).

5

The introducer assembly may further include an auto-retracting introducer assembly configured to surround at least a portion of the introducer tube, the introducer assembly including a distal introducer tip and a flexible proximal retraction member, the distal introducer tip extending from a distal end portion of the flexible proximal retraction member. In some embodiments, the singulation portion is configured to be manually actuated by a user so as to facilitate on-demand manual singulation to effect selection of one of the plurality of implants for delivery one at a time. The actuation portion may include an actuator that is configured to be manually actuated by a user to effect ejection of an implant of the plurality of implants out of the introducer tube toward a distal end of the trocar. The actuation portion may be configured to allow manual actuation an infinite (e.g., unlimited) number of times. The apparatus may include a trocar configured to extend within and through the introducer tube. The plurality of implants are configured to be positioned and advanced along the trocar. The trocar may include a plurality of separation regions formed by slits along a length of the trocar at spaced-apart locations along the length of the trocar, the separation regions configured to separate the plurality of implants from each other until a singulation actuator of the singulation portion interacts with a respective one of the plurality of implants upon manual actuation of the singulation portion by the operator.

In accordance with several embodiments, an implant delivery apparatus with an activation portion configured to facilitate actuation of an implant delivery mechanism an infinite (e.g., unlimited) number of times includes an external housing including an opening within an upper side of the external housing, an implant delivery actuator including a trigger button extending out of the opening, an actuation biasing member (e.g., a flat spring), and an actuator arm. Depressing of the trigger button causes the actuation biasing member to store energy sufficient to eject, or propel, an implant toward a distal end of the implant delivery apparatus. Release of the trigger button causes the actuation biasing member to release the stored energy. In some embodiments, depressing of the trigger button causes the flat spring to bend, thereby storing the energy. The flat spring may be positioned in contact with the actuator arm so as to cause the flat spring to bend as the trigger button is depressed.

In accordance with several embodiments, an implant delivery apparatus having a manual "on-demand" singulation portion includes an external housing including an opening on an upper side of the external housing, a trocar having a plurality of implants loaded thereon, wherein each implant is spaced apart at a separation distance along the trocar, and a frame fixed to the external housing, the frame including a singulation frame slot configured to facilitate selection and movement to a ready-to-fire position along the trocar of one of the plurality of implants. The apparatus further includes a singulation assembly including a lever extending out of the opening of the external housing, the lever configured to be retracted proximally (e.g., rearwardly) by a finger or thumb of an operator. The singulation assembly also includes a singulation arm comprising a proximal end and a distal end, wherein the proximal end is coupled to the singulation frame slot and wherein the distal end is coupled to a collet holder assembly. The collet holder assembly includes a singulation tube.

In some embodiments, a singulation biasing member is coupled to the lever, to the distal end of the singulation arm, and to the frame within the external housing. Proximal retraction of the lever of the singulation assembly causes the

6

collet holder assembly to retract proximally (e.g., rearwardly) and release of the lever causes the singulation tube of the collet holder assembly to engage a proximal end of one of the plurality of implants and advance the implant to the ready-to-fire position along the trocar. In some embodiments, the trocar includes a plurality of separation regions formed by slits in the trocar at spaced-apart locations along the length of the trocar. The separation regions may be configured to mechanically separate the plurality of implants from each other until the singulation tube engages the proximal end of a respective one of the plurality of implants and advances the implant to the ready-to-fire position along the trocar. The apparatus is configured to facilitate selection and delivery of two, three, or four implants without requiring different configurations.

According to several embodiments, the systems and methods described herein include one or more of the following advantages or benefits: (i) easy to assemble, (ii) inexpensive to manufacture (e.g., no overmolding, no adhesives, no lubricants), (iii) less concern about tight tolerances, (iv) similar use profile (e.g., singulated implants that look and feel the same to the clinician using the device), (v) improved surgical experience due to greater ease of use, equivalent stent delivery, less stent-to-stent or implantation to implantation variability, and less unit-to-unit variability; (vi) better recovery from under-implantation; (vii) no limit to the number of shots or implantation actuations available; and/or (viii) modularity of the design.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects, and advantages of the present disclosure will now be described with reference to the drawings of embodiments of the invention, which embodiments are intended to illustrate and not to limit the scope of the disclosure.

FIG. 1A is a schematic cross-sectional view of an eye.

FIG. 1B is an enlarged cross-sectional view of an anterior chamber angle of the eye of FIG. 1A.

FIG. 2A is a perspective view illustrating an embodiment of a multiple-implant delivery apparatus.

FIG. 2B is a close up perspective view illustrating an embodiment of an introducer assembly of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2C is a front view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2D is a rear view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2E is a side view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2F is another side view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2G is a top view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 2H is a bottom view illustrating an embodiment of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 3A is a close up top perspective view illustrating the introducer assembly of FIG. 2B.

FIG. 3B is a close-up bottom perspective view illustrating the introducer assembly of FIG. 2B.

FIG. 4 is a perspective exploded view of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 5A is a side perspective view of an embodiment of a left housing of the multiple-implant delivery apparatus of FIG. 2A.

7

FIG. 5B is a side perspective view of an embodiment of a right housing of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 6A is a right side view of an embodiment of the internal components of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 6B is a left side view of the internal components of FIG. 6A.

FIG. 7A is a right side view of an embodiment of a frame of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 7B is a left side view of the frame of FIG. 7A.

FIG. 8A is a right side view of an embodiment of a singulation assembly of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 8B is a left side view of the singulation assembly of FIG. 8A.

FIG. 9A is a right side perspective view of an embodiment of a singulation arm of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 9B is a left side view of the singulation arm of FIG. 9A.

FIG. 9C is a left and distal side perspective view of the singulation arm of FIG. 9A.

FIG. 10A is a right side perspective view of an embodiment of a tube set assembly of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 10B is a left side perspective view of the tube set assembly of FIG. 10A.

FIG. 10C is a close-up view of a portion of the tube set assembly of FIG. 10A.

FIG. 11A is a right side perspective view of an embodiment of an insertion tube subassembly of the tube set assembly of FIG. 10A.

FIG. 11B is a left side perspective view of the insertion tube subassembly of FIG. 11A.

FIG. 12A is a right side perspective view of an embodiment of an insertion tube of the tube set assembly of FIG. 10A.

FIG. 12B is a left side perspective view of the insertion tube of FIG. 12A.

FIG. 13 is a right side perspective view of an embodiment of a collet holder subassembly of the tube set assembly of FIG. 10A.

FIG. 14 is a perspective view of an embodiment of a singulation tube of the tube set assembly of FIG. 10A.

FIG. 15A is a perspective view of an embodiment of a trocar assembly of the tube set assembly of FIG. 10A.

FIG. 15B is a top view of an embodiment of a trocar of the trocar assembly of FIG. 15A.

FIG. 15C is a top view of an embodiment of a trocar of the trocar assembly of FIG. 15A.

FIG. 15D is a top close-up view of an embodiment of a trocar of the trocar assembly of FIG. 15A.

FIGS. 15E-1 to 15E-3 illustrate a front view, perspective view, and side view of an alternative embodiment of a trocar comprising polymeric material. FIG. 15E-3 shows a plurality of implants loaded on the trocar.

FIG. 15F illustrates a side view of an alternative embodiment of a trocar formed of a plurality of braided wires.

FIGS. 15G-1 and 15G-2 illustrate side views of an alternative embodiment of a trocar having a plurality of bend regions to facilitate singulation of implants.

FIGS. 15H-1 and 15H-2 illustrate side views of an embodiment of a “corkscrew” singulation assembly that includes a spiral wire configured to advance the implants along the trocar.

8

FIGS. 15I-1 to 15I-4 illustrate side and perspective views of two embodiments of an “ice-cube tray” singulation assembly.

FIG. 15J illustrates a side view of another embodiment of an ice cube tray singulation assembly having cut-out tabs on opposite sides of the insertion tube.

FIGS. 15K-1 to 15K-3 and 15L-1 to 15L-3 illustrate side, top and perspective views of two embodiments of spring wire singulation assemblies.

FIG. 15M illustrates an embodiment of a singulation assembly including multiple bioerodible spacers positioned between adjacent implants along the trocar.

FIGS. 15N-1 and 15N-2 illustrate side and perspective views of a singulation assembly including plurality of elongated snorkel stents, or implants, positioned along a trocar. FIG. 15N-3 illustrates a side view of an embodiment of one of the elongated snorkel stents.

FIG. 16A is right side view of an embodiment of the internal components of the multiple-implant delivery apparatus of FIG. 2A in a first singulation position.

FIG. 16B is a left side view of the internal components of the multiple-implant delivery apparatus in the first singulation position.

FIG. 16C is right side view of the internal components of the multiple-implant delivery apparatus of FIG. 16A in a second singulation position.

FIG. 16D is a left side view of the internal components of the multiple-implant delivery apparatus in the second singulation position.

FIG. 16E is right side view of the internal components of the multiple-implant delivery apparatus of FIG. 16A in a third singulation position.

FIG. 16F is a left side view of the internal components of the multiple-implant delivery apparatus in the third singulation position.

FIG. 17 is perspective view of an embodiment of an actuation assembly of the multiple-implant delivery apparatus of FIG. 2A.

FIG. 18 is a perspective view of an embodiment of an actuation biasing member of the actuation assembly of FIG. 17.

FIG. 19 is a perspective view of an embodiment of an actuator arm of the actuation assembly of FIG. 17.

FIGS. 20A-20D are partial cross-section views illustrating components of the multiple-implant delivery apparatus of FIG. 2A at various stages of an implant delivery cycle.

FIG. 21A is an example method of using the implant delivery apparatus.

FIG. 21B is a schematic and partial sectional view of a portion of an eye illustrating insertion of the multiple-implant delivery apparatus within the eye.

FIG. 22 is an enlarged schematic and partial sectional view of Schlemm’s canal and the trabecular meshwork of an eye illustrating the position and operation of an ocular implant delivered by the multiple-implant delivery apparatus of FIG. 2A.

DETAILED DESCRIPTION

I. Introduction

Embodiments of systems, devices and methods for delivering multiple (e.g. one, two, three, four, or more) ocular implants of various shapes and sizes are described herein. In the following description, numerous specific details are set forth to provide a thorough understanding of the embodiments; however, one skilled in the relevant art will recog-

nize, based upon the disclosure herein, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.

Reference throughout this description to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment described herein. Thus, the appearances of the phrases “in one embodiment” or “in certain embodiments” in various places throughout this description are not necessarily all referring to the same embodiments. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

FIG. 1A is a cross-sectional view of an eye 100. FIG. 1B is an enlarged sectional view of the eye showing the relative anatomical locations of a trabecular meshwork 121, an anterior chamber 120, and Schlemm’s canal 122. With reference to FIGS. 1A and 1B, the sclera 111 is a thick collagenous tissue that covers the entire eye 100 except a portion that is covered by a cornea 112. The cornea 112 is a thin transparent tissue that focuses and transmits light into the eye and through a pupil 114, which is a circular hole in the center of an iris 113 (colored portion of the eye). The cornea 112 merges into the sclera 111 at a juncture referred to as a limbus 115. A ciliary body 116 is vascular tissue that extends along the interior of the sclera 111 from the outer edges of the iris in the limbal region to a choroid 117. The ciliary body 116 is comprised of ciliary processes and ciliary muscle. Ciliary zonules extend from the ciliary processes to a lens 126. The choroid 117 is a vascular layer of the eye 100, located between the sclera 111 and a retina 118. An optic nerve 119 transmits visual information to the brain and is the anatomic structure that is progressively destroyed by glaucoma.

With continued reference to FIGS. 1A and 1B, the anterior chamber 120 of the eye 100, which is bound anteriorly by the cornea 112 and posteriorly by the iris 113 and the lens 126, is filled with aqueous humor. Aqueous humor is produced primarily by the ciliary processes of the ciliary body 116 and flows into the posterior chamber, bounded posteriorly by the lens 126 and ciliary zonules and anteriorly by the iris 113. The aqueous humor then flows anteriorly through the pupil 114 and into the anterior chamber 120 until it reaches an anterior chamber angle 125, formed between the iris 113 and the cornea 112.

As best illustrated by the drawing of FIG. 1B, in a normal eye, at least some of the aqueous humor drains from the anterior chamber 120 through the trabecular meshwork 121 via the canalicular route. Aqueous humor passes through the trabecular meshwork 121 into Schlemm’s canal 122 and thereafter through a plurality of collector ducts and aqueous veins 123, which merge with blood-carrying veins, and into systemic venous circulation. Intraocular pressure is maintained by an intricate balance between secretion and outflow of aqueous humor in the manner described above. Glaucoma is, in most cases, characterized by an increased outflow resistance of aqueous humor from the anterior chamber 120, which leads to an increase in intraocular pressure. Fluids are relatively incompressible, and thus intraocular pressure is distributed relatively uniformly throughout the eye 100.

As shown in FIG. 1B, the trabecular meshwork 121 lies adjacent a small portion of the sclera 111. Exterior to the sclera 111 is a conjunctiva 124. Traditional procedures that create a hole or opening for implanting a device through the

tissues of the conjunctiva 124 and sclera 111 involve extensive surgery, as compared to surgery for implanting a device, such as described herein, which ultimately resides entirely within the confines of the sclera 111 and cornea 112.

In accordance with some embodiments, an ophthalmic implant system is provided that comprises multiple ocular implants and a delivery instrument for delivering and implanting the multiple ocular implants within eye tissue. The multiple implants may be preloaded within the delivery instrument at the time of assembly, manufacture or packaging. These ocular implants can be configured to drain fluid from the anterior chamber of a human eye into a physiologic outflow pathway, such as Schlemm’s canal, aqueous collector channels, episcleral veins, the uveoscleral outflow pathway, the supraciliary space, and/or the suprachoroidal space. The physiologic outflow pathway can be an existing space or outflow pathway (such as Schlemm’s canal) or a potential space or outflow pathway (such as the suprachoroidal space). In some embodiments, the ocular implants are configured to be delivered to a location such that the implant communicates or allows fluid to communicate with an outflow pathway. While this and other systems and associated methods and apparatuses may be described herein in connection with glaucoma treatment (e.g., phakic or pseudophakic mild to moderate or refractory open angle glaucoma), the disclosed systems, methods, and apparatuses can be used to treat other types of ocular disorders in addition to glaucoma or to implant other devices (such as pressure sensors or analyte sensors (e.g., glucose sensors)).

While a majority of the aqueous leaves the eye through the trabecular meshwork and Schlemm’s canal, it is believed that a significant percentage of the aqueous in humans leaves through the uveoscleral pathway. The degree with which uveoscleral outflow contributes to the total outflow of the eye appears to be species dependent. As used herein, the term “uveoscleral outflow pathway” is to be given its ordinary and customary meaning to a person of ordinary skill in the art (and it is not to be limited to a special or customized meaning), and refers without limitation to the space or passageway whereby aqueous exits the eye by passing through the ciliary muscle bundles located at or near an angle of the anterior chamber and into the tissue planes between the choroid and the sclera, which extend posteriorly to the optic nerve. From these tissue planes, it is believed that the aqueous travels through the surrounding scleral tissue and drains via the scleral and conjunctival vessels, or is absorbed by the uveal blood vessels.

As used herein, the term “supraciliary space” is to be given its ordinary and customary meaning to a person of ordinary skill in the art (and it is not to be limited to a special or customized meaning), and refers without limitation to the portion of the uveoscleral pathway through the ciliary muscle and between the ciliary body and the sclera, and the term “suprachoroidal space” is to be given its ordinary and customary meaning to a person of ordinary skill in the art (and it is not to be limited to a special or customized meaning), and refers without limitation to the portion of the uveoscleral pathway between the choroid and sclera.

The following description will include references to distal and proximal ends of various components and right and left sides of various components. The terms “distal” and “proximal” are to be given their ordinary and customary meaning to a person of ordinary skill in the art (and are not to be limited to a special or customized meaning), and refer without limitation to opposite regions or ends of a particular structure. In some embodiments, the term “distal” is used to refer to a region or end farther away from a person using the

systems and devices described herein or performing the methods described herein and the term “proximal” is used to refer to a region or end closer to the person using the systems and devices described herein or performing the methods described herein; however, the meanings of the terms can be swapped.

The term “right side” should be understood to mean the side of the component that, upon assembly, faces the right housing of the multiple-implant delivery apparatus and the term “left side” should be understood to mean the side of the component that, upon assembly, faces the left housing of the multiple-implant delivery apparatus. However, these terms, as well as terms of orientation such as “top,” “bottom,” “upper,” “lower,” “front,” “rear,” and “end” are used herein to simplify the description of the context of the illustrated embodiments. Likewise, terms of sequence, such as “first” and “second,” are used to simplify the description of the illustrated embodiments. Because other orientations and sequences are possible, however, the claims should not be limited to the illustrated orientations or sequences. Those skilled in the art will appreciate, upon reading this disclosure, that other orientations of the various components described above are possible.

II. External Components of Multiple-Implant Delivery Apparatus

FIGS. 2A-2H illustrate various views of an embodiment of a multiple-implant delivery apparatus **200**, as described in the Brief Description of the Drawings section above. The multiple-implant delivery apparatus **200** can include an external housing **202**. The external housing **202** can include a distal end portion **220** and a proximal end portion **222**. The external housing **202** can extend between a distal terminus of the distal end portion **220** and a proximal terminus of the proximal end portion **222**. As shown in at least FIG. 2A, the proximal end portion **222** of the multiple-implant delivery apparatus **200** can be gradually tapered. In some embodiments, the distal end portion **220** is gradually tapered to form a somewhat nose-shaped cone **204**. The delivery apparatus **200** can include an introducer assembly **201** that extends from the cone **204**.

In some embodiments, the delivery apparatus **200** includes a forward portion **207** and a rearward portion **206**. The rearward portion **206** can include a curved and/or a reduced profile. In some embodiments, an upper portion of the rearward portion **206** of the apparatus **200** is generally rounded towards the proximal end portion **222**. In some embodiments, a lower portion of the rearward portion **206** of the apparatus **200** includes a cut-out region that extends from a lower surface of the delivery apparatus **200** towards the proximal end portion **222**. For example, the lower portion of the delivery apparatus **200** can have a convex region that extends from the distal end portion **220** towards the proximal end portion **222**. The convex region can extend to a lower edge **221** of the lower portion. In some embodiments, the lower portion of the apparatus **200** can include a concave region that extends from the lower edge **221** towards the proximal end portion **222**. For example, the concave region of the lower portion can extend upwardly from the lower edge **221** towards the upper portion at the proximal end portion **222**. In some configurations, the concave region can define a cutout region. The cutout region can provide a reduced profile to the apparatus **200**. In some configurations, the reduced profile allows for the apparatus **200** to include less material, be more lightweight, and/or be more comfortable to hold, among other benefits.

In some embodiments, the external housing **202** includes an opening **212**. The opening **212** can provide access to one or more actuators, such as buttons, sliders, and/or levers, among other actuation initiation mechanisms. For example, the delivery apparatus **200** can include a singulation actuator **214** and/or an implant delivery actuator **216**. In some embodiments, at least a portion of the singulation actuator **214** and/or the implant delivery actuator **216** extend through the opening **212**. In such configurations, the actuators **214**, **216** can be easily manipulated and/or accessible by the user. In some embodiments, the singulation actuator **214** facilitates on-demand manual singulation, which, as used herein, can mean isolation, separation, and/or selection of one of the multiple implants for delivery one at a time. The singulation actuator **214** interfaces with internal components (not shown) to effect singulation. In some embodiments, actuation of the implant delivery actuator **216** (e.g., pressing a button extending out of the opening **212**) causes the ejection of an implant (e.g., one implant manually singulated as a result of actuation of the singulation actuator **214**) out of an introducer tip of the introducer assembly **201** of the delivery apparatus and into a desired first location within the patient's internal eye tissue. In some embodiments, the singulation actuator **214** enables automatic singulation, isolation, and/or selection of respective implants. The implant delivery actuator **216** interfaces with internal components to effect delivery of the implants. In some embodiments, the implant delivery actuator **216** is configured to allow for an infinite number of actuations (e.g., infinite number of button presses) to cause movement of a collet sufficient to deliver an implant out of the introducer tip of the introducer assembly **201**. In accordance with several embodiments, the multiple-implant delivery apparatus **200** advantageously generates an implantation impulse to effect implant delivery by capturing and converting the energy used to press the implant delivery actuator **216** (e.g., de-pressing a button). In some embodiments, there is no pre-stored energy prior to actuation of the implant delivery actuator **216**, and thus no limit to the number of implant firing sequences or deliveries available.

The multiple-implant delivery apparatus **200** can be advantageously ergonomically shaped for easy gripping and manipulation. In some embodiments, the apparatus **200** can include a general overall shape similar to a conventional writing instrument, such as a fountain pen. In some embodiments, the multiple-implant delivery apparatus **200** can be grasped by the user between the thumb and the middle finger, with the index finger free to manipulate any portion of the apparatus **200**. The multiple-implant delivery apparatus **200** may include a finger rest **211**, as shown, for example, in FIG. 2A.

In some embodiments, the lower portion of the forward portion **207** of the delivery apparatus **200** can include a plurality of tactile ridges and/or recesses **218**. In some embodiments, the tactile ridges and/or recesses **218** provide a textured surface. In some embodiments, the tactile ridges and/or recesses **218** provide the user with a more stable and/or secure gripping surface to grip the delivery apparatus **200** in use.

In some embodiments, as described in more detail below, the external housing **202** is fabricated from a plurality of separate sections. For example, the external housing **202** can include one or more portions, such as half-sections, that can be coupled through various means, such as a snap-fit or press fit configuration or using an adhesive, or can be unitarily formed, among other arrangements. Although snap-fit or press-fit mechanisms of attachment are generally described

herein, these attachment mechanisms (for attachment of housing sections to each other and for attachment of members and components residing within the housing) can be replaced, substituted or enhanced with other attachments methods as desired and/or required (e.g., heat stake, glue or other adhesives, screws, welding, retaining by overhangs, and/or positioned by pressing a feature into plastic (with or without heat).

In some embodiments, a plurality of ocular implants is pre-loaded within the multiple-implant delivery apparatus **200** prior to packaging or delivery at the time of manufacture and assembly. In such embodiments, the multiple-implant delivery apparatus **200** can be used to deliver the multiple ocular implants at various desired locations within a mammalian (e.g., human) eye. For example, at least a portion of the introducer assembly **201** can be advanced through a preformed incision or opening in the eye (e.g., an incision in the cornea or limbus of the eye). In another embodiment, at least a portion of the introducer assembly **201** is advanced through external eye tissue (e.g., the cornea or limbus), creating an incision or opening through the eye as it is advanced into the eye tissue. As mentioned above, actuation of the implant delivery actuator **216** can actuate the multiple-implant delivery apparatus **200** and cause the ejection of an implant into a desired first location within the patient's internal eye tissue. In some embodiments, the multiple-implant delivery apparatus **200** can then be repositioned without removing at least a portion of the introducer assembly **201** from the incision and another implant can be delivered to a second location next to or spaced apart from the first location, and additional implants can be delivered to additional locations spaced apart from the second location. In some embodiments, the introducer assembly **201** can be removed from the incision and reinserted through eye tissue through a separate incision in order to deliver the implant to the second implantation site and/or third implantation site. In some configurations, the delivery of the multiple ocular implants advantageously can be performed during an outpatient procedure without extensive surgery.

As mentioned above, in some embodiments, the delivery apparatus **200** includes the introducer assembly **201**. The introducer assembly **201** can include (i) an auto-retracting insertion assembly that includes a distal introducer tip **208** and a proximal retraction member **232** and (ii) an insertion tube **356**. In some embodiments, at least a portion of the introducer assembly **201** can extend from the distal end portion **220** of the external housing **202** along an axis offset from the longitudinal axis **203** of the delivery apparatus **200**. In some embodiments, only the distal introducer tip **208** and/or the insertion tube **356** extends along an axis offset from the longitudinal axis **203**. For example, the introducer assembly **201** can extend at an angle α relative to the longitudinal axis **203**. In some embodiments, the angle α can be approximately 8 degrees. In some embodiments, the angle α can range from 1-15 degrees, from 1-3 degrees, from 3-5 degrees, from 5-7 degrees, from 7-9 degrees, from 9-11 degrees, from 11-13 degrees, from 13-15 degrees or ranges extending therebetween, or can be any value within the recited ranges.

The angled introducer assembly **201** advantageously provides beneficial ergonomics and more comfortable hand positions of the clinician operator during use. For example, the angled introducer assembly **201** can allow the clinician operator to more easily reach certain portions of the eye. In some embodiments, the angled introducer assembly **201** can allow the clinician operator to rotate the delivery apparatus **200** about an arc to efficiently and more easily access

implantation locations spaced apart from each other (e.g., three locations spaced apart at various clock hours (e.g., two clock hours from each other) along a circumference of Schlemm's canal). In some embodiments, the angling of the introducer assembly advantageously allows the clinician operator to sweep out a wider arc to use as much (e.g., up to 50% or more) of the conventional outflow system as possible through a single incision or opening into the eye.

As mentioned above, the introducer assembly **201** can include the insertion tube **356**. At least the distal portion of the insertion tube **356** can extend from an opening at a distal terminus of the distal end portion **220** of the external housing **202**. In some embodiments, the introducer tip **208** surrounds and/or is guided by the insertion tube **356**, which has a lumen. The insertion tube **356** can include a lateral viewing slot described in more detail below (not visible in figure) to facilitate visualization of an implant positioned in a "ready-to-fire" position along a trocar (not visible in figure) extending along and within the lumen of the insertion tube **356**. The lateral slot may exhibit any of the structural and/or functional features of the slots described in U.S. Publication No. 2013/0253528 (e.g., Paragraphs [0118]-[0125] and FIG. 19). In some embodiments, the insertion tube **356** can assist in more easily accessing certain portions of the eye. In some embodiments, a plurality of ocular implants can be pre-loaded into the insertion tube **356** along the trocar.

As shown in at least FIGS. 3A and 3B, the proximal retraction member **232** of the auto-retracting insertion assembly can extend from a mating component on the distal terminus of the distal end portion **220** of the external housing **202**. The proximal end of the proximal retraction member **232** may be fixedly or removably coupled to the mating component on the distal terminus of the distal end portion **220**. The distal end of the retraction member **232** includes a generally cone-shaped interface component **231** specifically designed to interface with a boundary of a pre-formed incision or opening in eye tissue (e.g., cornea or limbus) and prevent continued advancement of the proximal retraction member within the eye. This interface component **231** of the retraction member **232** may be advantageously shaped and sized to facilitate insertion within incisions or openings of between 1 mm and 4 mm (e.g., between 1 mm and 3 mm, between 2 mm and 4 mm, less than 2 mm).

In some embodiments, the distal introducer tip **208** forms the distal end portion of the auto-retracting insertion assembly. For example, as shown in the illustrated embodiments, the distal introducer tip **208** can extend from the interface component **231** of the retraction member **232**. In some embodiments, the distal introducer tip **208** is integrally formed with the proximal retraction member **232**. For example, the proximal end portion of the distal introducer tip **208** may reside within and be fixedly coupled to (e.g., adhered to, molded to) the interface component **231** of the retraction member **232**. The distal introducer tip **208** can include a hollow needle, among other types of needles. For example, the distal introducer tip **208** can include an interior lumen that can allow the insertion tube **356** to pass through. In some embodiments, the length of the retraction member **232** can be sized such that at least a portion of the distal introducer tip **208** is configured to always surround at least a portion of the length of the insertion tube **356**, thereby maintaining coaxial alignment between the insertion tube **356** and the distal introducer tip **208**. The distal tip of the distal introducer tip **208** may be beveled to facilitate insertion within eye tissue.

In some embodiments, the proximal retraction member **232** includes a flexible material, such as silicone elastomer,

15

plastic, rubber, or other materials. The proximal retraction member 232 can be configured to bend in use. The proximal retraction member 232 may include multiple openings 235 positioned along its length at locations designed to facilitate bending of the retraction member 232 in a desired or predetermined bending configuration. As shown best in FIG. 3B, a bottom side of the proximal retraction member 232 may include a tube engagement member 234 configured to engage and receive the insertion tube 356. The tube engagement member 234 may include two feet as shown with a slot formed between the two feet. The slot may have a general curve as shown or may have a keyhole shape or configuration. In some embodiments, the tube engagement member 234 is positioned and configured to cause bending of the proximal retraction member 234 in a particular configuration. The proximal retraction member 232 may also be pressed at a location corresponding to the location of the tube engagement member 234 to force the slot of the tube engagement member 234 onto and around the tube, thereby resulting in greater force for insertion. In various embodiments, the column of the proximal retraction member 232 has a tapered width and/or a uniform width. In some embodiments, the proximal retraction member 232 is tapered in a distal direction. For example, the proximal retraction member 232 can include a width that is wider on a proximal side than at a distal side of the proximal retraction member 232. The shape of the proximal retraction member 232 can desirably allow the introducer assembly 201 to be more smoothly inserted into the eye.

In some embodiments, when the delivery apparatus 200 enters the eye, such as at the anterior chamber, at least a portion of the introducer assembly 201 is advanced to the trabecular meshwork. When the interface component 231 of the proximal retraction member 232 reaches a portion of the eye, such as the trabecular meshwork, the retraction member 232 can yield and/or buckle to form a bent or curved “inchworm” configuration. For example, the distal side of the retraction member 232 can slide rearwardly along the insertion tube 356 and a central portion of the retraction member 232 can extend radially outwardly from the distal introducer tip 208 (see FIG. 2B). The retraction member 232 is specifically engineered so that, when the clinician operator is entering the anterior chamber, there is enough force transmitted to push the distal introducer tip 208 and insertion tube 356 into and through the incision formed in the eye tissue (e.g., corneal incision) and into the anterior chamber and yet when the clinician operator wants to advance the insertion tube 356 across the anterior chamber to the trabecular meshwork, the proximal retraction member 232 buckles and yields to “retract” the distal introducer tip 208 and allow the insertion tube 356 to be advanced across the anterior chamber without being surrounded by the distal introducer tip 208. In some embodiments, the forces generated by components of the retraction member 232 are just high enough to get the distal introducer tip 208 through the wound and then the interface component 231 of the retraction member 232 bottoms out on the wound and buckles. The force profile may advantageously be linear or substantially linear or substantially constant during “retraction” (e.g., from the point of buckling or bending to the point of full insertion into the eye). In other embodiments, the force profile may transition from very high during insertion to almost zero through use of a mechanical lockout that would be locked during insertion until some part of the distal introducer tip 208 bottomed out (e.g., on the surface of the cornea) and then the mechanical lockout would be releases,

16

allowing the retraction member 232 to bend and allowing the force to drop to almost zero.

In some embodiments, when the retraction member 232 slides along an exterior surface of the insertion tube 356, the insertion tube 356 can slide through at least a portion of the distal introducer tip 208 and/or the interface component 231 of the retraction member 232 to facilitate delivery of one or more implants according to one or more methods described herein.

In some embodiments, the interface component 231 of the retraction member 232 advantageously acts as a stop for the insertion tube 356 against the cornea or other portion of the eye. In some embodiments, the interface component 231 of the retraction member 232 advantageously helps to seal, limit or prevent leakage of aqueous humor from the anterior chamber of the eye as the interface component 231 sits against the insertion site.

In some embodiments, the introducer assembly 201 includes a stop member 233. In some embodiments, the stop member 233 surrounds at least a portion of the insertion tube 356 extending out of the exterior housing 202 of the delivery apparatus 200. In some embodiments, the insertion tube 356 extends out of a distal end of the stop 233. The stop 233 can stop the retraction member 232 from sliding further proximally along the insertion tube 356 in use. For example, as the interface component 231 of the retraction member 232 slides in a proximal direction, the stop 233 can contact at least a proximal flange portion of the interface component 231. The contact between the stop 233 and the interface component 231 can limit or prevent further proximal movement of the retraction member 232. The stop 233 may advantageously help support the insertion tube 356, thereby keeping the insertion tube in place and inhibiting movement of the insertion tube 356. In some embodiments, distal component 236 of the retraction member 232 can be coupled with distal component 238 of the retraction member 232 to maintain the retraction member 232 in a retracted position (e.g., for training or rethreading an implant back on the trocar outside the eye).

III. Internal Operation of Multiple-Implant Delivery Apparatus

FIG. 4 is an exploded perspective view of the multiple-implant delivery apparatus 200. The external components of the multiple-implant delivery apparatus 200 include a left housing 302, a right housing 304, and the introducer assembly 201 (described above).

As shown, the external housing 202 is formed of two separate half-sections (left housing 302 and right housing 304). The left housing 302 can include a left section of the opening 212 and the right housing 304 can include a right section of the opening 212. In alternative embodiments, the external housing 202 could be separated into top and bottom half-sections instead of right and left half-sections. In yet other alternative embodiments, the external housing 202 is formed of more than two sections configured to be attached together to form a contiguous unit.

FIG. 5A illustrates a side perspective view of the left housing 302 and FIG. 5B illustrates a side perspective view of the right housing 304. The left housing 302 and the right housing 304 can be assembled to form a unitary body. The left housing 302 and the right housing 304 can be coupled through various means, such as a snap-fit or press fit configuration or using an adhesive, or can be unitarily formed, among other arrangements. The left housing 302 and/or the right housing 304 can include various mating

17

features to couple the left and right housings **302**, **304** and/or features for receiving, supporting and/or aligning the internal components of the multiple-implant delivery apparatus **200**. For example, the left housing **302** can include a rim **306** that protrudes away from the outer surface of the left housing **302** and extends along at least a portion of an inner perimeter of the left housing **302**. The rim **306** is configured to couple with a corresponding recess **308** that extends along at least a portion of an inner perimeter of the right housing **304**.

In some embodiments, the left housing **302** includes a plurality of snap-fit or press-fit receiving members **307**. One or more of the snap-fit or press-fit receiving members **307** can form receptacles that are configured to align with and to receive one or more snap-fit or press-fit engaging members **309** (e.g., flanges, protrusions) that extend from the right housing **304** or other component of the multi-implant delivery apparatus **200**. One or more of the snap-fit or press-fit receiving members **307** and/or one or more of the snap-fit or press-fit engaging members **309** can be configured to pass through an opening formed in at least one of the internal components of the multiple-implant delivery apparatus **200** to support the internal components within the external housing **202**. Other configurations are contemplated such as the left housing **302** and/or the right housing **304** having one or more of the mating features described above (e.g., the right housing **304** can include the snap-fit or press-fit receiving members **307** and the left housing **302** can include the snap-fit or press-fit engaging members **309**). The left and right housings **302**, **304** can be coupled to surround at least a portion of the internal components of the multiple-implant delivery apparatus **200**. In certain embodiments, there is an audible click when snap-fit receiving members **307** and snap-fit engagement members **309** and/or the rim **306** and recess **308** are fully engaged. In alternative embodiments, the left and right housings **302**, **304** can be connected or otherwise coupled to each other via adhesion, screws, glue, welding (e.g., sonic welding), and/or the like. In some embodiments, a proximal tail **223** of the multiple-implant delivery apparatus **200** may be configured to detach from and reattach to the rest of the body (e.g., as indicated by the dashed line) via snap-fit or press-fit coupling mechanisms or configurations.

In various embodiments, the left housing **302** and the right housing **304** are composed of any rigid or semi-rigid material, such as plastic, polymer, metal, composites, or the like. In one embodiment, the left housing **302** and the right housing **304** are molded from Lexan® polycarbonate. In some embodiments, at least a portion of the left housing **302** and/or the right housing **304** is composed of a flexible material, such as silicone or similar elastomeric or flexible polymers (including but not limited to acrylonitrile butadiene styrene (ABS), a blend of polycarbonate and ABS, polystyrene, polypropylene, and/or polyethylene).

With reference to FIG. 4 and FIGS. 6A-6B, the internal components of the multiple-implant delivery apparatus **200** include a frame **310**, a singulation assembly **330** (including the singulation actuator **214**, a singulation biasing member **332**, and a singulation arm **334**), a tube set assembly **350** (including an insertion tube subassembly **352** having an insertion tube carrier **354** and an insertion tube **356**), a collet holder assembly **358** having a collet holder **360** and a singulation tube **362**, and a trocar assembly **364**), an actuation assembly **370** (including the implant delivery actuator **316**, an actuator biasing member **372**, and an actuator arm **374**), and the introducer assembly **201** (described above).

18

The internal components can be secured to or within the external housing **202** during assembly of the multiple-implant delivery apparatus **200** using various methods of fixation (e.g., adhesion, bonding, gluing, snap-fitting, and the like). The interaction of the internal components and the operation of the multiple-implant delivery apparatus **200** will be discussed in more detail later.

In certain embodiments, the multiple-implant delivery apparatus **200** is disposable or configured for a single use and includes one or more safety mechanisms that prevent reuse. For example, the safety mechanism can be an internal component that renders the instrument inoperable if re-sterilized. In accordance with several embodiments, the safety mechanism is that plastic parts do not survive sterilization with an autoclave. In other embodiments, the multiple-implant delivery apparatus **200** is reloaded with implants, sterilized, and re-used on the same or a different patient.

A. Frame

FIGS. 7A and 7B illustrate an example of the frame **310**. The frame **310** can have an overall shape that corresponds to the shape of the external housing to allow the frame **310**, among the other internal components of the multiple-implant delivery apparatus **200**, to fit and be secured within the external housing **202**. For example, the frame **310** can have a proximal portion **402**, a distal portion **404**, a left side **403**, and a right side **405**. The proximal portion **404** can be raised relative to the distal portion **404**.

The frame **310** can include certain attachment features that secure the frame **310** to the external housing. As shown in FIGS. 7A and 7B, the frame **310** can include one or more openings, slots, receptacles or apertures **406**. The openings **406** can be positioned along various portions of the frame **310**, such as along the distal portion **404** and/or the proximal portion **402**. The frame **310** can include one, two, three, four, five, six, or seven or more openings **406**. The openings **406** can pass entirely through a width of the frame **310**. The openings **406** can allow certain attachment features, such as the snap-fit or press-fit receiving members **307** and/or the snap-fit or press-fit engagement members **309** to pass through the frame **310** and secure the frame **310** within the external housing **202**. Such configurations can help to limit movement of the frame **310** within the external housing **202** when assembled. In some embodiments, the frame **310** includes certain attachment members or features that are configured to secure other internal components to the frame **310**.

In some embodiments, at least one of the left and right sides **403**, **405** of the frame **310** includes an implant delivery actuator receptacle **408**. The receptacle **408** can be shaped to receive and/or secure at least a portion of the implant delivery actuator **216**, as described in more detail below. The receptacle **408** can have a generally circular shape, a generally rectangular shape, or other shapes. The shape of the receptacle **408** can allow the implant delivery actuator **216** to pivot about a center of the receptacle **408**. The receptacle **408** can be defined by at least one arc-shaped wall that extends outwardly from the right side **405** of the frame **310**. The receptacle **408** can be positioned at the proximal portion **402** of the frame **310**. In some embodiments, the receptacle **408** is positioned on at least a portion of an upper region of the proximal portion **402** of the frame **310**.

As shown, a proximal terminus end of the proximal portion **402** of the frame **310** can include an implant delivery actuator slot **410**. The implant delivery actuator slot **410** can be configured to receive and/or secure at least a portion of the implant delivery actuator **216**, such as a proximal end

portion and/or a proximal terminus end of the actuator **216**. The implant delivery actuator slot **410** can be shaped such that at least a portion of the implant delivery actuator **216** sits within the slot **410**. The slot **410** can be defined by one or more prongs. For example, at least a first prong **410A** can be formed along a wall of the frame **310** and at least a second prong **410B** can be spaced apart from the first prong to define the slot **410**. The slot **410** can be shaped to limit or prevent lateral movement of the implant delivery actuator **216** when the actuator is assembled to the frame **310**. In some embodiments, an upper wall of the slot **410** advantageously helps to limit or prevent upward movement of the proximal portion of the implant delivery actuator **216**. Thus, as explained below, in some embodiments, only a portion of the implant delivery actuator **216** rotates about the receptacle **408** when the implant delivery actuator **216** is manipulated.

In some embodiments, the frame **310** includes a singulation frame slot **412**. The singulation frame slot **412** can assist in singulation of one or more implants loaded in the multiple-implant delivery apparatus **200**. In some embodiments, the singulation frame slot **412** is positioned at the proximal portion **402** of the frame **310**. The singulation frame slot **412** may be positioned on the right side **405** of the frame **310**. The singulation frame slot **412** can pass entirely through the frame **310**.

The singulation frame slot **412** can be shaped to receive at least a portion of the singulation arm **334** (e.g., proximal singulation slot member **456** as shown in FIGS. 9A-9C). In some embodiments, the singulation frame slot **412** includes a plurality of platforms. For example, the singulation frame slot **412** can include one, two, three, four, five, six, seven, eight, or nine or more platforms.

As shown in FIGS. 7A and 7B, the singulation frame slot **412** can include a first platform **414A**, a second platform **414B**, a third platform **414C**, and a fourth platform **414D**. The first, second, third, and fourth platforms **414A**, **414B**, **414C**, **414D** can extend from an internal distal side of the slot **412**. In some embodiments, proximal and/or distal ends of each of the first, second, third, and fourth platforms **414A**, **414B**, **414C**, **414D** are positioned offset from one another. For example, the distal end of the first platform **414A** can be positioned distal of the distal end of the second platform **414B**, the distal end of the second platform **414B** can be positioned distal of the distal end of the third platform **414C**, the distal end of the third platform **414C** can be positioned distal of the distal end of the fourth platform **414D**. Similarly, the proximal end portion of the first platform **414A** can be positioned proximal of the proximal end portion of the second platform **414B**, the proximal end portion of the second platform **414B** can be positioned proximal of the proximal end portion of the third platform **414C**, the proximal end portion of the third platform **414C** can be positioned proximal of the proximal end portion of the fourth platform **414D**. As explained in more detail below, the positioning of each of the first, second, third, and fourth platforms **414A**, **414B**, **414C**, **414D** can desirably allow at least a portion of the singulation arm **334** to move along one platform to the next platform in an arc-like manner during singulation of an implant.

In some embodiments, the proximal end portion of at least one of the first, second, third, and fourth platforms **414A**, **414B**, **414C**, **414D** is at least partially tapered or slanted in a distal direction. The sharp tapered or slanted portions of the proximal end portions of at least one of the first, second, third, and fourth platforms **414A**, **414B**, **414C**, **414D** can allow the singulation arm **334** to slide between adjacent

platforms without requiring as much force perpendicular to the platform. In some embodiments, the tapered or slanted portions include one, two, or more tapered regions to allow the distal singulation arm collet holder member **458** to return to its original position.

In some embodiments, the frame **310** includes a singulation frame arm slot **416**. The slot **416** can be generally L-shaped, among other shapes. The slot **416** can be positioned in at least a portion of the distal portion **404** of the frame **310**. In some embodiments, the singulation frame arm slot **416** is sized and adapted to receive and/or support at least a portion of the singulation arm **334**. The slot **416** can allow the singulation arm **334** to slide and/or rotate within the slot **416**.

In some embodiments, the frame **310** includes an actuator arm attachment member **418**. The actuator arm attachment member **418** can extend outwardly from a side of the frame **310**, such as from the right side **405** of the frame **310**. In some embodiments, the actuator arm attachment member **418** is positioned on the distal portion **404** of the frame **310**. In some embodiments, the actuator arm attachment member **418** is configured to secure the actuator arm **374** by passing through the actuator arm connection opening **542** in the actuator arm **374** when the actuator arm **374** is assembled with the frame **310**. In some embodiments, the actuator arm attachment member **418** is sized and/or configured to snap to or otherwise reside within at least a portion of the actuator arm **374**, such as the actuator arm connection opening **542**. As explained in more detail below, the actuator arm **374** can be configured to pivot about the actuator arm attachment member **418**.

In some embodiments, the frame **310** includes at least one tube set assembly attachment member **420**. In the illustrated embodiment, the frame **310** includes at least two tube set assembly attachment members **420A**, **420B** and can optionally include additional tube set assembly attachment members **420**. The tube set assembly attachment members **420** can extend outwardly from a side of the frame **310**, such as from the right side **405** of the frame **310**. In some embodiments, the tube set assembly attachment members **420** is positioned on the distal portion **404** of the frame **310**. In some embodiments, the tube set assembly attachment members **420** is configured to secure the tube set assembly **350** within the external housing **202** by passing through at least an opening **470** in at least one component of the tube set assembly **350**, such as the insertion tube carrier **354**, when at least a portion of the tube set assembly **350** is assembled with the frame **310**. In some embodiments, the tube set assembly attachment members **420** is sized and/or configured to snap to or otherwise reside within at least a portion of the tube set assembly **350**, such as the opening **470**.

In some embodiments, the frame **310** includes an actuator arm protrusion platform **422**. The actuator arm protrusion platform **422** can extend outwardly from a side of the frame **310**, such as from the right side **405** of the frame **310**. In some embodiments, the actuator arm protrusion platform **422** is positioned on the distal portion of the frame **310**. In some embodiments, the actuator arm protrusion platform **422** is configured to support at least a portion of the actuator arm **374**, such as a distal facing side **548A**, and an upwardly facing side **548B**, as explained in more detail below. In some embodiments, the actuator arm protrusion member **544** is configured to rest on at least a portion of the platform **422**.

In some embodiments, the frame **310** includes a singulation actuator track **424**. The singulation actuator track **424** can define a space formed between an upper track member **424A** and a lower track member **424B** that extend outwardly

from at least one side of the frame 310, such as the left side 403. The singulation actuator track 424 can be configured to receive and/or secure at least a portion of the singulation actuator 214, such as a proximal portion of the singulation actuator 430. The singulation actuator track 424 can include a singulation track extension portion 426 that extends laterally outwards from the lower track member 424B. The singulation track extension portion 426 can include an upwardly extending portion that extends upwardly from an outer end of the extension portion 426 to help to secure the singulation actuator 214 to the frame 310. The extension portion 426 can help limit or prevent lateral and/or distal movement of the singulation actuator 214 relative to the frame 310. As shown in at least FIG. 7B, the right side of the singulation frame adjustment slot 412 can be positioned within the singulation actuator track 424 on the left side 403 of the frame 310.

B. Singulation Assemblies and Methods

FIGS. 8A and 8B illustrate an example of the singulation assembly 330. The singulation assembly 330 can include the singulation actuator 214, the singulation biasing member 332, and/or the singulation arm 334, among other components. The singulation actuator 214 can include a proximal portion 430, a central transition region 432, and a distal portion 434. The proximal portion 430 can be generally trapezoidal shaped, rectangular, square, or other shape. The proximal portion 430 can be configured to be secured within the singulation actuator track 424 on a side of the frame 310 when assembled. The securement between the proximal portion 430 and the singulation actuator track 424 can help to limit lateral movement of the singulation actuator 214, yet allow proximal movement of the singulation actuator 214. For example, in some embodiments, the singulation actuator track 424 is configured to allow the singulation actuator 214 to slide in a proximal-distal direction along the track 424.

In some embodiments, at least a portion of the proximal portion 430, such as a top surface of the proximal portion 430 is positioned at a higher elevation than at least a portion of the distal portion 434, such as a top surface of the distal portion. The proximal portion 430 and the distal portion 434 can be connected by the central transition region 432. The central transition region 432 can smoothly transition between the proximal portion 430 and the distal portion 434. The central transition region 432 can include a straight portion and a tapered portion that gradually transitions downwardly from the straight portion to the distal portion 434.

In some embodiments, the distal portion 434 includes an actuator lever, such as a handle 436. The handle 436 can extend upwardly and/or rearwardly from the distal-most terminus end of the distal portion 434. In some embodiments, the handle 436 is configured to extend through the opening 208 of the external housing 202 when assembled to allow the user to easily grasp or otherwise manipulate the handle 436. As shown, in some embodiments, the distal portion 434 is generally U-shaped such that a portion of the distal portion 434 is configured to be positioned on the left side 403 of the frame 310 and a portion of the distal portion 434 is configured to be positioned on the right side 405 of the frame 310. Such configurations can desirably provide rigidity and/or structural support to the handle 436. Other shapes and configurations are contemplated.

In some embodiments, the singulation actuator 214 includes a singulation spring protrusion 438. The singulation spring protrusion 438 can be positioned at the region where the tapered region of the central transition region 432 transitions to the distal portion 434. In some embodiments,

the singulation spring protrusion 438 is positioned at the distal portion 434 or other portion of the singulation actuator 214. The singulation spring protrusion 438 can extend downwardly and/or outwardly from a main body of the singulation actuator 214. The singulation spring protrusion 438 can be configured to be coupled with the singulation biasing member 332.

The singulation biasing member 332 can include a singulation spring 332 or other biasing material. The singulation spring 332 can include a proximal spring portion 440 and a distal spring portion 442. In some embodiments, the proximal spring portion 440 of the spring 332 is in a generally compressed state in an initial position or configuration in use. For example, the plurality of adjacent coils of the spring 332 can contact one another in the initial position or configuration of the proximal spring portion 440. In some embodiments, the distal spring portion 442 of the spring 332 is in a generally tensioned state in an initial position or configuration in use. For example, the plurality of adjacent coils of the spring 332 can be spaced apart in the initial position or configuration of the distal spring portion 442.

As shown in FIGS. 8A and 8B, the singulation spring protrusion 438 can be coupled with the singulation spring 332 at a transition region 444 of the spring 332. The transition region 444 can be positioned at the portion of the spring 332 that transitions between the distal spring portion 442 and the proximal spring portion 440. In some embodiments, a ratio of a length of the proximal portion 440 to a length of the distal portion 442 is approximately 1:1, 1:2, 1:3, 1:4 or other ratio. In some embodiments, the spring 332 or portions of the spring 332 is angled. For example, in some embodiments, a terminus proximal end of the spring 332 is positioned at a higher elevation than a terminus distal end of the spring 332 when assembled. The spring 332 includes one or more attachment members, such as rings, that can be configured to attach to other components of the multiple-implant delivery apparatus 200. The spring 332 can include a distal ring 448 positioned at the terminus distal end of the spring 332 that can be configured to be coupled with a protrusion 447 positioned at the distal end portion of the left side 403 of the frame 310. The spring 332 can include a proximal ring 446 positioned at the terminus proximal end of the spring 332 that can be configured to be coupled with the singulation arm 334 as described below. The rings 446, 448 may be replaced with other connection or fastening members.

FIGS. 9A-9C illustrate an embodiment of the singulation arm 334. The singulation arm 334 can include a main body portion 450, a distal singulation spring connector 452, a singulation arm connection member 454, a proximal singulation slot member 456, and a distal singulation arm collet holder member 458. In some embodiments, the main body portion 450 is generally triangular shaped. In some embodiments, a distal end portion 462 of the singulation arm 334 has a height or width that is larger than a height or width of a proximal end portion 460 of the singulation arm 334. In some embodiments, the proximal singulation slot member 456 extends outwardly from a left side of the proximal end portion 460. The proximal singulation slot member 456 can be generally rectangular, among other shapes. The proximal singulation slot member 456 can have a flat top surface and/or a flat bottom surface. In some embodiments, the proximal singulation slot member 456 is shaped to easily slide between adjacent platforms 414 of the singulation frame slot 412 of the frame 310 in use.

As mentioned above, the distal singulation spring connector 452 can extend outwardly from the left side of the

main body portion **450** of the singulation arm **334** and can be configured to be coupled with the proximal ring **446** of the spring **332**. In some embodiments, the distal singulation spring connector **452** is spaced apart from the main body portion **450** by the singulation arm connection member **454**. The singulation arm connection member **454** can be configured to reside within the singulation arm slot **416** of the frame **310** to allow the distal singulation spring connector **452** to be positioned at least partially on the left side **403** of the frame **310** and the proximal singulation slot member **456** and the distal singulation arm collet holder member **458** to be positioned on the right side **405** of the frame **310** when assembled.

In some embodiments, the distal singulation arm collet holder member **458** extends outwardly towards the right from an extension of the distal end portion **462** of the singulation arm **334**. The distal singulation arm collet holder member **458** is configured to contact or otherwise be positioned within a rear collet holder recess **484** of the collet holder **360** when assembled (see FIG. 6A). When a respective implant of the multiple-implant delivery apparatus **200** is in a "ready to fire" position, the distal singulation arm collet holder member **458** is advantageously always located in the same place regardless of which singulation slot or platform the singulation arm **334** is in or in contact with so that the clinician sees the same implant position prior to each successive implant delivery, thereby providing ease-of-use benefits and peace of mind for the clinician.

In some embodiments, the singulation assembly **330** is configured to interface with the tube set assembly **350** when assembled. As shown in FIGS. 10A-10C, the tube set assembly **350** can include the insertion tube subassembly **352** having an insertion tube carrier **354** and an insertion tube **356**; a collet holder assembly **358** having a collet holder **360** and a singulation tube **362**; and a trocar assembly **364**.

FIGS. 11A and 11B illustrate the insertion tube subassembly **352**. The insertion tube subassembly **352** can include the insertion tube carrier **354** and the insertion tube **356**. The insertion tube carrier **354** can include at least one opening **470**. In some embodiments, the insertion tube carrier **354** includes two, three, four, five, or six or more openings **470**. As described above, in some embodiments, the openings **470** are configured to receive the tube set assembly attachment members **420** to secure the tube set assembly **350** within the external housing **202** when at least a portion of the tube set assembly **350** is assembled with the frame **310**. In some embodiments, the tube set assembly attachment members **420** is sized and/or configured to snap to, be heat staked to, be screwed to, or otherwise reside within or be received by or engaged with the openings **470**. The insertion tube subassembly **352** can be fixedly connected to the frame **310** in use.

In some embodiments, the insertion tube carrier **354** includes a carrier notch **473**. The carrier notch **473** can be positioned at approximately a midpoint along a bottom edge of, or anywhere along the insertion tube carrier **354**. The carrier notch **473** can be configured to align with a corresponding collet holder notch **488** of the collet holder **358** when assembled. The alignment of the carrier notch **473** with the collet holder notch **488** can indicate an initial position of the collet holder subassembly relative to the insertion tube subassembly **352**.

In some embodiments, the insertion tube carrier **354** includes a plurality of tabs **472** that extend outwardly from the insertion tube carrier **354**, for example, towards a right side of the carrier **354**. As shown, the insertion tube carrier **354** can include a first tab **472A**, a second tab **472B**, and a

third tab **472C**. The first tab **472A** can be positioned at a distal end portion of the insertion tube carrier **354**. The first tab **472A** can limit distal movement of the collet holder assembly **358** relative to the insertion tube carrier **354**. For example, as explained in more detail below, a distance between the front end of the collet holder **358** and the first tab **472A** when the tube set assembly **350** is in an initial position defines the distance the collet holder **358** can translate relative to the insertion tube carrier **354** during implant delivery after the implant delivery actuator **216** is actuated. For example, when the implant delivery actuator **216** is actuated, the collet holder **360** can translate distally and contact the first tab **472A** such that the distal end **361** of the collet holder **360** abuts the first tab **472A**.

In some embodiments the second tab **472B** forms a pronged arrangement including at least two spaced apart prongs defining a slot between the prongs. The pronged arrangement can be configured to pass through a collet holder opening **480** when the collet holder **360** is assembled with the insertion tube carrier **354**. The pronged arrangement can allow at least a portion of the trocar assembly **364**, such as the backup tube **502** to pass through the spaced apart prongs. The pronged arrangement can desirably help to limit vertical displacement of the trocar assembly **364** when the tube set assembly **350** is assembled. In some embodiments, the second tab **472B** is configured to act as a stop to limit proximal movement of the collet holder **358** relative to the insertion tube carrier **354** in use. For example, as discussed below, the collet holder **358** can move proximally during singulation. A distance between a front end of the collet holder opening **480** and the second tab **472B** when assembled can define a distance the collet holder **358** moves proximally during singulation. The second tab **472B** can act as a stop to provide the maximum proximal distance the collet holder **358** can translate relative to the insertion tube carrier **354**. The distances can be designed to be very precise.

In some embodiments, the third tab **472C** is positioned at a proximal terminus end of the insertion tube carrier **354**. The third tab **472** can extend at least partially through the collet holder opening **480** when the collet holder **360** is assembled with the insertion tube carrier **354**. In some embodiments, the third tab **472** is configured to define a backstop to limit proximal movement of the trocar assembly **364** in use.

In some embodiments, the insertion tube **356** is fixedly coupled with the insertion tube carrier **354** by various attachment means, such as by soldering, heat staking, welding, laser welding, laser machining, gluing, or other processes. In some embodiments, a proximal end portion of the insertion tube **356** is coupled with the insertion tube carrier **354**, for example, at a distal portion of the right side of the carrier.

FIGS. 12A and 12B illustrate an embodiment of the insertion tube **356** in more detail. In some embodiments, the insertion tube **356** includes a slot **474** formed at the distal end **476** of the insertion tube **356**. The slot **474** can be substantially V-shaped, U-shaped, or other shapes. The slot **474** can be configured to allow the user to visualize the implant before and/or after the implant is delivered by the delivery apparatus **200** into the patient's eye. Delivery of an implant can be actuated multiple times until it is in a desired final implantation position, as advantageously determined by visualization through the slot **474**.

The insertion tube **356** can define a lumen configured to surround at least a portion of the trocar assembly **364** and/or the singulation tube **362** (see FIG. 10C). The insertion tube

25

356 can include a plurality of openings 478. The openings 478 can include one, two, three, four, five, or more openings. The number of openings 478 can correspond to the number of pre-loaded implants 901 in the multiple-implant delivery apparatus 200. The openings 478 can be spaced apart axially along an outer circumferential surface of the insertion tube 356. In some configurations, the openings 478 can desirably allow a user to visualize the implants 901 within the tube during manufacture to verify that the desired number of implants 901 are pre-loaded and pre-loaded at the proper location. As shown, the openings 478 may be positioned circumferentially opposite (180 degrees or approximately 180 degrees circumferentially away from) the slot 474.

FIG. 13 shows an embodiment of the collet holder sub-assembly 358. The collet holder subassembly can include a collet holder 360 and a singulation tube 362. The collet holder 360 can include an opening 480, as described above, that can receive at least a portion of the insertion tube carrier 354, such as at least one of the tabs (e.g., the second tab 472B and/or the third tab 472C). The collet holder opening 480 can be positioned in a distal portion of the collet holder 360 and can pass through a width of the collet holder 360. The collet holder 360 can include a rear collet holder recess 484. The rear collet holder recess 484 can be positioned at a proximal end portion of the collet holder 360. The rear collet holder recess 484 can extend from a top surface 486 of the collet holder 360 downwardly into the collet holder. The rear collet holder recess 484 can be configured to retain at least a portion of the singulation arm 334, such as the distal singulation arm collet holder member 458.

In some embodiments, the collet holder 360 includes a central collet holder recess 482. The central collet holder recess 482 can be positioned in a central region of the collet holder 360. The central collet holder recess 482 can extend from a top surface 486 of the collet holder 360 downwardly into the collet holder. The central collet holder recess 482 can be configured to retain at least a portion of the actuator arm 374, such as the actuator arm protrusion member 544, when the implant delivery actuator 214 is actuated.

The collet holder 360 can include the collet holder notch 488 as briefly referred to above. The collet holder notch 488 can be positioned at approximately a midpoint or distal to the midpoint of the collet holder opening 480 along a bottom edge of the collet holder 360. The collet holder notch 488 can be configured to align with the corresponding carrier notch 473 of the insertion tube carrier 354 when assembled. The alignment of the carrier notch 473 with the collet holder notch 488 can indicate an initial position of the collet holder subassembly relative to the insertion tube subassembly 352.

In some embodiments, the singulation tube 362 is fixedly coupled with the collet holder 360 by various attachment means, such as by soldering, heat staking, welding, laser welding, laser machining, gluing, or other processes. In some embodiments, a proximal end portion of the singulation tube 362 is coupled with the collet holder 360, for example, at a distal portion of the left side of the collet holder. FIG. 14 illustrates an embodiment of the singulation tube 362 in more detail. In some embodiments, the singulation tube 362 includes a slot 490 formed at the distal end 492 of the singulation tube 362. The slot 490 can be substantially V-shaped, U-shaped, or other shapes. The slot 490 can be formed by forming tines in the distal end 492 of the singulation tube 362 (e.g., by laser cutting or machining) to form two or more symmetrical or asymmetrical flanges about a central longitudinal axis of the singulation tube 362. The distal end 492 with formed tines advantageously allows the singulation tube 362 to be retracted over a maximum

26

cross-sectional dimension of a next implant positioned along the trocar 500 when the singulation tube is being retracted proximally during singulation after delivery of a previous implant and then transition back to a configuration that is configured so that a distal terminus of the distal end 492 contacts a distal end of the next implant and propels the next implant out of the delivery apparatus 200 when the implant delivery actuator 214 is actuated to deliver the next implant. The distal end 492 may comprise shape memory or flexible material. In some embodiments, the shape of the slot 490 is configured such that the distal terminus of the distal end 492 is configured to contact a distal end of an implant every time the implant delivery actuator 214 is actuated (e.g., button is pressed). In some embodiments, the distal end 492 and the slot 490 are advantageously configured to allow for an infinite number of actuations.

As shown in FIG. 10C, the singulation tube 362 can define a lumen to surround at least a portion of the trocar assembly 364. In some embodiments, the singulation tube 362 has an outer diameter sized to fit within the insertion tube 356.

FIG. 15A illustrates an embodiment of the trocar assembly 364. The trocar assembly 364 can include a trocar 500 and/or a backup tube 502. FIGS. 15B-15D illustrate embodiments of a splayed trocar configured to facilitate manual singulation of multiple implants on demand (e.g., not automatic).

FIG. 15B illustrates a close-up view of the trocar 500. In accordance with several embodiments, the trocar 500 defines an elongate member (e.g., a wire or rod) on which the multiple implants 901 can be loaded and along which the implants 901 can be advanced. In some embodiments, one or more singulation features are present (e.g., formed or coupled) at various positions along the trocar 500 to singulate, or separate, the multiple implants 901 so that they are not stacked up directly against one another (e.g., no double-stacking). The singulation features function to keep the implants 901 apart so that they are not touching one another. Without the singulation features (and thus with the implants 901 stacked up directly against one another), the tolerances within the instrument and the tolerances of the implants themselves have to be extremely precise in order to facilitate singulation of each successive individual implant for delivery. In some embodiments, the singulation features are formed by slitting the trocar (e.g., by forming a small incision along a central longitudinal axis of the trocar and entirely through the trocar as shown, for example, in FIG. 15B) and then splaying the portion of the trocar 500 with the slit 504 to form a splay 506 (e.g., eyelet or opening) within the trocar 500 (as shown, for example, in FIG. 15B). Splaying the trocar 500 can be performed by inserting a wire or other device through the slit 504 and forcing the slit 504 open. The splays 506 may be formed while maintaining a generally straight geometry. The eyelet or opening can be heat set to maintain the eyelet or opening in a resting, relaxed or unstressed configuration. When sufficient force is applied to the rear of the implant 901 (e.g., by a collet or other advancement member), the implant, the heat set shape of the splay or eyelet is overcome and the implant is advanced over the splay or eyelet. In some embodiments, the splayed trocar 500 is formed by laser cutting, machining or etching a solid elongate member (e.g., cylindrical trocar) to form the splays 506 (e.g., eyelets or openings) at various positions along the length of the trocar 500. The trocar 500 can be made of stainless steel, MP35 superalloy, tungsten, nickel titanium alloy (e.g., nitinol), plastic, or other elastic, flexible, and/or shape memory materials.

In some embodiments, multiple (e.g. one, two, three, four, or more) singulation features (e.g., eyelets, splays, protrusions, balloons) are included along the length of the trocar **500** (depending on the number of implants intended to be included and delivered). An implant **901A** can be positioned (e.g., pre-loaded) before (distal of) the first splay **506A** so that it is in a “ready-to-fire” position upon initial actuation by a user. A second implant **901B** can be loaded or positioned between the first and second splays **506A**, **506B**, a third implant **901C** can be loaded or positioned between the second and third splays **506B**, **506C**, and so on (as illustrated, for example, in FIG. **15D**). In some embodiments, the number of splays may correlate to the number of implants. In some embodiments, an implant is positioned between each splay **506**. Thus, each of the implants can be separated and can be easily singulated by manually actuating the singulation actuator **214** without having to be as precise with tolerances and singulation distances. With the trocar **500** having the singulation features, instead of the singulation member having to be positioned between the proximal end of a first stent and the distal end of the second stent stacked up right against the first stent, the singulation member advantageously only has to be positioned within the length of the splay (e.g., eyelet or opening). As one example, the singulation tolerance may be less than 0.004” without the singulation features and may be less than 0.016” with the singulation features (a 4:1 difference in singulation tolerance). The splayed trocar **500** can be used with any number of implant shapes as long as they have a central lumen sized to be advanced along the outer diameter of the trocar. In some embodiments, the implants are ophthalmic implants designed to shunt aqueous humor between an anterior chamber of an eye and Schlemm’s canal to facilitate reduction of intraocular pressure to treat glaucoma.

In some embodiments, the trocar **500** is angled or curved in certain embodiments. The trocar **500** can be rigid, semi-rigid, or flexible. In certain embodiments, some portions of the trocar **500** are flexible and other portions are rigid. In embodiments where the trocar **500** can be stiff, the implant can be, but need not be relatively flexible. In certain embodiments, the trocar **500** and the backup tube **502** are advantageously constructed of stainless steel. In other embodiments, the trocar **500** and the backup tube **502** can be constructed of other suitable materials, such as other metals, plastics, or polymers.

The backup tube **502** can surround at least a portion of the trocar **500** to provide additional rigidity to the trocar **500**. The backup tube **502** includes a hollow tube having an inner diameter sized to receive the trocar **500**. In certain embodiments, backup tube **502** has an inner diameter of about 0.0035 inches; however, the backup tube **502** can have any inner diameter sized so as to receive the trocar **500**. As shown, the backup tube **502** can include a chamfered distal end **503**. In certain embodiments, the backup tube **503** is advantageously laser welded to the trocar **500** upon assembly. In some embodiments, the backup tube **502** can be bonded to the trocar **500** using other methods of fixation (for example, curing, welding, press-fitting, adhesive). The trocar **500** and backup tube **502** may be cut by laser or centerless ground.

As mentioned above, the singulation actuator **214** facilitates the singulation (e.g., isolation, separation, and/or selection) of one of the multiple implants for delivery one at a time. The singulation actuator **214** interfaces with internal components (described above) to effect singulation. Several alternative embodiments can be used to effect singulation. For example, FIGS. **15E** to **15N-3** illustrate various embodi-

ments that could be implemented instead of the splayed trocar singulation embodiment illustrated in FIGS. **15A-15D**. In some embodiments, the singulation features are formed by including nubs or protrusions at various positions along the trocar **500** that are flexible or resilient enough to allow the implants to be advanced over them when sufficient force is applied to the implants.

FIGS. **15E-1** to **15E-3** illustrate an embodiment of a singulation design that is similar to the splayed trocar design illustrated and described above in connection with FIGS. **15A-15D** but the singulation features are polymer “balloons” instead of splays along the trocar **500E**. In this “polymer balloon” embodiment, the trocar **500E** is made out of a tubular polymer material and the “balloon” areas (i.e., separation areas) **551** are enlarged with a triple bump (e.g., bumps spaced 120 degrees apart circumferentially as shown in FIGS. **15E-1** to **15E-3**) or a double bump (e.g., bumps spaced 180 degrees apart on opposite sides of the trocar **500E**, not shown). The “polymer balloon” embodiment may include the structural or functional features described in connection with the splayed trocar embodiment of FIGS. **15A-15D**.

FIG. **15F** illustrates an embodiment of a twisted trocar singulation design. In this embodiment, the trocar **500F** comprises a plurality of wires twisted together. The wires are twisted tightly in most locations, but in the specified implant separation areas **551**, the wires are configured (e.g., twisted or arranged in such a manner) to allow the outer diameter of the trocar **500F** to enlarge at the separation areas. During singulation, as a respective implant **901** moves forward (e.g., distally), the twisted wires compress, thereby allowing the implant **901** to pass over the separation area.

FIGS. **15G-1** and **15G-2** illustrate an embodiment of a “bent trocar” singulation design. This singulation design includes a trocar **500G** having multiple singulation features in the form of bend regions **553** along the length of the trocar **500G** that are spaced so as to effectively separate the plurality of implants **901**. The “bent trocar” design provides a mechanical separation of the implants **901** that is strong enough to prevent movement while the multiple-implant delivery apparatus **200** is not in use, but weak enough to deflect flat (straighten out) during singulation (e.g., when sufficient force is exerted on the implant in the distal direction to straighten out the adjacent bend region). FIG. **15G-2** shows the implants **901** after being singulated to the next distal position by the singulation actuator **214**. Each implant **901** in FIG. **15G-2** has been advanced distally over the respective bend region that was distally adjacent to each respective implant **901** in FIG. **15G-1**.

FIGS. **15H-1** and **15H-2** illustrate an embodiment of a “corkscrew” singulation design. The “corkscrew” singulation design includes a secondary wire **555** that is spiral-shaped and is configured to wrap around the trocar **500**. The spiral wire **555** effectively separates the implants **901** and can advance the implants **901** forward when the wire **555** is turned, or rotated. FIG. **15H-2** shows the implants after having been moved distally along the trocar **500** by the wire **555**.

FIGS. **15I-1** to **15I-4** illustrate an embodiment of an “ice cube tray” singulation design. The “ice cube tray” embodiment includes a semi-circular “tube” portion **560** with cut-out tabs **565** bent inward. The tabs **565** are positioned in front of and behind each implant **901**, thereby securing each respective implant’s initial position along the trocar **500**. During singulation, each implant **901** can cause the adjacent tab(s) **565** to bend downward due to the force applied on the implant by the singulation actuator **214**, thereby allowing

each implant **901** to move forward, or distally. The semi-circular tube portion **560** can be affixed to a preassembled multiple-implant delivery apparatus prior to shipping or after shipping but prior to surgery. The semi-circular “tube” portion **560** may be coupled to or form a portion of the insertion tube **356**. As shown in FIGS. **15I-3** and **15I-4**, the semi-circular tube portion **560** can be attached or coupled to a corresponding tubular member **556** to form a complete insertion tube **556** to surround the trocar **500**. The insertion tube **356** may replace or incorporate any of the structural or functional features of the insertion tube **356** described herein. In one embodiment, the force sufficient to cause the implants **901** to bend the adjacent tab(s) **565** inward can be provided by a pusher member **567**. FIG. **15J** illustrates an embodiment of a singulation design that is similar to the “ice cube tray” design, but the tube **560'** is fully enclosed and the opposite side of the tube **560'** also includes alternating cut-out tabs **565'**. The advancement of the implants **901** for this alternating cut-out tabs embodiment may operate similarly to that described for the “ice-cube tray” embodiment.

FIGS. **15K-1** to **15K-3** illustrate an embodiment of a spring wire singulation implementation. A spring wire **575** is positioned along the trocar **500** that includes the plurality of implants **901**. The spring wire **575** is bent at multiple spaced-apart locations in a configuration designed to keep the implants **901** separate in their natural state, but the spring wire **575** can be rotated so that the bent regions or portions **577** are rotated out of the way so as not to obstruct advancement of the implant(s) **901** during singulation, as shown in FIG. **15K-3**. The bent regions can form a triangular shape or a “shark-fin” shape or other angled shape. The embodiment illustrated in FIGS. **15L-1** to **15L-3** is similar to the spring wire singulation implementation but the wire **575'** has a flat wire shape (e.g., an elongated U shape) that can bend flat over itself out of the way during singulation. FIG. **15L-3** illustrates the wire **575'** rotated so that the bent regions or portions **577'** are rotated out of the way so as not to obstruct advancement of the implant(s) **901** during singulation.

FIG. **15M** illustrates an embodiment of a singulation implementation that includes bioerodible spacers **570** positioned along the length of the trocar **500** at spaced-apart locations. The spacers **570** may be made, for example, from a rapidly erodible biocompatible polymer. Before entering the eye, the spacers **570** keep the implants **901** in their specified starting location. After entering the eye the spacers **570** rapidly erode, thereby allowing the implants to be singulated freely.

FIGS. **15N-1** to **15N-3** illustrate an embodiment of a singulation implantation in which the implants themselves are configured (e.g. sized and designed) to facilitate singulation. Each implant **901N** includes an elongated snorkel **9011** extending proximally from a rear flange **9012** of the implant **901N**. The elongated snorkel **9011** on this stent design provides a built-in separation of the implants **901N**. The elongated snorkel **9011** provides a zone for a component of the singulation actuator **214** to land as the singulation actuator **214** is retracted, and then as it moves forward it pushes on the rear flange **9012** of the implant **901N**. FIG. **15N-3** illustrates an embodiment of the implant **901N**. The length of the elongated snorkel **9011** can vary depending on the desired spacing between the implants **901N**.

FIGS. **16A-16F** illustrate configurations of components of the multiple-implant delivery apparatus **200** at various stages before, during, and after singulation of an implant to help illustrate how singulation is carried out in the illustrated embodiment. FIGS. **16A** and **16B** illustrate an example first

or initial position or configuration of several of the components of the multiple-implant delivery apparatus **200**. For example, this first or initial position or configuration may be how the multiple-implant delivery apparatus **200** is shipped for initial use. In this initial configuration, the delivery apparatus **200** is ready to deliver the first implant. FIGS. **16C** and **16D** illustrate an example second position or configuration of the internal components during singulation of a second implant after delivery of the first implant. In this second position or configuration, the delivery apparatus **200** has isolated the second implant for delivery but has not yet advanced the second implant to the “ready-to-fire” position along the trocar **500**. FIGS. **16E** and **16F** illustrate a third position or configuration of the components after singulation of the second implant has been completed and the second implant has been moved to the “ready-to-fire” position, which is the same position on the trocar **500** as the “ready-to-fire” position that the first implant was in for the initial position or configuration of FIGS. **16A** and **16B**. In some embodiments, the transition between the second and third position or configuration is performed in a continuous manner (e.g., fluid or uninterrupted) and not considered as separate, discrete steps or configurations from an operator’s perspective.

First, the handle **436** of the singulation actuator **214** can receive a force, such as a manual rearward force **600** initiated by a thumb or finger of a user. As the singulation actuator **214** translates proximally (e.g., rearwardly or toward the user, or operator) relative to the frame **310**, the distal singulation spring connector **452** translates proximally. As mentioned above, the distal spring portion **442** of the spring **332** is in tension and the proximal spring portion **440** of the spring **332** is also in tension when in a normal resting configuration.

As the proximal spring portion **440** of the spring **332** reaches its minimum tension when the singulation actuator **214** continues to move proximally within the singulation actuator track **424** of the frame **310**, the proximal portion **430** of the singulation actuator **214** pushes the singulation arm **334** (e.g., the proximal singulation slot member **456**) proximally. Once the proximal singulation slot member **456** slides along one of the platforms **414** (e.g., the first platform **414A**) within the singulation frame slot **412**, the proximal singulation slot member **456** reaches the proximal end of the platform **414A** (see FIGS. **16C** and **16D**). Due to the force applied by the proximal spring portion **440**, the proximal singulation slot member **456** slides downwardly along the tapered rear edge of the platform **414A**. The tapered rear edge of the first platform **414A** can cause the proximal singulation slot member **456** to slide distally into contact with the distal end portion of the second platform **414B**. By releasing the singulation actuator **214**, the tension stored in the distal spring portion **442** causes the singulation actuator **214** to return to its original position (e.g., the ready-to-fire position as shown in FIGS. **16E** and **16F**).

At the same time, when the singulation actuator **214** moves proximally, the distal singulation arm collet holder member **458** moves proximally within the rear collet holder recess **484** and contacts a proximal wall of the rear collet holder recess **484**. When the singulation actuator **214** continues to move proximally within the singulation actuator track **424** of the frame **310**, the proximal portion **430** of the singulation actuator **214** pushes the singulation arm **334** (e.g., the distal singulation arm collet holder member **458** and the proximal singulation slot member **456**) proximally. This causes the collet holder subassembly **358** to slide proximally relative to the insertion tube carrier **354** as the

31

distal singulation arm collet holder member **458** contacts the proximal wall of the rear collet holder recess **484**. When the collet holder subassembly **358** slides proximally relative to the insertion tube carrier **354**, the splayed distal end **492** of the singulation tube **362** and the singulation tube slot **490** pass over the next implant loaded on the trocar **500** until the distal terminus of the singulation tube **362** is positioned proximal to the proximal end of the next implant (as shown in the close-up detailed view of FIG. **16C** showing the distal end of the delivery apparatus **200**).

When the singulation actuator **214** is released (e.g., such as when a manual rearward force **600** is removed), the collet holder **360** moves distally a distance until the collet holder notch **488** and the insertion tube carrier notch **473** are in alignment. In this position, the distal end of the singulation tube **362** is positioned immediately adjacent the proximal end of the implant and in a “ready to fire” position (as shown in the close-up detailed view of FIG. **16E** showing the distal end of the delivery apparatus **200**). The insertion tube **356** is not shown in FIGS. **16A-16E** so that internal components can be viewed.

C. Implant Delivery Assemblies

The multiple-implant delivery apparatus **200** can include the implantation actuator assembly **370**. The implantation actuator assembly **370** can include the implant delivery actuator **216**, an actuator biasing member **372**, and an actuator arm **374**. As described above, in some embodiments, at least a portion of the implant delivery actuator **216** extends through the opening **212**, such as a trigger portion **510** (e.g., button) of the implant delivery actuator **216**. The implant delivery actuator **216** can be easily manipulated and/or accessible by the user. In some embodiments, actuation of the implant delivery actuator **216** causes the ejection of an implant (e.g., one implant singulated as a result of actuation of the singulation actuator **214**) out of the insertion tube **356** of the delivery apparatus and into a desired first location within the patient’s internal eye tissue. The implant delivery actuator **216** interfaces with internal components, such as other components or subassemblies of the multiple-implant delivery apparatus **200** or other components of the actuator assembly **370**, to effect delivery of the implants. In some embodiments, the implant delivery actuator **216** is configured to allow for an infinite number of actuations since it does not rely on pre-stored energy.

FIG. **17** illustrates an embodiment of the implantation actuator assembly **370** including the implant delivery actuator **216** and the actuator biasing member **372**. The implant delivery actuator **216** can include a trigger portion **510** (e.g., button), and a main body portion **514**. The trigger portion **510** can be positioned at a distal end portion **512** of the implant delivery actuator **216**. The trigger portion **510** can include a recess and/or surface to receive a user’s finger or thumb to allow the user to easily find and press on the implant delivery actuator **216**. The trigger portion **510** can extend upwardly from the main body portion **514**. The trigger portion **510** can extend through the opening **212** to be easily accessed by the user.

In some embodiments, the implant delivery actuator **216** includes a trigger connection portion **516**. The trigger connection portion **516** can connect the trigger portion **510** with the main body portion **514**. The trigger connection portion **516** can extend laterally from the main body portion **514** and/or upwardly from the main body portion **514**. In some embodiments, the trigger connection portion **516** includes a biasing member opening **518**. The biasing member opening **518** can be defined by the space formed between a top wall **520** of the opening and a biasing member platform **522** of

32

the implant delivery actuator **216**. In some embodiments, the biasing member opening **518** is configured to receive and support the actuator biasing member **372**.

As shown in FIG. **18**, the actuator biasing member **372** can be generally T-shaped. For example, the actuator biasing member **372** can have a first portion **372A** and a second portion **372B**. The first portion **372A** can extend in a generally longitudinal direction. The second portion **372B** can extend laterally outwardly from either side of the first portion **372A** at one end of the first portion **372A**. The actuator biasing member **372** can include a spring, such as a leaf spring, a metal, plastic, or other material. The spring can advantageously be pre-loaded so that the spring already has about 30% of the spring force included. In some embodiments, a single spring can be used for both activation and trigger reset. The energy required to effect delivery of an implant may be provided by the actuator biasing member **372** (e.g., spring). The energy is generated in the actuator biasing member **372** (e.g., spring) from the pressing of the trigger portion **510** (e.g., button).

Referring back to FIG. **17**, the biasing member **372** can be configured to be positioned within the biasing member opening **518**. As shown, the actuator biasing member **372** can be positioned such that at least a portion of the first portion **372A** is positioned on one side (e.g., a distal side) of the top wall **520** and at least a portion of the second portion **372B** and/or the first portion **372A** is positioned on the other side (e.g., a proximal side) of the top wall **520**. The portion positioned on the proximal side of the top wall **520** can be configured to be supported at least partially by the biasing member platform **522**. In some embodiments, at least a portion of the top wall **520**, such as a proximal edge **520A** of the top wall **520** exerts a force on the biasing member **372** when assembled.

In some embodiments, the distal portion **512** includes an actuator foot **552** that extends outwardly from the main body portion **514** and is configured to contact a top wall of the lower arm portion **532** of the actuator arm **374**, as explained below.

In some embodiments, the main body portion **514** of the implant delivery actuator **216** extends in a proximal direction from the trigger portion **510**. The main body portion **514** can have a distal region **514A** and a proximal region **514B**. The distal region **514A** can have a generally straight top surface and a generally tapered bottom surface. The proximal region **514B** can have a vertical thickness that is generally thinner than the distal region **514B**. The proximal region **514B** and the distal region **514A** can be connected by an implant delivery actuator pivot member **524**. In some embodiments, the connection between the proximal region **514B** and the distal region **514B**, and/or the shape of the proximal and/or distal regions **514B**, **514A** is configured to allow the proximal region **514B** to flex in use relative to the distal region **514B**. The implantation delivery actuator pivot member **524** can extend outwardly from the main body portion **514**. The implantation delivery actuator pivot member **524** can be generally circular, among other shapes. The implantation delivery actuator pivot member **524** can be configured to fit within the implant delivery actuator receptacle **408** to allow the implant delivery actuation assembly **370** to pivot in use. As described above, the implant delivery actuator **216** can include a proximal portion **513** having a proximal terminus end **515**. The proximal terminus end **515** can be shaped to fit within the implant delivery actuator slot **410** when assembled with the frame **310**. Such configurations can help to limit lateral movement of the implant delivery actuator slot **410** when assembled. In some embodi-

33

ments, when the implant delivery actuator **216** has been actuated, proximal terminus end **515** of the actuator **216** contacts the portion of the frame **310** positioned between the first prong **410A** and second prong **410B**. As described below, when the user releases the implant delivery actuator **216**, the deflection in the proximal region **514B** caused by contact with the portion of the frame **310** helps to cause the implant delivery actuator **216** to rotate back into an initial position.

FIG. **19** illustrates an embodiment of the actuator arm **374**. The actuator arm **374** can be generally U-shaped, among other shapes. The actuator arm **374** can form a spring, or other biasing member to bias the delivery implant actuator **216**. The actuator arm **374** can include an upper arm portion **530** and a lower arm portion **532** connected by an arced portion **534**. The upper arm portion **530** can include a top surface **536** and a bottom surface **533**. The top surface **536** can include an upward extension portion **538** that extends upwardly from the top surface **536**. The upward extension portion **538** can have a generally triangular shape. The upward extension portion **538** can be configured to contact a distal end portion **540** of the first portion **372A** of the actuator biasing member **372** when activating the implantation actuator assembly **370**.

In some embodiments, the upper arm portion **530** includes an actuator arm connection opening **542**. The actuator arm connection opening **542** passes through a width of the actuator arm **374**. The actuator arm connection opening **542** is configured to receive the actuator arm attachment member **418** when the actuator arm **374** is assembled with the frame **310**. In some embodiments, the actuator arm attachment member **418** is sized and/or configured to snap to or otherwise reside within at least a portion of the actuator arm **374**, such as the actuator arm connection opening **542**. As explained in more detail below, the actuator arm **374** can be configured to pivot about the actuator arm attachment member **418**.

In some embodiments, the lower arm portion **532** includes an actuator arm protrusion member **544**. The actuator arm protrusion member **544** can extend outwardly from the lower arm portion **532**. The actuator arm protrusion member **544** can be generally rectangular shaped. The actuator arm protrusion member **544** can be positioned generally above the central collet holder recess **482** in an initial position when the actuator arm **374** is assembled with the frame **310**. When assembled to the frame **310**, the actuator arm protrusion member **544** can contact the actuator arm protrusion platform **422**. In some embodiments, the actuator arm protrusion platform **422** is adapted to support at least a portion of the actuator arm **374**.

In some embodiments, the lower arm portion **532** includes a stepped region **548** at the distal end portion of the lower arm portion **532**. For example, the stepped region **548** includes a distal facing side **548A** and an upwardly facing side **548B**. The upwardly facing side **548B** is generally aligned with a top surface of the actuator arm protrusion member **544**. The distal facing side **548A** is configured to contact a proximal end portion of the platform **422** of the frame **310** when assembled.

As mentioned above, the singulation actuator **214** facilitates the user-initiated singulation, isolation, and/or selection of one of the multiple implants for delivery one at a time. The singulation actuator **214** interfaces with internal components (described above) to effect singulation. After singulation, the next implant may be actuated for delivery.

FIGS. **20A-20D** illustrate an embodiment of how various components of the multiple-implant delivery apparatus **200**

34

interact to effect delivery of an implant. The same process can occur for delivery of any or all of the implants. FIG. **20A** illustrates the components prior to initiation of implant delivery when the components are in an initial resting configuration.

As mentioned above, the implant delivery actuation assembly **370** is configured to allow for an infinite (e.g., unlimited) number of actuations. With reference to FIG. **20B**, when the implant delivery actuator trigger portion **510** receives a downward force **550** (e.g., press by a user's thumb or finger), the implantation delivery assembly (e.g., the implant delivery actuator **216** and the actuator biasing member **372**) attempts to rotate about the pivotable connection between the implant delivery actuator **216** (the implant delivery actuator pivot member **524**) and the frame **310** at the implant delivery actuator receptacle **408** (shown best in FIG. **7A**). As mentioned above, when this occurs, the proximal region **514B** (shown best in FIG. **17**) of the implant delivery actuator **216** deflects slightly when the proximal terminus end **515** (shown best in FIG. **17**) of the implant delivery actuator **216** presses against the portion of the frame **310** positioned between the first prong **410A** and second prong **410B** (shown best in FIG. **7A**).

In some embodiments, downward movement of the trigger portion **510** causes at least a portion of the distal end portion **540** of the first portion **372A** (e.g., distal terminus or a portion spaced one or two millimeters away from the distal terminus) of the actuator biasing member **372** (shown best in FIG. **18**) to contact and/or press down on the upwardly extending portion **538** of the actuator arm **374**. In some embodiments, continued downward movement of the implant delivery actuator trigger portion **510** causes the actuator lever, or foot, **552** to contact a top, or upper, wall of the lower arm portion **532** of the actuator arm **374**. Such contact can cause the implant actuator arm protrusion member **544** to slide into the central collet holder recess **482** and the distal facing side **548A** of the stepped region **548** to contact a proximal end portion of the platform **422** of the frame **310**.

As shown in FIG. **20B**, the actuator arm **374** is unable to rotate due to the mating of the sear surfaces between the proximal end portion of the platform **422** of the frame **310** and the distal facing side **548A** of the stepped region **548** at the distal end portion of the lower arm portion **532** of the actuator arm **374**. With reference to FIG. **20C**, in some embodiments, continued downward movement of the trigger portion **510** generates enough pressure on the lower arm portion **532** of the actuator arm **374** to cause it to flex, which causes the distal facing side **548A** of the lower arm portion **532** of the actuator arm **374** to slide downwardly along the proximal end portion of the platform **422** until the sear surfaces disengage, thereby allowing the actuator arm **374** to rotate about the pivotable connection between the actuator arm **374** (the actuator arm connection opening **542**) and the frame **310** at the actuator arm attachment member **418**. Once the distal facing side **548A** passes the proximal end portion of the platform **422**, the spring force in the biasing member **372** that is captured and converted to energy from the downward movement of the trigger portion **510** causes the implantation actuator arm protrusion member **544** to move forward a short distance, contacting a proximal side wall of the central collet holder recess **482** (see FIG. **20D**). This causes distal movement of the collet holder **360**, causing the singulation tube **362** to move distally and push an implant towards the end of the trocar **500** to deliver the implant into the proper position within eye tissue. At this stage, the collet holder notch **488** is positioned distally of the carrier notch

35

473 (not shown). The trigger portion 510 can be actuated an infinite (e.g., unlimited) number of times to push the implant into the proper position in the eye since the mechanism does not rely on a pre-stored, limited supply of energy. When the trigger portion 510 is released, the proximal region 514B of the implant delivery actuator 216 unbends and the actuator foot 552 lifts a bottom surface 533 of the upper arm portion 530 of the actuator arm 374 (see FIG. 19), rotating the actuator arm 374 into the initial position or configuration and allowing the collet holder 360 to translate proximally into the initial position or configuration. At this point, the collet holder notch 488 and the carrier notch 473 are aligned again. Accordingly, no collet return spring is required to reset the collet holder 360 to the initial position or configuration. The collet holder 360 is held in a forward position until the operator releases the trigger portion 510; it does not automatically return to the initial position or configuration.

IV. Example Method of Use of Multiple-Implant Delivery Apparatus

FIG. 21A illustrates an example method 700 of operating the multiple-implant delivery apparatus 200. At block 702, a user or operator, such as a doctor or clinician, can insert a portion of the multiple-implant delivery apparatus into the patient's eye and advance it to a location adjacent a desired implantation site. The user can grip the multiple-implant delivery apparatus 200 and rotate their hand so that their elbow is positioned by a side of the user's body. In some embodiments, the apparatus is rotated significantly in the user's hand (e.g., rotated right between 10 and 90 degrees with respect to a central longitudinal axis of the apparatus 200).

FIG. 21B illustrates the insertion of the multiple-implant delivery apparatus 200 within the eye 100. In one embodiment, a small self-sealing incision or opening is made in the cornea 112 at or near the limbus or in other external surface area of the eye. In certain embodiments, the introducer assembly 201 is inserted from a site transocularly situated from the desired implantation site, at block 702. The distal introducer tip 208 is then advanced through the incision or opening across the anterior chamber 120 toward the desired implantation site within the trabecular meshwork 121 (e.g., under gonioscopic guidance). Although FIG. 21B illustrates an ab interno method of insertion, it should be appreciated that ab externo methods of insertion are also contemplated.

Referring back to FIG. 21A, at block 704, upon reaching the desired implantation site adjacent or in the trabecular meshwork 121, the user presses the implant delivery actuator 216 to deliver the pre-loaded implant (e.g., a first pre-loaded implant). The advancement position can be determined by visualization (e.g., imaging or fiberoptic) or tactile methods or by depth markings or a depth stop. At this point, the first implant is ready to be delivered to the desired implantation site upon depression of the implant delivery actuator 216 by the user. The implant is delivered according to methods described above.

Optionally, the operator can deliver a second implant according to a method 705 of delivering the second implant. At block 706, the operator can manually actuate the singulation actuator 214 according to methods described above to select the second implant 901B. At block 708, the operator can remove the trocar 500 from the first implantation site in the internal eye tissue. The multiple-implant delivery apparatus 200 can then be moved to facilitate delivery to a second desired implantation site for the second implant 901B within the same eye by simply withdrawing the apparatus 200

36

within the eye and repositioning and/or reorienting the apparatus 200 (e.g., rotating the operator's hand to rotate the apparatus 200 back left to a straight configuration with elbow against the operator's side). Thus, the multiple-implant delivery apparatus 200 can advantageously deliver two ocular implants at multiple locations within the eye without necessitating removal of the apparatus 200 from the eye to reload another implant. Once the multiple-implant delivery apparatus 200 is positioned in the correct position, the implant can be delivered at block 710 by actuating the implant delivery actuator 216 according to methods described above. In some embodiments, the implant delivery actuator 216 is configured to allow for an infinite number of actuations of a single implant until the second implant 901B is properly positioned within the eye.

Optionally, the operator can deliver a third implant according to a method 707 of delivering the third implant. At block 712, the operator can manually actuate the singulation actuator 214 according to methods described above to select the third implant 901C. At block 714, the operator can remove the trocar 500 from the second implantation site in the internal eye tissue. The multiple-implant delivery apparatus 200 can then be moved to facilitate delivery to a third desired implantation site for the third implant 901C within the same eye by simply withdrawing the apparatus within the eye and repositioning and/or reorienting the apparatus 200 (e.g., operator lifting his or her elbow outwardly away from the operator's side and then rotating the operator's hand to the left by between 10 and 90 degrees to rotate the apparatus 200 left). Thus, the multiple-implant delivery apparatus 200 can advantageously deliver three ocular implants at multiple locations within the eye without necessitating removal of the apparatus from the eye to reload another implant. Once the multiple-implant delivery apparatus 200 is positioned in the correct position, the implant can be delivered at block 716 by actuating the implant delivery actuator 216 according to methods described above. In some embodiments, the implant delivery actuator 216 is configured to allow for an infinite number of actuations until the third implant 901C is properly positioned within the eye.

Optionally, the operator can deliver a fourth implant according to a method 709 of delivering the fourth implant. At block 718, the operator can manually actuate the singulation actuator 214 according to methods described above to select the fourth implant 901D. At block 720, the operator can remove the trocar 500 from the third implantation site in the internal eye tissue. The multiple-implant delivery apparatus 200 can then be moved to a fourth desired implantation site for delivery of the fourth implant 901D within the same eye by simply withdrawing the apparatus within the eye and repositioning and/or reorienting the apparatus 200. Thus, the multiple-implant delivery apparatus 200 can advantageously deliver four ocular implants at multiple locations within the eye without necessitating removal of the apparatus from the eye to reload another implant. Once the fourth implant 901D is selected, the singulation actuator 214 includes a stop to prevent the operator from pulling back on the actuator to select another implant. Once the multiple-implant delivery apparatus 200 is positioned in the correct position, the fourth implant can be delivered at block 722 by actuating the implant delivery actuator 216 according to methods described above. In some embodiments, the implant delivery actuator 216 is configured to allow for an infinite number of actuations until the fourth implant is properly positioned within the eye.

At block 724, the multiple-implant delivery apparatus 200 can be removed from the eye (after delivery of one implant,

two implants, three implants or four implants). In some embodiments, the singulation and positioning steps can be reversed (i.e., singulation can be performed after relocating the apparatus to a location adjacent a subsequent desired implantation site). In accordance with several embodiments, every successive implantation appears (e.g., location of implant in visualization slot) and feels the same to the clinician using the multiple-implant delivery apparatus **200** regardless of how many implants are included in the apparatus or actually implanted because of the manual actuation of the singulation.

V. Example Implants and Implantation Locations

FIG. **22** illustrates an example implant implanted in the eye. The aqueous humor flows from the anterior chamber **120**, through an inlet lumen **908**, and then out through one, two or more outlet ports **906** (e.g., four outlet ports **906A**, **906B**, **906C** and a fourth outlet port not seen opposite outlet port **906C**) to be directed in both directions along Schlemm's canal **122**. In some embodiments, the implant **901** includes an axial outlet port in communication with the inlet lumen **908** that is located along a distal end **902B** to potentially direct flow in an axial direction if the distal end **902B** is not obstructed. Alternatively, flow could be directed in only one direction through a single outlet port **906A** or flow could be directed in two directions through two outlet ports **906A** and **906B**, depending on a rotational position of the implant **901** within Schlemm's canal or other physiologic outflow pathway upon implantation. More than two outlet ports **906** can be efficaciously used, as needed or desired to increase outflow or reduce the potential for obstruction of the outlet ports to flow within Schlemm's canal **122**. For example, in some embodiments, four outlet ports can be oriented at 90 degrees with respect to the inlet lumen **908** and with respect to adjacent outlet ports such that an outlet port is positioned at every 90 degree rotation of the implant **901**. The use of four or more outlet ports may increase the likelihood that at least two outlet ports are oriented to facilitate flow within Schlemm's canal **122** without rotational adjustment or orientation after delivery or implantation. The proximal end of the distal portion **907** can abut the inner wall **912** of Schlemm's canal **122**, and the distal end of the proximal portion **904** can abut the trabecular meshwork **121** upon delivery. Accordingly, the implant **901** can be secured in place by the proximal and distal portions of the implant **901** abutting opposite sides of the trabecular meshwork **121**. In some embodiments, the distal end **902B** is in contact with the outer wall **914** of Schlemm's canal **122**. As described above, the implant delivery actuator **216** of the multiple-implant delivery apparatus **200** may be actuated multiple times until the implant **901** is in the desired final implantation location. In alternative embodiments, the implant **901** can be implanted such that an outlet of the implant **901** is positioned in a physiologic outflow pathway other than Schlemm's canal **122**.

At least some of the disclosed embodiments include implants that provide a fluid flow path for conducting aqueous humor from the anterior chamber of an eye to a physiologic outflow pathway to reduce intraocular pressure, preferably below episcleral venous pressure without hypotony. The implants can have an inflow portion and an outflow portion. The outflow portion of the implant preferably is disposed at or near a distal end of the implant. When the implant is implanted, the inflow portion may be sized and configured to reside in the anterior chamber of the eye and the outflow portion may be sized and configured to

reside in a physiologic outflow pathway. In some embodiments, the outflow portion may be sized and configured to reside in Schlemm's canal. In other embodiments, the outflow portion may be sized and configured to reside at least partially in the supraciliary region of the uveoscleral outflow pathway or the suprachoroidal space.

In some embodiments, the implants have a generally sharpened forward end and are self-trephinating, i.e., self-penetrating, so as to pass through tissue without pre-forming an incision, hole or aperture. The sharpened forward end can be, for example, conical or tapered. The tip can be sufficiently sharp to pierce eye tissue. The tip also can be sufficiently blunt so as not to substantially penetrate eye tissue. The taper angle of the sharpened end can be, for example, about $30^{\circ} \pm 15^{\circ}$ in some embodiments. The radius of the tip can be about 70 to about 200 microns. Where an outlet opening is formed at the distal end of the implant, the distal portion can gradually increase in cross-sectional size in the proximal direction, preferably at a generally constant taper or radius or in a parabolic manner. In some embodiments including an outlet opening at the distal end, the diameter of the axial outlet opening formed at the distal end may be between 40 and 200 microns (e.g., 40 microns, 60 microns, 80 microns, 100 microns, 120 microns, 140 microns, 160 microns, 180 microns). Additionally, in such embodiments, an annulus may be formed between an edge defined by the outer circumference of the axial outlet opening and an edge defined by the intersection of the distal tip surface and the conical or tapered section of the distal portion. The width of this annulus may advantageously be sufficiently small such that, after the trocar has created a pilot hole in eye tissue (e.g., trabecular meshwork), the distal portion can expand eye tissue surrounding the pilot hole as the implant is advanced into the eye tissue. The eye tissue can then retract around an intermediate portion of the eye implant. If the annulus width is not sufficiently small, the distal portion may potentially push, rather than expand, the eye tissue.

The implants may be of varied lengths and sizes to optimize flows. In some embodiments, the implant has sufficient length such that the outflow portion resides in a physiologic outflow pathway and the inflow portion is exposed to the anterior chamber. In some embodiments, the length of the implant from the portion residing in the anterior chamber to the portion residing in the physiologic outflow pathway may be about 0.001 mm to about 5 mm, about 0.01 mm to about 1 mm, about 0.1 mm to about 0.5 mm, or overlapping ranges thereof. In some embodiments, the length of the implant is about 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, or 0.50 mm.

The implant is preferably made of one or more biocompatible materials. Suitable biocompatible materials include, for example, polypropylene, polyimide, glass, nitinol, polyvinyl alcohol, polyvinyl pyrrolidone, collagen, chemically-treated collagen, polyethersulfone (PES), poly(styrene-isobutyl-styrene), Pebax, acrylic, polyolefin, polysilicon, polypropylene, hydroxyapatite, titanium, gold, silver, platinum, other metals, ceramics, plastics and a mixture thereof. The implants can be manufactured by sintering, micro machining, laser machining, and/or electrical discharge machining. However, other suitable manufacturing methods can be used.

If desired, more than one implant of the same or different type may be implanted. Additionally, implantation may be performed in combination with other surgical procedures, such as cataract surgery. All or a portion of the implant may be coated, e.g. with heparin, preferably in the flow path, to

reduce blood thrombosis or tissue restenosis. In accordance with several embodiments, the same multiple-implant delivery apparatus **200** may be used to deliver different sizes or types of implants without having to reconfigure or readapt the apparatus. For example, implants with different maximum outer cross-sectional dimensions may be used in the multiple-implant delivery apparatus. The different sizes or types of implants may be implanted during a single procedure at a single time or in separate procedures at separate times. For example, if desired, a multiplicity of implants having different flow capacities and/or sizes may be implanted.

Although the multiple-implant delivery apparatus **200** has been described herein with respect to delivery of ocular implants within internal eye tissue, the multiple-implant delivery apparatus **200** could be used or adapted to singulate and deliver multiples of other types of implants within the body. Examples of the ocular implants that could be used with the multiple-implant delivery apparatus **200** include implants, stents, or shunts of the shape, size and scale of those described and illustrated in U.S. Pat. Nos. 7,135,009; 9,301,875; 7,867,186; and U.S. Publication No. 2014/0276332.

VI. Kits and Additional Language

According to some embodiments, a kit (e.g., system or collection of items for a common purpose) for addressing ocular disorders is provided. The term “kit” as used herein should be given its ordinary meaning and should include any system, grouping and/or collection of devices, systems, components, features, materials and/or the like provided for a common goal. In one embodiment, the kit comprises one or more of the following: a delivery apparatus (such as the multiple-implant delivery apparatus **200** described herein), a plurality of drainage implants (such as the drainage implants described herein), an incising member, and a sensor (such as a pressure sensor, an intraocular pressure sensor, an analyte sensor, a glucose sensor, or any other sensor configured for placement within an eye). In some embodiments, the drainage implants are pre-loaded within or on the delivery apparatus during manufacture and assembly prior to shipping. In other embodiments, the drainage implants are not pre-loaded. The kit can further comprise instructions for using the various devices, components and/or other features of the kit for a particular procedure or treatment protocol. For example, such instructions for use can include details regarding the order in which the devices, systems or other components are used, the duration of use and/or the like.

While certain embodiments of the disclosure have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods, systems, and devices described herein may be embodied in a variety of other forms. For example, embodiments of one illustrated or described implant can be combined with embodiments of another illustrated or described implant. Moreover, the implants described above can be utilized for other purposes. For example, the implants can be used to drain fluid from the anterior chamber to other locations of the eye or outside the eye. In some embodiments, the systems comprise various features that are present as single features (as opposed to multiple features). Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Multiple features or components are provided in some embodiments. Some embodi-

ments have been described in connection with the accompanying drawings. However, it should be understood that the figures are not necessarily drawn to scale. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Furthermore, various omissions, substitutions and changes in the form of the methods, systems, and devices described herein may be made without departing from the spirit of the disclosure.

Conditional language, for example, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The section headings used herein are merely provided to enhance readability and are not intended to limit the scope of the embodiments disclosed in a particular section to the features or elements disclosed in that section. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

Any methods disclosed herein need not be performed in the order recited. The methods disclosed herein include certain actions taken by a practitioner; however, they can also include any third-party instruction of those actions, either expressly or by implication. For example, actions such as “advancing a delivery apparatus” include “instructing advancing a delivery apparatus.” The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers or an amount that is within less than or equal to 10% of the stated amount. For example, “about 200 microns” includes “200 microns.” Terms or phrases preceded by a term such as “substantially” or “generally” include the recited term or phrase. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. For example, “substantially V-shaped” includes “V-shaped” and “generally rounded” includes “rounded.”

For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

What is claimed is:

1. An implant delivery apparatus configured to deliver a plurality of implants for treating an ocular disorder, the apparatus comprising:

41

an external housing comprising an opening on an upper side of the external housing;

an auto-retracting introducer assembly configured to facilitate introduction of a distal portion of the implant delivery apparatus into an eye of a subject, the introducer assembly comprising a distal introducer tip and a flexible proximal retraction member extending from a distal end of the external housing;

a singulation assembly configured to facilitate on-demand singulation of each of the plurality of implants upon manual actuation by an operator of a lever extending out of the opening of the external housing; and

an implantation actuator assembly configured to effect delivery of each of the plurality of implants following singulation, the implantation actuator assembly comprising an implant delivery actuator comprising a trigger button extending out of the opening of the external housing that is configured to be actuated a plurality of times by the operator.

2. The apparatus of claim 1, further comprising an insertion tube extending from a distal end of the external housing at an angle relative to a longitudinal axis of the implant delivery apparatus, the insertion tube configured to retain the plurality of implants therein.

3. The apparatus of claim 1, further comprising a collet holder assembly comprising a collet holder and a singulation tube extending from the collet holder, wherein a distal end of the singulation tube comprises multiple tines configured to facilitate retraction of the distal end of the singulation tube over a maximum cross-sectional dimension of a respective implant during singulation, and wherein the distal end of the singulation tube is configured to engage a proximal end of the respective implant following singulation and advance the respective implant to a ready-to-fire position.

4. The apparatus of claim 3, wherein the singulation assembly further comprises a singulation arm coupled to the lever and to the collet holder and a singulation biasing member coupled to the singulation arm and to a fixed frame within the external housing, wherein proximal retraction of the lever of the singulation assembly causes the collet holder to retract proximally and such that release of the lever causes the singulation tube to engage the proximal end of the respective implant following singulation and advance the respective implant to the ready-to-fire position.

5. The apparatus of claim 4, wherein the implantation actuator assembly further comprises an actuator arm and an actuator biasing member, wherein the energy sufficient to effect delivery of each respective implant is provided by the actuator biasing member, and wherein the energy provided by the actuator biasing member is generated from pressing of the trigger button of the implant delivery actuator by the operator.

6. The apparatus of claim 4, wherein the frame comprises a singulation frame slot comprising a plurality of platforms sized and shaped to facilitate singulation of the plurality of implants.

7. The apparatus of claim 5, wherein the actuator biasing member comprises a flat spring.

8. The implant delivery apparatus of claim 7, wherein depressing of the trigger button causes the flat spring to bend, thereby storing the energy.

9. The implant delivery apparatus of claim 7, wherein the flat spring is positioned in contact with the actuator arm so as to cause the flat spring to bend as the trigger button is depressed.

10. The apparatus of claim 4, wherein the singulation biasing member comprises a spring.

42

11. An implant delivery apparatus configured to deliver a plurality of implants for treating an ocular disorder, the apparatus comprising:

an external housing comprising an opening on an upper side of the external housing;

a singulation assembly configured to facilitate on-demand singulation of each of the plurality of implants upon manual actuation by an operator of a lever extending out of the opening of the external housing; and

an implantation actuator assembly configured to effect delivery of each of the plurality of implants following singulation, the implantation actuator assembly comprising an implant delivery actuator comprising a trigger button extending out of the opening of the external housing that is configured to be actuated a plurality of times by the operator.

12. The apparatus of claim 11, further comprising a collet holder assembly comprising a collet holder and a singulation tube extending from the collet holder, wherein a distal end of the singulation tube comprises multiple tines configured to facilitate retraction of the distal end of the singulation tube over a maximum cross-sectional dimension of a respective implant during singulation, and wherein the distal end of the singulation tube is configured to engage a proximal end of the respective implant following singulation and advance the respective implant to a ready-to-fire position.

13. The apparatus of claim 12, wherein the singulation assembly further comprises a singulation arm coupled to the lever and to the collet holder and a singulation biasing member coupled to the singulation arm and to a fixed frame within the external housing, wherein proximal retraction of the lever of the singulation assembly causes the collet holder to retract proximally and such that release of the lever causes the singulation tube to engage the proximal end of the respective implant following singulation and advance the respective implant to the ready-to-fire position.

14. The apparatus of claim 13, wherein the implantation actuator assembly further comprises an actuator arm and an actuator biasing member, wherein the energy sufficient to effect delivery of each respective implant is provided by the actuator biasing member, and wherein the energy provided by the actuator biasing member is generated from pressing of the trigger button of the implant delivery actuator by the operator.

15. The apparatus of claim 14, wherein the actuator biasing member comprises a flat spring.

16. The implant delivery apparatus of claim 15, wherein depressing of the trigger button causes the flat spring to bend, thereby storing the energy.

17. The implant delivery apparatus of claim 15, wherein the flat spring is positioned in contact with the actuator arm so as to cause the flat spring to bend as the trigger button is depressed.

18. The apparatus of claim 13, wherein the frame comprises a singulation frame slot comprising a plurality of platforms sized and shaped to facilitate singulation of the plurality of implants.

19. The apparatus of claim 13, wherein the singulation biasing member comprises a spring.

20. The apparatus of claim 11, further comprising an insertion tube extending from a distal end of the external housing at an angle relative to a longitudinal axis of the implant delivery apparatus, the insertion tube configured to retain the plurality of implants therein.